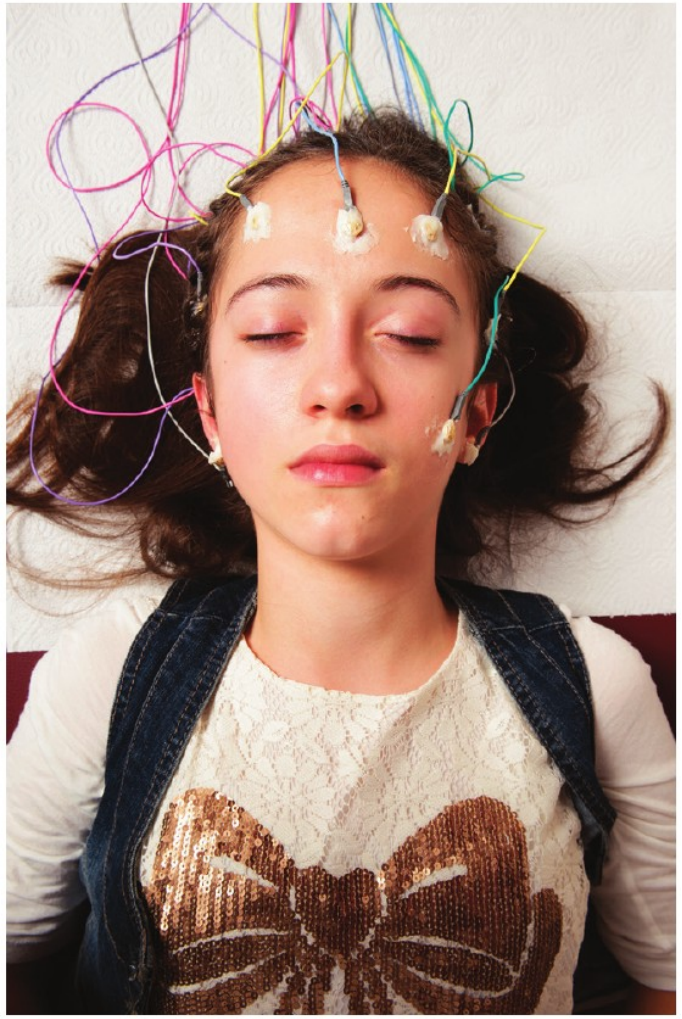


Carlson and Birkett (2022), Fig. 9.1, p. 268.

BEHAVIORAL NEUROSCIENCE 392

Sleep and Biological Rhythms

The Rhythmic Brain: Why We Sleep and How the Body Keeps Time

Fall 2026

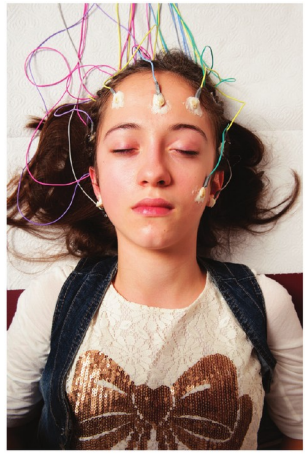
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Sleep changes responsiveness while the brain stays active

Sleep is a reversible state in which the brain changes how it responds to the surrounding world. A sleeper usually moves less and requires a stronger stimulus to respond than an alert animal. These changes occur while neurons continue to communicate, regulate the body, and generate organized patterns of activity.

Reduced movement alone cannot identify sleep. An animal may remain still while watching for prey, conserving energy, or preparing to escape. Researchers therefore measure responsiveness as well as posture and movement. They also examine whether preventing normal rest produces additional sleep afterward, a response called **sleep rebound**.

During sleep, sensory information can still influence the nervous system. A sufficiently strong sound or touch can activate arousal circuits and restore waking behavior. The quiet octopus recording provides a visible example of reduced movement, while the distinction between quiet wakefulness and sleep depends on the animal's responses and the sequence of states measured across time.



Carlson and Birkett (2022), Fig. 9.1, p. 268.



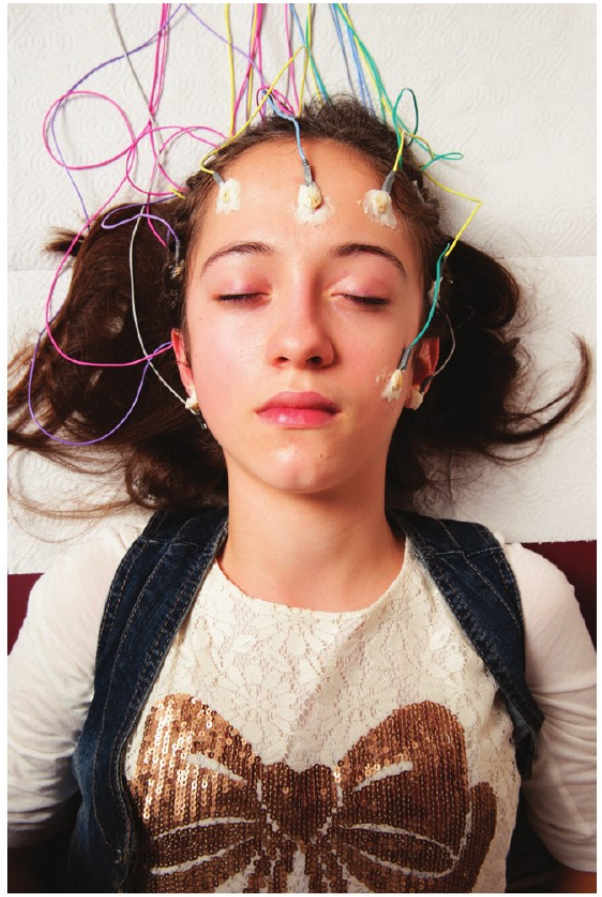
Medeiros et al. (2021), Video S2.

Several signals are needed to identify a sleep stage

Polysomnography combines several measurements to identify sleep and its stages. An **electroencephalogram**, or EEG, records voltage changes across scalp electrodes. Much of this signal reflects electrical currents associated with the coordinated activity of cortical neurons. The recording describes population activity, rather than the action potentials of one identified cell.

An **electro-oculogram**, or EOG, measures changes associated with eye movements. An **electromyogram**, or EMG, records muscle activity. Combining these signals allows a researcher to distinguish an alert person from a person whose cortical recording looks relatively active but whose postural muscles are strongly inhibited during rapid eye movement sleep.

Breathing, oxygen levels, heart rate, and body movement provide additional information. A pause in airflow can interrupt sleep even when the person does not remember awakening. A sleep laboratory therefore records several signals on the same timeline, allowing a change in breathing to be compared with a change in muscle tone or brain activity.



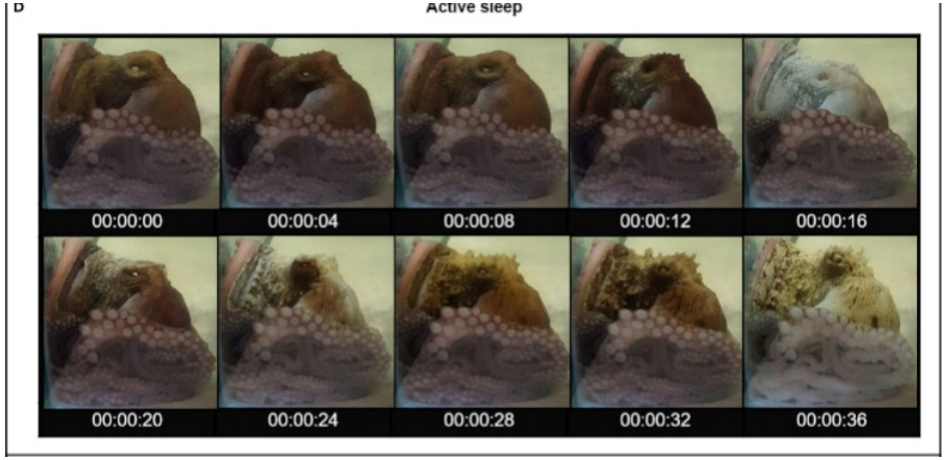
Carlson and Birkett (2022), Fig. 9.1, p. 268.

An octopus alternates between quiet and active sleep

Medeiros and colleagues recorded octopuses across long periods and classified their behavior using pupil opening, body posture, skin patterns, eye movement, and mantle movement. The animals entered a **quiet sleep** state with relatively pale, stable skin and closed pupils. These periods were often followed by brief episodes of a different, more active state.

During **active sleep**, skin color and texture changed rapidly, the eyes moved, and the body produced small movements while the animal remained in place. Active episodes were much shorter than quiet sleep and frequently followed it. The recording captures these visible changes in an octopus without requiring the animal to perform a trained task.

The researchers also presented visual or vibratory stimuli and measured how long the animals took to respond. Responsiveness differed across the observed states, with especially long response delays during active sleep. This behavioral evidence distinguishes the sleep states from an octopus that is merely motionless while remaining alert to a nearby event.



Medeiros et al. (2021), Fig. 2, selected panels.



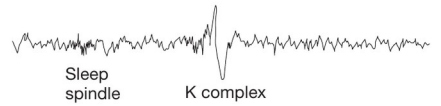
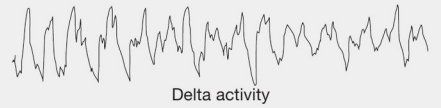
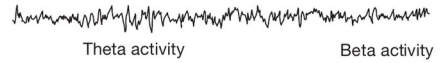
Medeiros et al. (2021), Video S4.

NREM stages have distinct electrical patterns

Human **non-rapid eye movement sleep**, or NREM sleep, includes three stages. During **N1**, the transition from waking to sleep, cortical activity becomes less like the irregular pattern of alert waking and includes more theta activity. Muscle tone remains present, and brief jerks can occur as the person becomes drowsy.

N2 includes sleep spindles, short bursts of rhythmic activity produced through interactions between thalamus and cortex. A spindle usually contains roughly twelve to fifteen cycles per second. N2 also includes **K complexes**, large isolated waveforms that can occur spontaneously or follow a sound without producing a complete awakening.

N3 contains prominent slow, high-amplitude waves and is often called slow-wave sleep. The stages describe recurring patterns of physiology rather than three different locations in the brain. A person can move between stages during the night, and a change in the EEG must be interpreted together with eye movements and muscle activity.

Waking	Alpha (8–12 Hz) and beta (13–30 Hz) activity	
Stage 1	NREM sleep, consists of theta activity (3.5–7.5 Hz)	
Stage 2	NREM sleep, contains sleep spindles and K complexes	
Stage 3	Slow-wave sleep, consists of delta activity (<3.5 Hz)	
REM	REM sleep, consists of theta and beta activity	

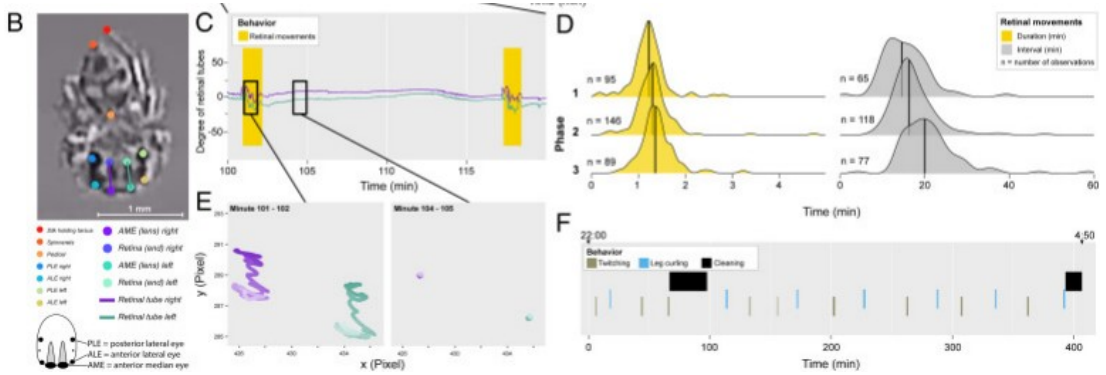
Carlson and Birkett (2022), Table 9.1, p. 270.

Retinal movements occur during a spider's rest

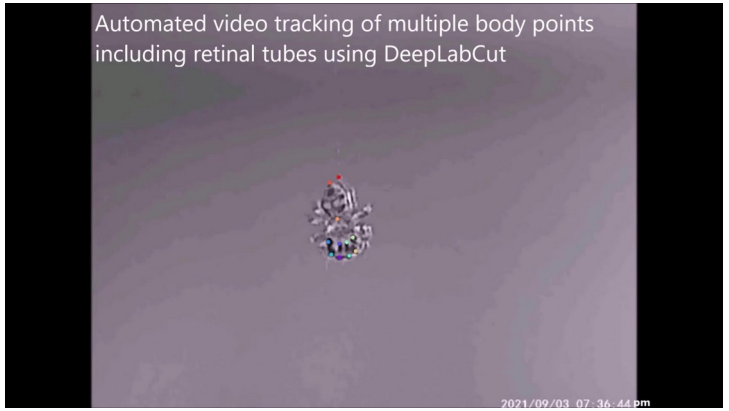
The principal eyes of jumping spiders contain movable retinal tubes behind their outer lenses. In newly emerged spiderlings, the outer body covering is temporarily translucent, making the retinal movements visible through the head. Rößler and colleagues used nighttime video to examine whether these movements occurred while the spiders were hanging in their characteristic resting position.

The researchers observed recurring bouts of retinal movement during rest. Automated tracking followed landmarks on the head and retinal tubes, allowing movement within the head to be separated from rotation of the whole spider. The source recording contains the original tracking overlay, so the changing landmark positions can be compared with the visible retinal structures.

The observed bouts had regular durations and intervals, and both tended to increase over the night. These are behavioral observations of a REM-like state, rather than an EEG classification of human REM sleep. The measurements establish when the retinal tubes move and how those movements are organized during the spider's resting period.



Rößler et al. (2022), Fig. 1, selected panels.



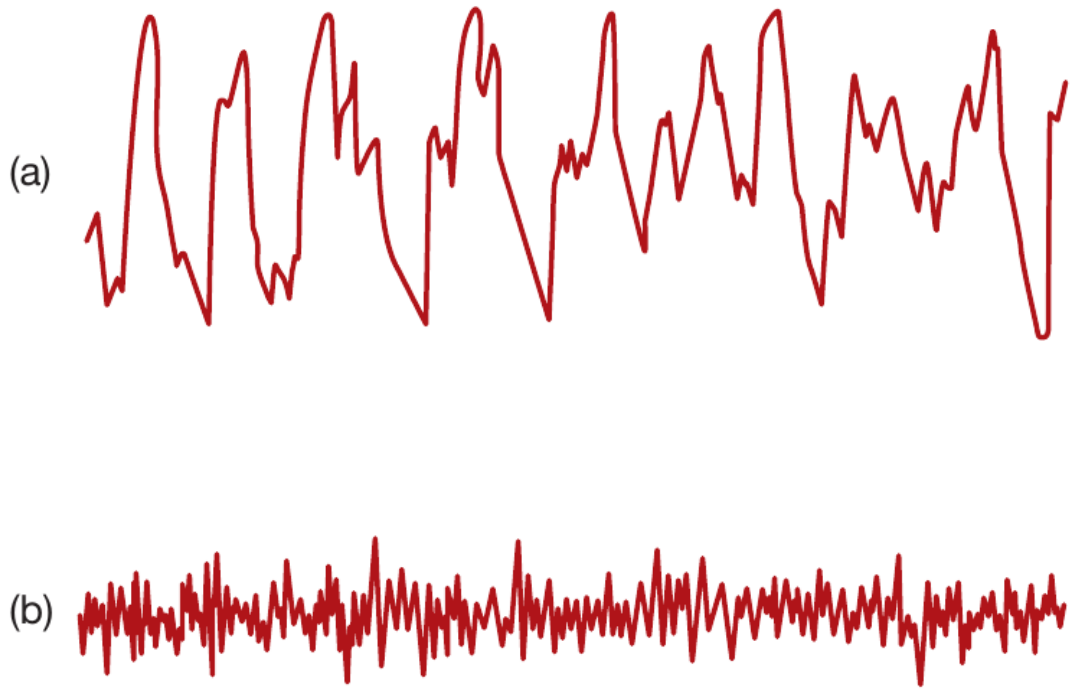
Rößler et al. (2022), Movie S1.

Slow waves reflect coordinated changes in cortical activity

A large EEG wave can arise when many cortical neurons change their electrical activity in a coordinated way. During a slow oscillation, populations alternate between relatively active **UP states** and quieter **DOWN states**. These changes influence when incoming signals can produce activity and when neurons participate in an ongoing cortical pattern.

When activity is less synchronized, electrical signals from different locations partly cancel in the scalp recording. The EEG can therefore have a smaller amplitude while many neurons remain active. A low-amplitude waking pattern does not mean that the cortex is doing less work than it does during a large slow wave.

Slow-wave activity also varies across cortical locations and across the night. A region can show stronger slow waves after substantial use or prolonged waking, while another region shows a different pattern. Researchers can measure local activity alongside the overall sleep stage to examine how a global sleeping state contains differences among individual cortical regions.



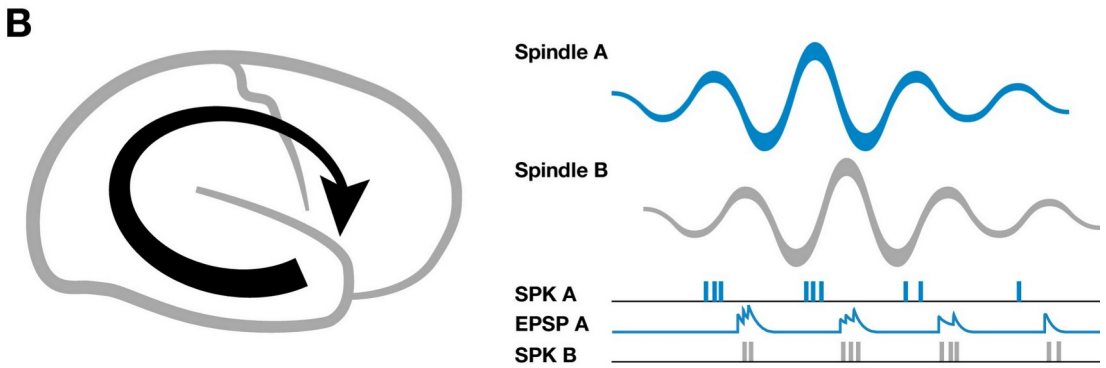
Carlson and Birkett (2022), Fig. 9.2, p. 269.

A sleep spindle can travel across the cortex

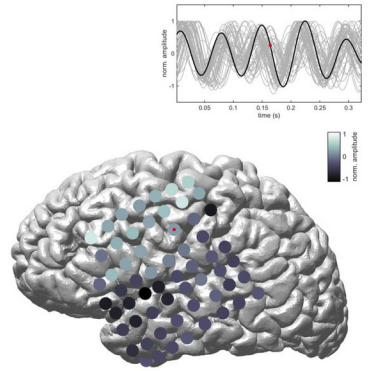
A **sleep spindle** is a brief oscillation produced by interacting thalamic and cortical circuits. Its activity need not rise and fall at exactly the same moment everywhere. Muller and colleagues analyzed electrical recordings from electrodes placed directly over the cortex of sleeping patients and examined how spindle activity changed across the recording sites.

Some spindle cycles formed rotating patterns. Activity reached one cortical location before another, creating a reproducible progression across the electrode array. The source animation displays recorded, filtered signals at the electrode positions. The changing colors represent the phase of those electrical signals, rather than a physical substance moving across the surface of the brain.

Small timing differences can affect how neurons influence one another through their connections. An input arriving when a receiving population is relatively excitable can have a different effect from the same input arriving during a quieter phase. A traveling spindle therefore creates an ordered sequence of opportunities for communication across several cortical locations.



Muller et al. (2016), Fig. 2, selected panels.



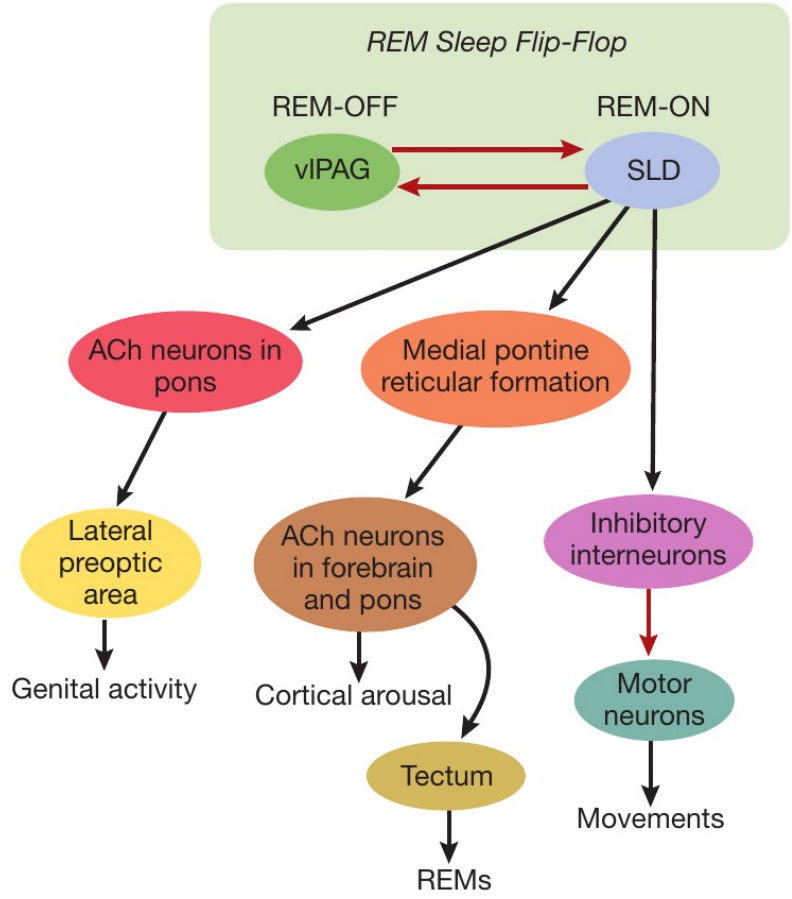
Muller et al. (2016), Video 1.

REM sleep combines activation with muscle inhibition

Rapid eye movement sleep, or REM sleep, combines relatively activated cortical activity with a marked reduction in postural muscle tone. Brainstem circuits send inhibitory signals toward motor pathways, reducing the ability of many skeletal muscles to produce sustained movement. The eyes and breathing muscles do not follow exactly the same pattern as the large postural muscles.

In the **sublaterodorsal nucleus**, or SLD, REM-active neurons help recruit medullary inhibitory pathways. Neurons downstream release **GABA** and **glycine** onto motor-control circuits, making spinal motor neurons less likely to fire. The resulting loss of sustained postural tone is called **atonia**, while separate pathways still permit breathing, eye movements, and brief twitches.

The distinction between cortical activation and motor inhibition explains why a sleep stage cannot be identified from the EEG alone. A person in REM sleep can have an active-looking cortical record while producing little postural muscle activity. Eye movements and the EMG supply the additional information needed to separate that pattern from ordinary waking.



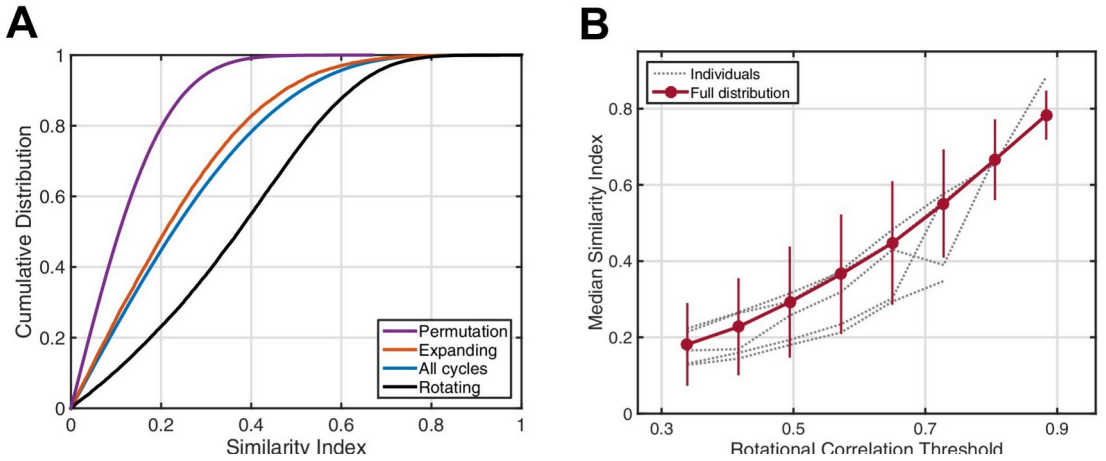
Carlson and Birkett (2022), Fig. 9.24, p. 288.

Some spindle patterns recur with precise spatial timing

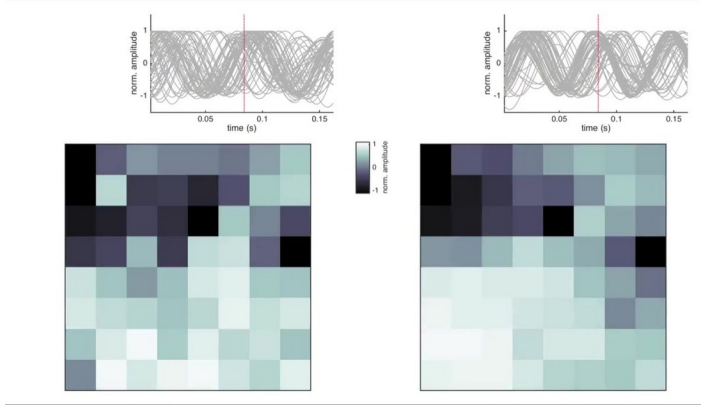
The cortical pattern of one spindle can be compared with spindles occurring later in the same sleeping period. Muller and colleagues found examples with very similar spatial timing despite their separation by several minutes. The repeated feature was the order and phase of activity across the electrode array, rather than an identical voltage at every instant.

The source comparison places two recorded waves side by side. Each recording is normalized so that its spatial progression can be compared with the other. The original signals above the maps show that the animation represents a sequence of measured oscillations, with different cortical locations reaching corresponding phases at closely matched relative times.

Repeated spatial timing could repeatedly bring the same connected populations into favorable windows for communication. The anatomical connections between areas impose travel times on their signals, while spindle phase changes the receiving populations' excitability. Matching these times is a proposed mechanism for coordinating distributed activity; the recordings measure the timing pattern itself.



Muller et al. (2016), Fig. 3, selected panels.



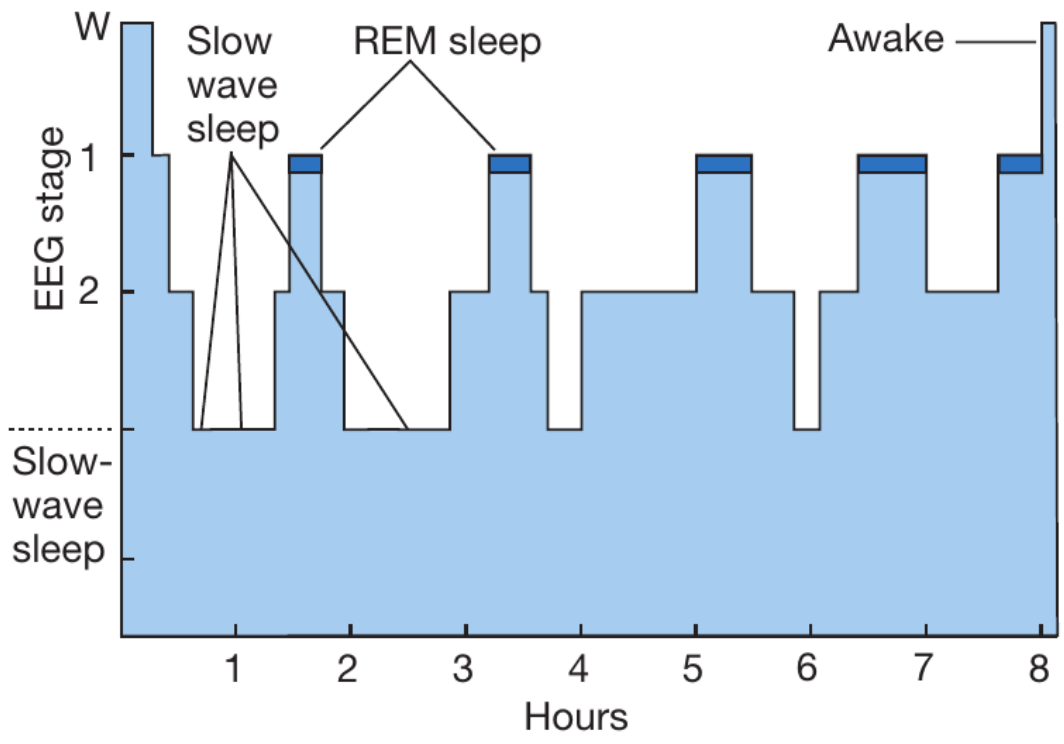
Muller et al. (2016), Video 2.

Sleep architecture changes across the night

A **hypnogram** plots sleep stage against time. Early in a typical human night, sleep progresses through NREM stages and includes substantial N3 sleep. REM sleep appears later in the cycle, followed by a return to NREM or a brief awakening. This sequence repeats rather than remaining at one depth until morning.

The proportions change across the night. Slow-wave sleep is concentrated more strongly in the earlier cycles, while REM episodes tend to become longer later. A person awakened near the beginning and near the end of the same night can therefore be sampled from quite different physiological conditions, even when both awakenings occur after a period of apparent stillness.

A cycle often lasts around an hour and a half, but its duration varies across people and across the night. Brief awakenings are also common and may not be remembered. Examining the full hypnogram reveals whether sleep is consolidated, repeatedly interrupted, shifted to an unusual time, or missing an expected stage pattern.



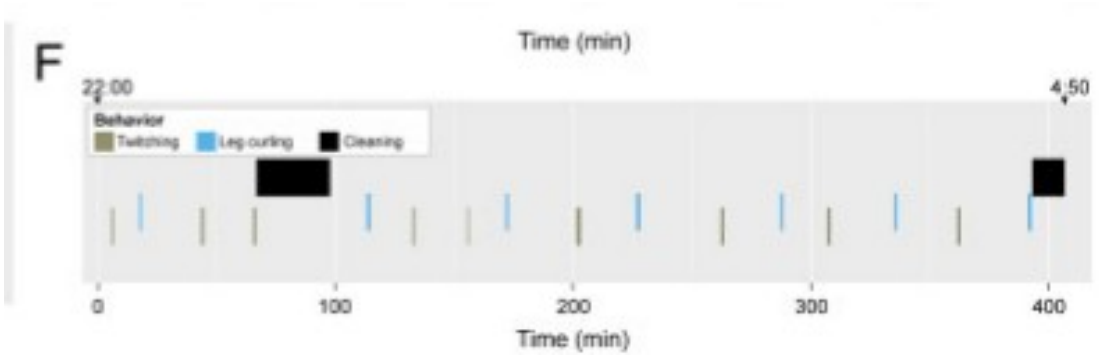
Carlson and Birkett (2022), Fig. 9.3, p. 271.

Twitches and retinal movements can occur together

Retinal movements in sleeping spiderlings can occur with movements elsewhere in the body. Rößler and colleagues described leg curling and small twitches during the same REM-like bouts. The recording follows an original bout in which the retinal tubes move while the legs change position, allowing the timing of these visible events to be compared.

Leg curling differs from coordinated actions such as cleaning a leg or extending a limb to change the resting position. The researchers scored these behavior categories separately. This separation prevents every small movement during the night from being counted as the same event and allows the recurring twitching bouts to be distinguished from other resting behaviors.

The body covering of an adult spider is less transparent, so retinal movements are harder to observe directly. However, adults also produced characteristic curling and twitching behaviors. Observations at different ages therefore provide complementary measurements: spiderlings make internal eye movements visible, while adult recordings establish that the associated body movements are not restricted to newly emerged animals.



Rößler et al. (2022), Fig. 1, selected panels.



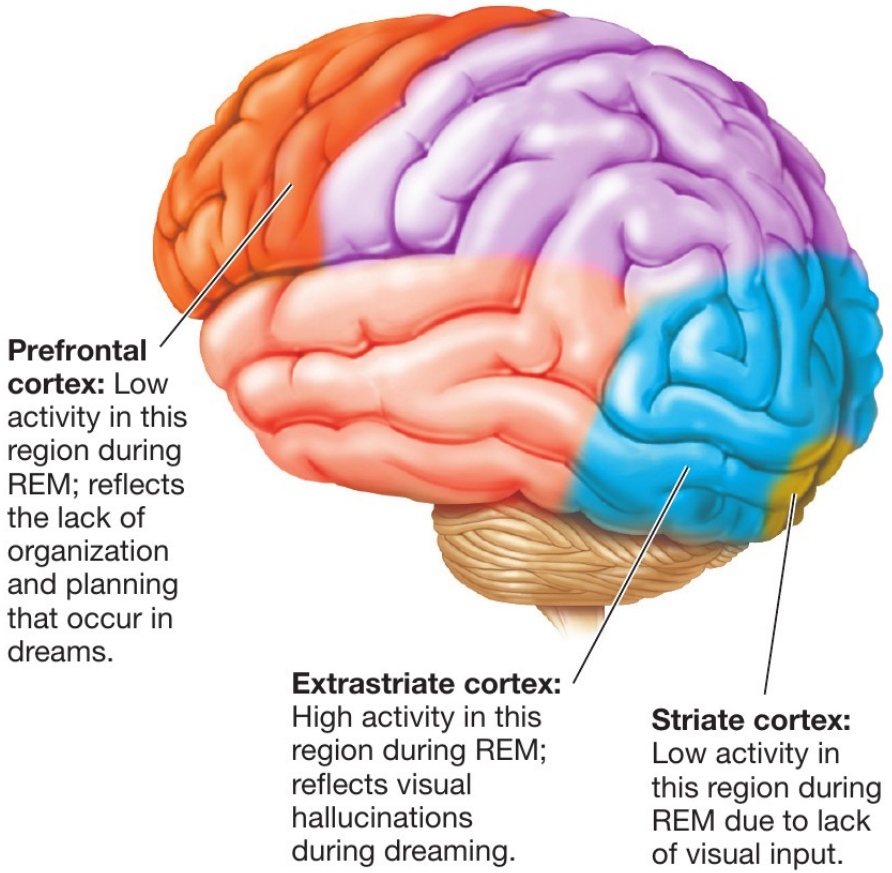
Rößler et al. (2022), Movie S3.

Dream reports depend on when a sleeper is awakened

A **dream report** is a person's description of an experience after awakening. Reports from REM sleep often contain vivid scenes, emotions, and a sequence of events. Experiences can also be reported after NREM awakenings. Dreaming therefore cannot be defined simply as the presence of rapid eye movements or the complete absence of NREM sleep.

During REM sleep, visual association regions can be active while some prefrontal regions show reduced activity compared with waking. This pattern can help explain why a dream contains convincing visual events while unusual transitions or contradictions are accepted. The activity pattern concerns several interacting regions, rather than one small structure that generates every part of a dream.

The timing of awakening changes what can be reported. A person may describe an experience immediately after an experimental awakening but forget much of it later in the morning. Researchers distinguish the occurrence of an experience, its storage in memory, and the ability to describe it after waking because those processes can vary independently.



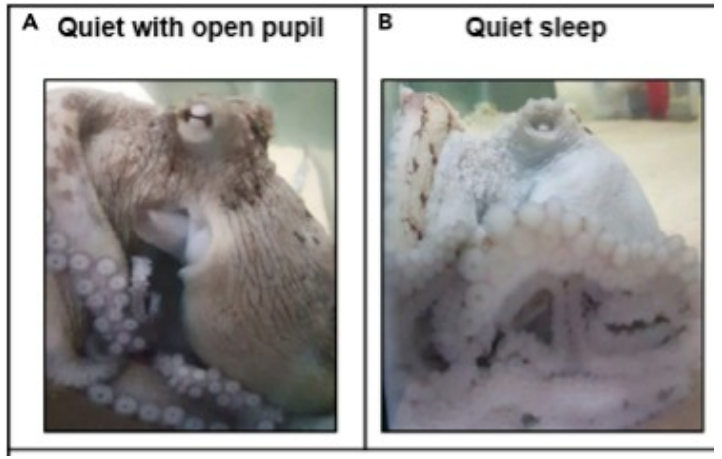
Carlson and Birkett (2022), Fig. 9.4, p. 271.

Quiet behavior alone does not establish sleep

A motionless octopus does not always occupy the same physiological state. Medeiros and colleagues distinguished quiet behavior with an open pupil from quiet sleep with a closed pupil, as well as other recognizable patterns. Their observations combined several features so that a brief pause in movement was not automatically treated as a sleep episode.

The recording here contains the open-pupil quiet state. Comparing it with quiet sleep reveals a difference that a simple movement counter could miss. The eye remains available for visual sampling even though the body is relatively still, while skin pattern and mantle movements provide additional information about the animal's behavior at that moment.

Stimulus testing adds a functional comparison to those visible features. An animal that responds quickly to an approaching event differs from one that requires stronger stimulation or responds after a longer delay. Combining the state descriptions with these response measurements allows quiet wakefulness and sleep to be separated without assigning them solely from the animal's posture.



Medeiros et al. (2021), Fig. 2, selected panels.



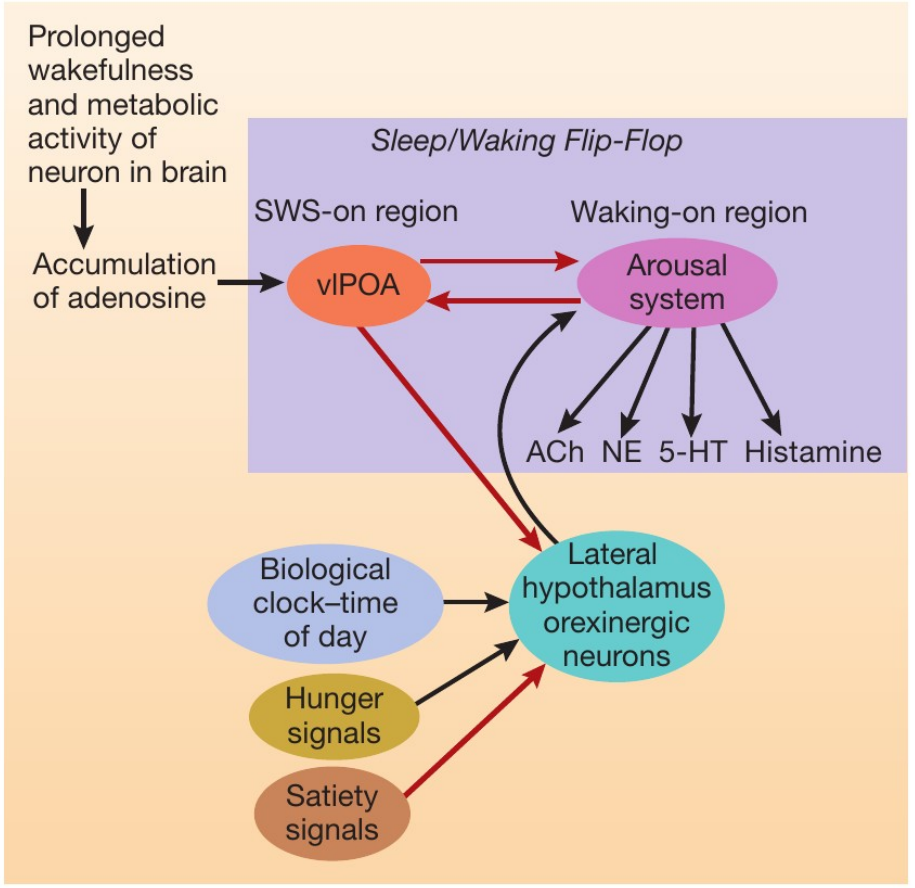
Medeiros et al. (2021), Video S1.

Sleep pressure and clock time provide different signals

Sleep pressure increases with time spent awake and usually declines during sleep. This is a homeostatic process, meaning that the system changes in a direction that helps restore a regulated condition. After unusually prolonged waking, an animal can show more sleep or stronger slow-wave activity during the next opportunity to sleep.

A **circadian clock** provides a different signal: information about biological time within the daily cycle. It can promote alertness at one time of day and favor sleep at another. A person can therefore feel temporarily more alert late in a long waking period even though the accumulated need for sleep has not disappeared.

The two signals interact. Falling asleep depends on both the pressure built during waking and the clock's influence at that moment. After a daytime nap, pressure may be lower at the usual bedtime; after a night shift, pressure may be high while the circadian system is beginning to promote daytime alertness.



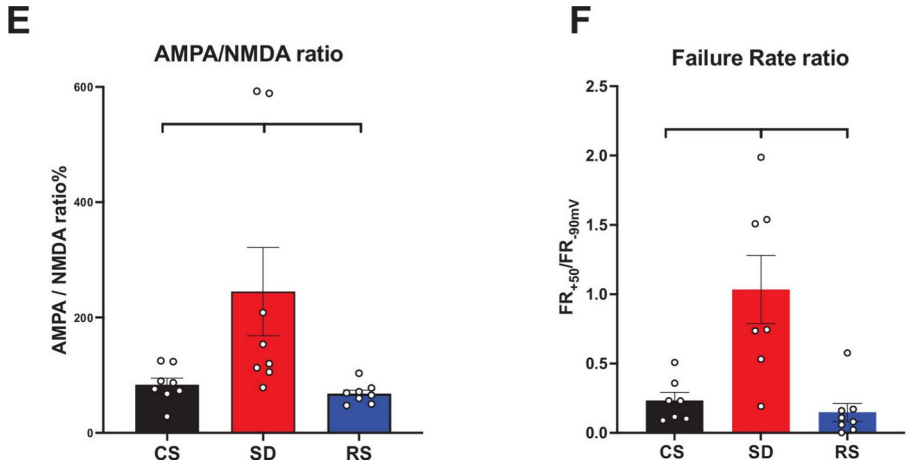
Carlson and Birkett (2022), Fig. 9.20, p. 286.

Sleep deprivation changes excitatory synapses

Glutamate excites many cortical neurons by activating receptors including AMPA and NMDA receptors. AMPA receptors contribute a rapid excitatory response. NMDA receptors behave differently because their current depends strongly on the receiving cell's voltage. Comparing the two responses can reveal changes in the functional properties of an excitatory synapse.

Vogt and colleagues compared mouse motor-cortex neurons after normal sleep, sleep deprivation, or deprivation followed by recovery sleep. The sleep-deprived group had a higher AMPA-to-NMDA response ratio. The original treadmill recording documents one method used to maintain waking; the synaptic measurements were made afterward in brain slices from the different experimental conditions.

The investigators also estimated the fraction of **silent synapses**, which have an NMDA response but little detectable AMPA response under the recording conditions. Sleep deprivation reduced this fraction, while recovery sleep restored a pattern closer to the sleeping controls. The measured changes concern the responsiveness of particular cortical inputs, rather than a uniform increase in every synapse throughout the brain.



Vogt et al. (2025), Fig. 1, selected panels.



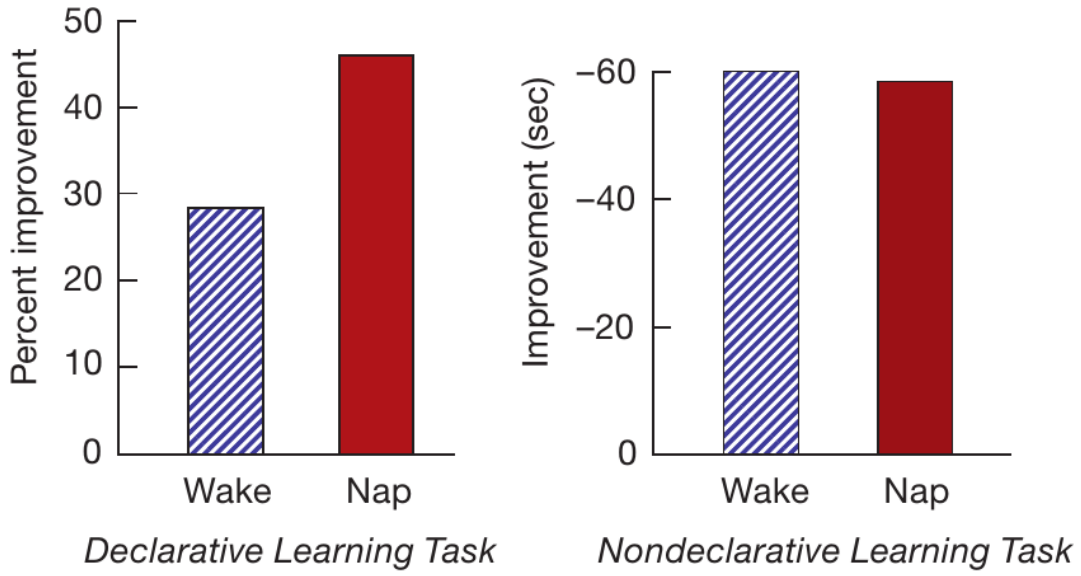
Vogt et al. (2025), Video 1.

Sleep can change how newly learned information is retained

Memory consolidation refers to changes that make information more stable or reorganize how it is represented after initial learning. Learning an event first requires activity during experience, but the resulting representation can continue changing afterward. Sleep provides conditions in which recently active circuits can be reactivated while the brain is less occupied with incoming events.

The hippocampus helps rapidly encode relationships among elements of an experience, while distributed cortical regions represent features of those elements. Communication between these systems during subsequent sleep can influence later recall. A memory does not have to be moved as a complete object from one brain location to another for its representation to become more stable.

To test a sleep-related contribution, researchers compare later performance after controlled sleep or waking intervals. They must also consider when learning and testing occur because circadian alertness can influence performance. Measuring the same learned material before and after the interval helps distinguish the amount initially learned from the information retained later.



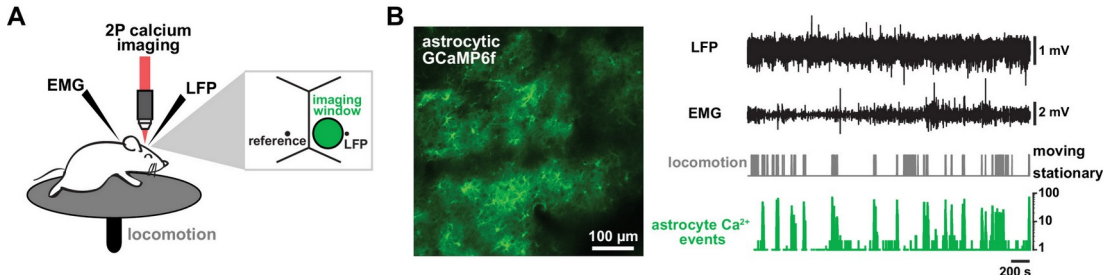
Carlson and Birkett (2022), Fig. 9.7, p. 277.

Astrocyte calcium activity changes with behavioral state

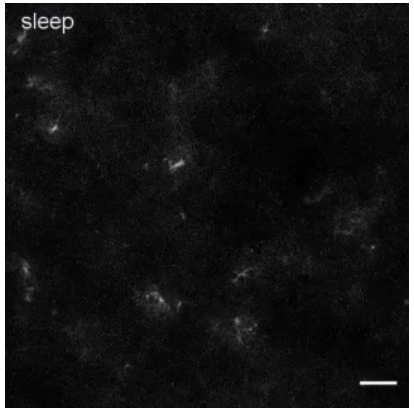
Astrocytes are glial cells that interact with neurons, blood vessels, and other astrocytes. They do not communicate through the same rapid action-potential patterns as neurons, but they can produce changes in intracellular calcium. A fluorescent calcium indicator allows these changes to be followed in living brain tissue over time.

Vaidyanathan and colleagues imaged cortical astrocytes in mice while also recording local electrical activity, muscle activity, and movement. Astrocyte calcium events were more frequent during waking than during NREM sleep, when slow-wave activity was stronger. The state labels in the source recording allow the fluorescent events to be compared with the independently measured behavioral condition.

The relationship also changed around state transitions. Astrocyte events could be examined before and after the mouse entered sleep or became active, rather than averaging the entire experiment into one value. The later manipulations in this study separately changed astrocyte signaling pathways to test which features of sleep those pathways could influence.



Vaidyanathan et al. (2021), Fig. 1, selected panels.



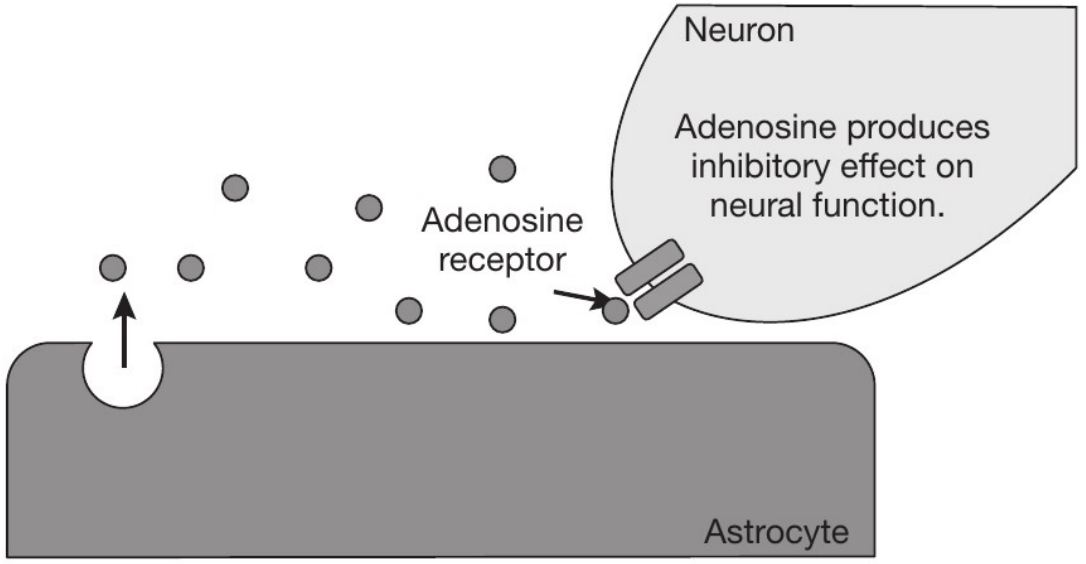
Vaidyanathan et al. (2021), Video 1.

Adenosine changes the balance between sleep and waking

Adenosine is a molecule produced through cellular metabolism that can alter neuronal activity through cell-surface receptors. Its extracellular concentration can increase in some brain regions during sustained waking. Activating adenosine receptors can reduce activity in arousal systems and influence sleep-promoting circuits, helping translate recent waking history into a change in the likelihood of sleep.

Different adenosine receptors act through different intracellular pathways. Their effects depend on the receiving cells and their connections. Inhibiting a wake-promoting neuron can favor sleep, while changing an inhibitory input to another circuit can have a different consequence. Adenosine therefore acts through identifiable receptors and circuits rather than functioning as a single chemical switch for the whole brain.

Caffeine blocks adenosine receptors, reducing some of adenosine's effects on alertness and sleep. The drug does not remove the time already spent awake or directly repay accumulated sleep need. As its receptor-blocking action declines, the remaining homeostatic pressure can again strongly influence how sleepy the person feels and how readily sleep occurs.



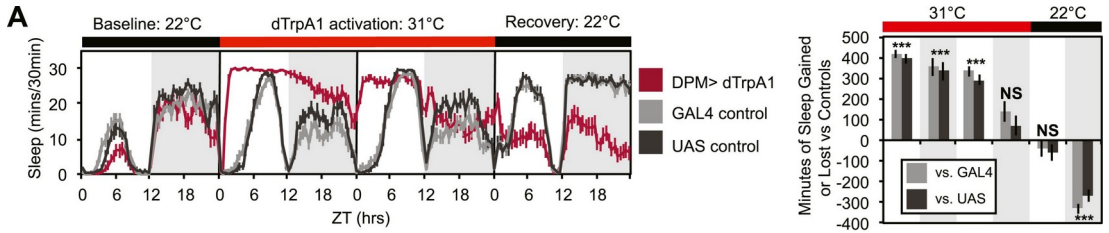
Carlson and Birkett (2022), Fig. 9.9, p. 279.

Activating memory-related fly neurons promotes sleep

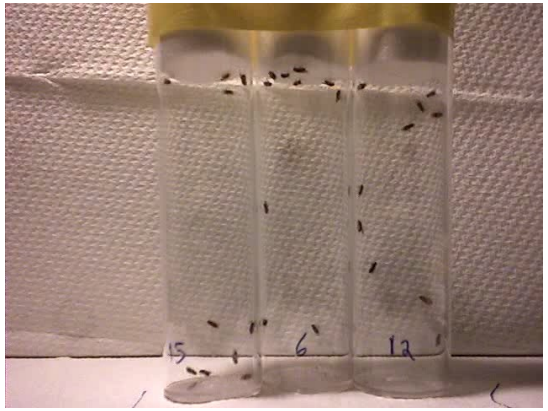
The **mushroom bodies** are major learning-related structures in the fly brain. Haynes and colleagues investigated a pair of dorsal paired medial neurons that innervate these structures and are required for memory consolidation. Activating these neurons also increased sleep, allowing the researchers to examine how a memory-related circuit affects behavioral state.

The investigators used a heat-sensitive channel to increase activity in the targeted neurons. At the activating temperature, the experimental flies became stationary more readily than the genetic controls. The original recording compares the experimental and control groups under the same warm condition, allowing the behavioral effect of targeted channel expression to be distinguished from heating alone.

A gentle tap caused the experimental flies to move, after which they soon returned to a resting position while controls continued exploring. This response demonstrates that the flies could still perform ordinary movements when stimulated. The increased rest therefore differed from an inability of the muscles or movement system to respond after the neuronal manipulation.



Haynes et al. (2015), Fig. 1, selected panels.



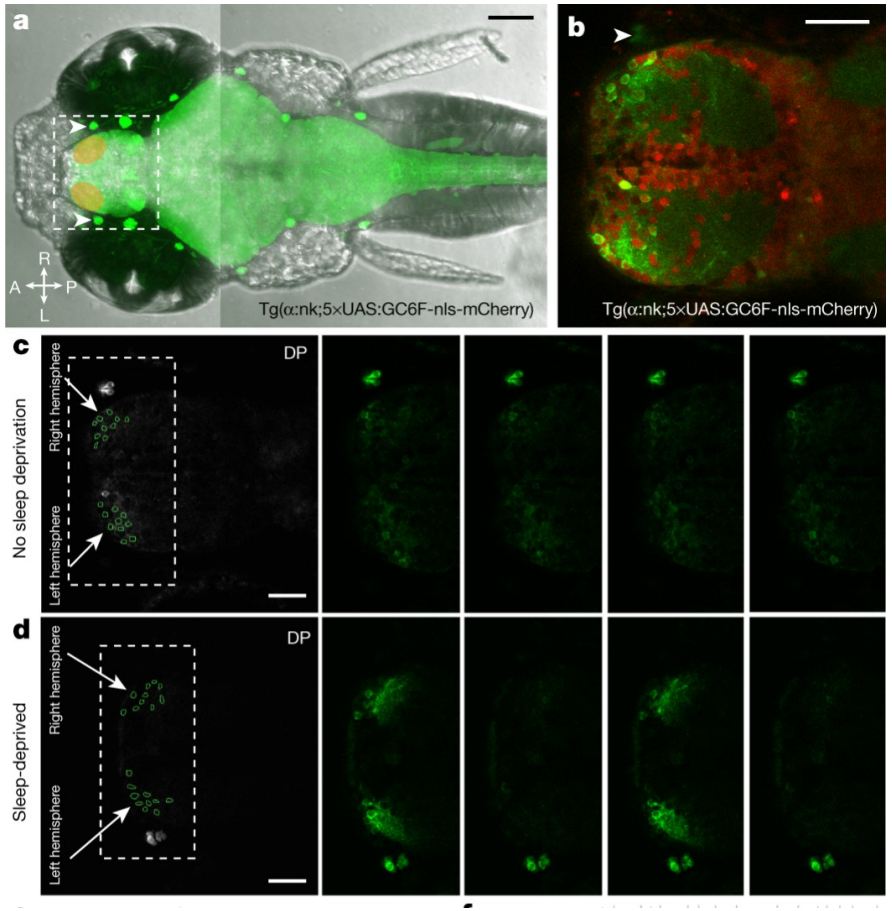
Haynes et al. (2015), Video 1.

Zebrafish show more than one neural sleep signature

Larval zebrafish are transparent enough for researchers to image activity across much of the brain. Leung and colleagues combined these recordings with measurements of muscle, eye, and cardiac activity. This **fluorescence-based polysomnography** allowed several body and brain signals to be aligned in the same animal as its behavioral state changed.

The investigators identified slow bursting sleep and propagating wave sleep. During slow bursting sleep, populations showed coordinated activity changes. Propagating wave sleep began with a wave of activity through brain regions and was accompanied by a marked loss of muscle tone, changes in eye activity, and slower, more variable heartbeats.

These combined measurements reveal differences that an immobility score alone would not capture. The fish brain does not have a mammalian neocortex where an ordinary scalp EEG could be interpreted in exactly the same way. Recording several conserved physiological features provides a way to compare sleep organization across vertebrates while retaining the actual signatures measured in each species.



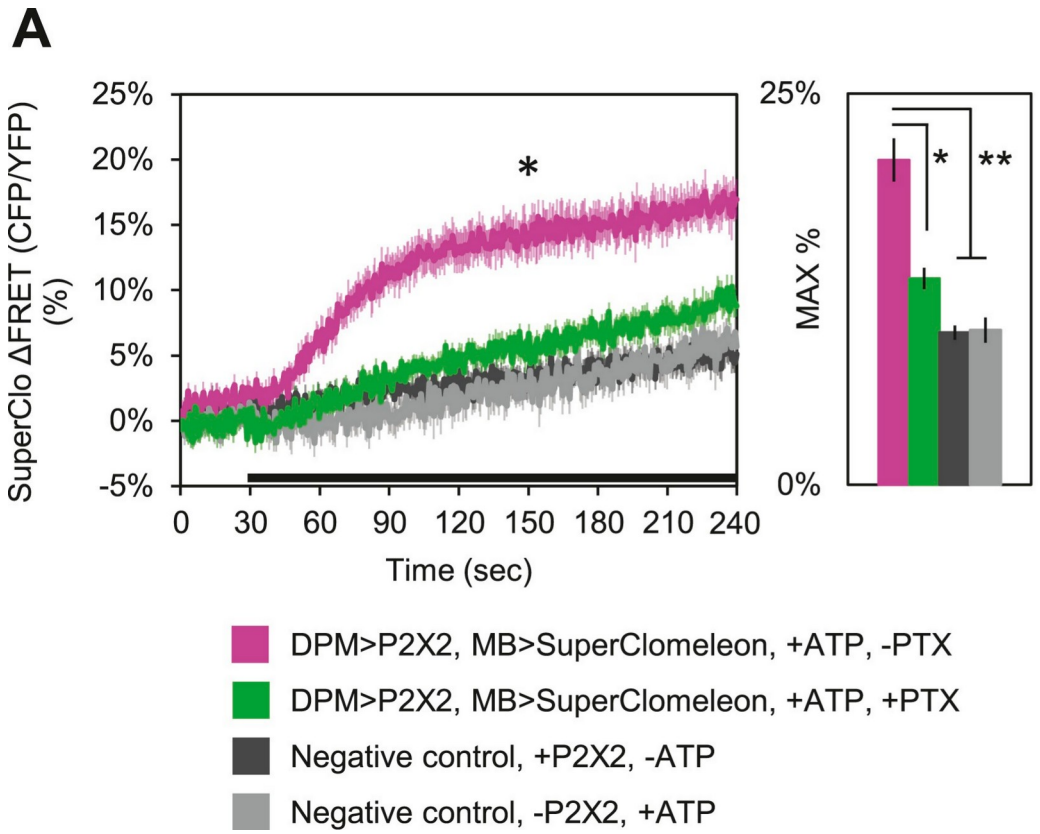
Leung et al. (2019), Fig. 1, selected panels.

Inhibition helps shared sleep and memory circuits operate

The sleep-promoting effect of dorsal paired medial neurons depends partly on inhibitory communication with mushroom-body neurons. Haynes and colleagues activated the input neurons while monitoring a chloride-sensitive fluorescent reporter in the receiving cells. Chloride increased in those cells, consistent with activation of GABA receptors that allow chloride ions to cross the membrane.

Blocking GABA-sensitive chloride channels reduced the recorded response. The investigators also reduced particular GABA receptors in wake-promoting mushroom-body neurons. Those flies slept less or had shorter nighttime sleep episodes, showing that the receiving cells' receptors were necessary for the normal pattern of sustained sleep in this circuit.

The anatomical arrangement allows the same pair of input neurons to participate in both behavioral-state regulation and memory-related processing. Inhibiting wake-promoting mushroom-body neurons reduces their contribution to keeping the animal active. The receptor manipulations identify part of that inhibitory route, rather than assuming that every neuron involved in forming a memory must increase the activity of its targets.



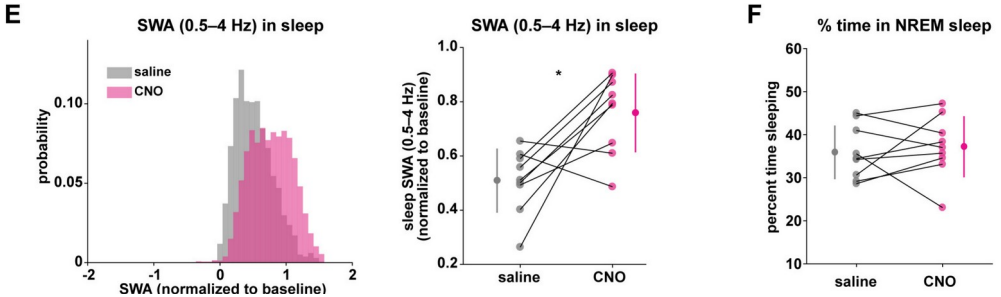
Haynes et al. (2015), Fig. 5, selected panels.

Astrocyte Gi signaling changes NREM sleep depth

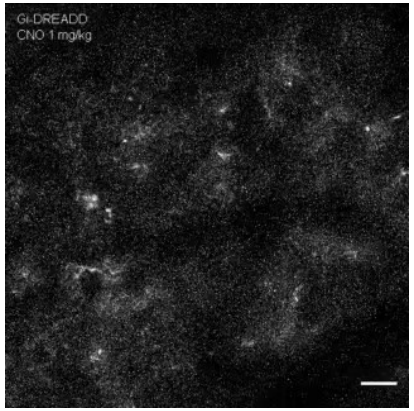
Vaidyanathan and colleagues expressed an engineered receptor in cortical astrocytes so that a drug could activate a selected intracellular pathway. The receptor used **Gi signaling**, a family of signals that changes cell function after receptor activation. A second fluorescent protein reported calcium events in the same astrocytes while electrical recordings identified sleep and waking.

Activating the engineered receptor increased the frequency of astrocyte calcium events. During NREM sleep, the mice also developed stronger slow-wave activity than after a saline injection. The source recording follows the calcium response after drug administration, making the cellular measurement visible alongside the separate electrical measurements used to assess the sleeping brain.

The mice did not spend more total time asleep after this manipulation. Instead, their existing sleep contained stronger slow-wave activity, a commonly used measure of NREM sleep intensity. Increasing astrocyte signaling therefore changed a property within sleep without requiring a longer sleep period or a larger fraction of the recording to be classified as sleep.



Vaidyanathan et al. (2021), Fig. 3, selected panels.



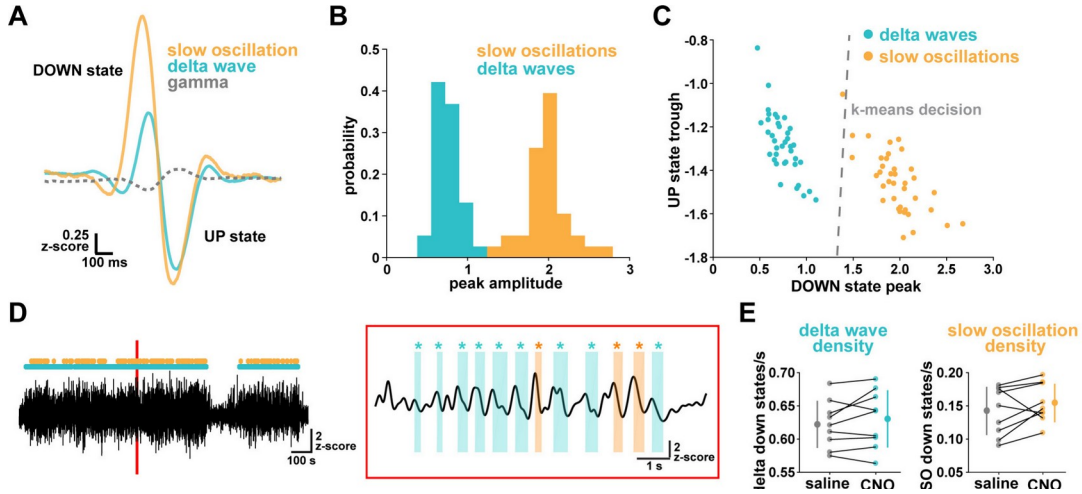
Vaidyanathan et al. (2021), Video 2.

Sleep depth and sleep duration can change separately

Sleep **duration** measures how long an animal sleeps, while sleep **intensity** describes properties of the sleeping state, such as slow-wave activity. These measurements can change independently. A mouse can spend the same number of minutes asleep while cortical populations produce larger or more frequent slow waves during those minutes.

The astrocyte Gi manipulation increased local delta-wave activity during NREM sleep. The researchers separated different classes of slow events instead of combining every low-frequency fluctuation into one category. This analysis showed that an increase in the overall slow-wave signal could arise from a particular change in the events composing it.

The investigators also compared electrical activity near the manipulated visual cortex with activity in frontal cortex. Local and distant recordings did not show identical effects. Sleep has a whole-animal behavioral organization, but the strength and timing of its electrical rhythms can vary across cortical regions according to their local cellular activity.



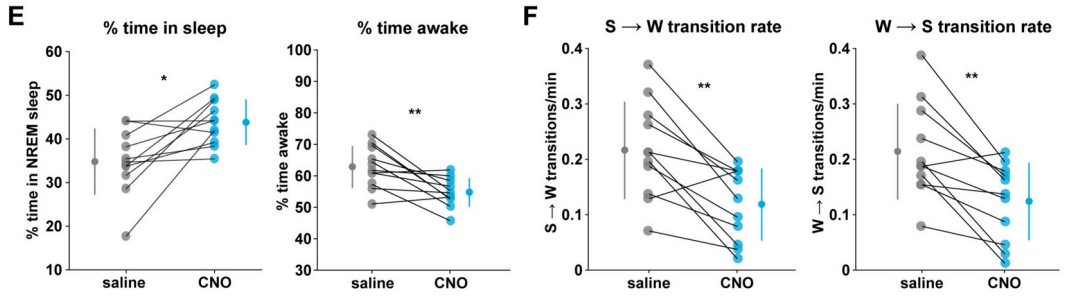
Vaidyanathan et al. (2021), Fig. 4, selected panels.

Astrocyte Gq signaling changes sleep duration

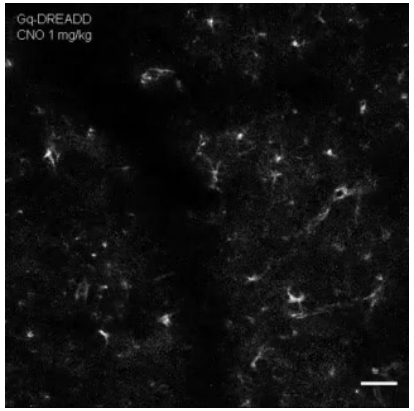
A second engineered receptor activated **Gq signaling** in astrocytes. After the activating drug, calcium events initially increased, then became almost absent for an extended period. Fluorescence could remain elevated even when separate events disappeared. The source recording follows this changing response, so the manipulation cannot be described as a sustained increase in ordinary calcium events.

During the later period with suppressed calcium dynamics, mice spent more time asleep and made fewer transitions between sleep and waking. Individual sleep and wake bouts became longer. During the brief initial period of increased calcium activity, sleep was instead reduced, showing why the time after receptor activation changes the observed behavioral result.

Despite the later increase in sleep time, slow-wave activity within sleep did not increase. Compared with the Gi experiment, this manipulation altered a different feature of sleep organization. The two receptor pathways produced different calcium responses and different combinations of sleep duration, state transitions, and electrical intensity in the same general class of glial cell.



Vaidyanathan et al. (2021), Fig. 6, selected panels.



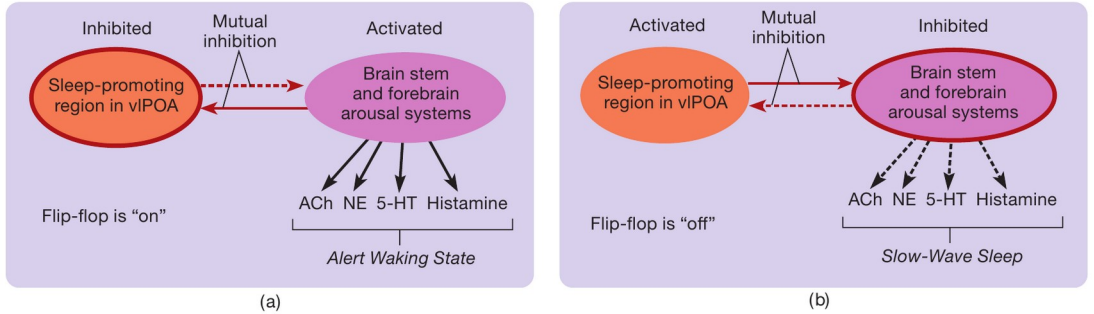
Vaidyanathan et al. (2021), Video 3.

Sleep and wake circuits inhibit one another

The **ventrolateral preoptic area**, or VLPO, contains neurons that become active during sleep. Many release the inhibitory transmitter GABA onto wake-promoting systems. By reducing activity in those systems, VLPO neurons decrease the arousal signals reaching the cortex and other brain regions, making sustained waking less likely.

Wake-promoting systems also inhibit sleep-promoting neurons. This **mutual inhibition** creates a circuit that can settle into either of two relatively stable conditions: strong arousal activity suppressing sleep neurons, or strong sleep-neuron activity suppressing arousal systems. Increasing one side's activity tends to weaken the opposing side rather than maintaining an intermediate state indefinitely.

Homeostatic pressure and circadian inputs influence which side becomes dominant. As waking continues, sleep-promoting influences can overcome the active arousal network and produce a relatively rapid transition into sleep. Additional stabilizing signals help prevent small fluctuations from repeatedly switching the brain between states when an animal needs to remain awake.



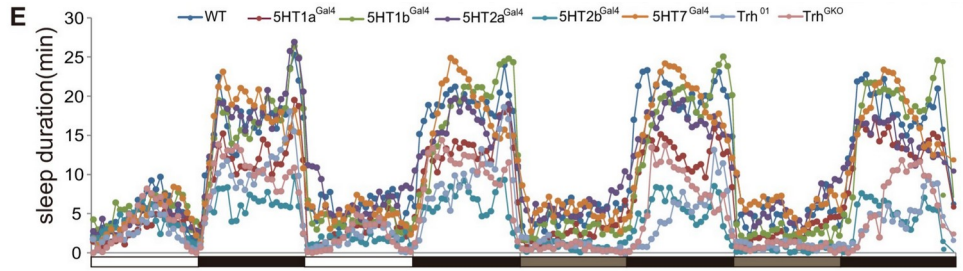
Carlson and Birkett (2022), Fig. 9.17, p. 285.

Serotonin-producing fly neurons reach many brain regions

Serotonin is a chemical transmitter produced by a limited population of neurons whose axons reach many other cells. Qian and colleagues labeled serotonin-producing neurons in the fly brain. The source movie rotates through this labeled anatomy; its fluorescence identifies cells and projections rather than recording moment-to-moment serotonin release during sleep.

Flies with reduced serotonin synthesis slept less and showed an impaired rebound after sleep deprivation. The investigators also examined serotonin receptors, the receiving-cell proteins that detect released serotonin. Removing the **5-HT2B receptor** disrupted normal sleep regulation, indicating that the behavioral effect depended on a particular receptor rather than serotonin production alone.

A small group of neurons in the dorsal fan-shaped body required this receptor for normal sleep homeostasis. Restoring or removing receptor function in selected cells helped locate the relevant receiving population. A widely projecting transmitter system can therefore regulate sleep through a much smaller subset of targets with the necessary receptor machinery.



Qian et al. (2017), Fig. 1, selected panels.



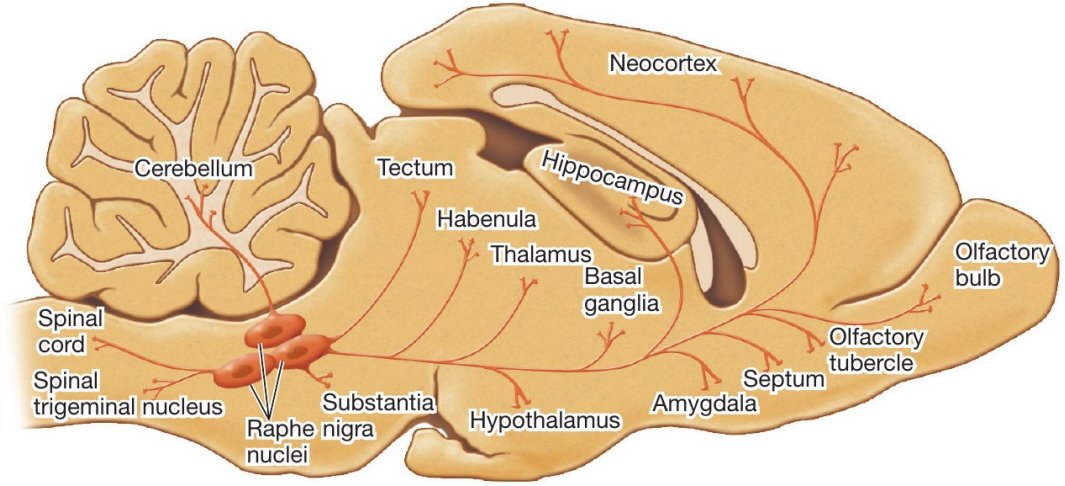
Qian et al. (2017), Video 1.

Arousal depends on several chemical systems

Several brainstem and hypothalamic systems help maintain arousal. **Serotonin-producing raphe neurons** project broadly through the brain, while **histamine-producing neurons** in the tuberomammillary nucleus arise in the hypothalamus. Their activity changes the responsiveness of receiving circuits, influencing how readily sensory inputs, internal needs, and ongoing behavior sustain waking.

Many mammalian serotonin and histamine neurons are most active during waking and become less active during sleep, especially REM sleep. Their effects depend on which receptor is present on the receiving cell. One receptor may reduce excitability, while another changes intracellular signaling in a way that increases the cell's response to other inputs.

Species and circuits can consequently show different effects of the same transmitter. The sleep-promoting serotonin receptor identified in flies occupies a particular circuit, whereas broadly projecting mammalian raphe neurons participate in several functions. Describing the source neurons, receptor, and target cells explains the effect more accurately than assigning serotonin one universal behavioral action.



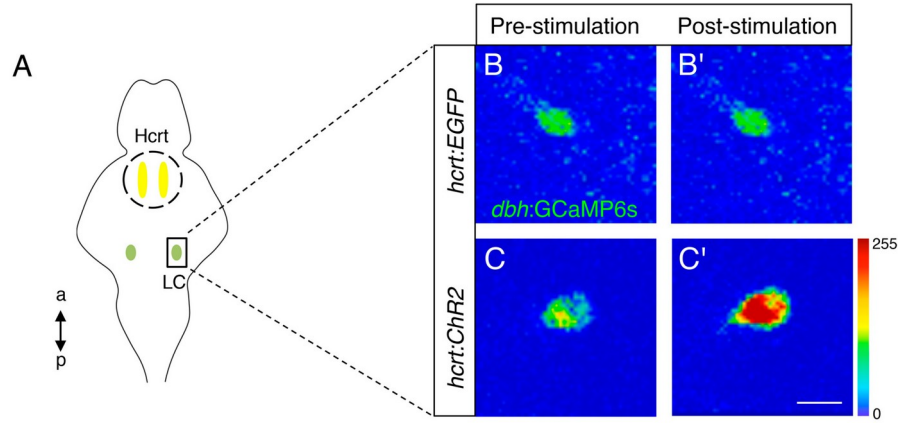
Carlson and Birkett (2022), Fig. 9.14, p. 282.

A light-only control tests apparent neural responses

Hypocretin, also called **orexin**, is a peptide released by hypothalamic neurons involved in arousal. Singh and colleagues studied how these neurons influence the locus coeruleus in zebrafish. Locus coeruleus neurons were monitored with a fluorescent calcium indicator, allowing their responses to be measured while the hypocretin system was manipulated.

The researchers used **channelrhodopsin**, a light-sensitive ion channel, to activate selected hypocretin neurons. Before interpreting a response as a consequence of neuronal activation, they tested animals expressing a fluorescent label without the light-sensitive channel. The recording here shows this light-only control, which received illumination without the intended optogenetic activation of the input neurons.

Imaging was paused during the stimulation flashes and resumed afterward. This timing reduced direct interference between the stimulating light and the fluorescence measurement. Comparing control animals with channelrhodopsin animals separated an optical or handling effect from the change expected when hypocretin neurons actually increased their activity and influenced their targets.



Singh et al. (2015), Fig. 7, selected panels.



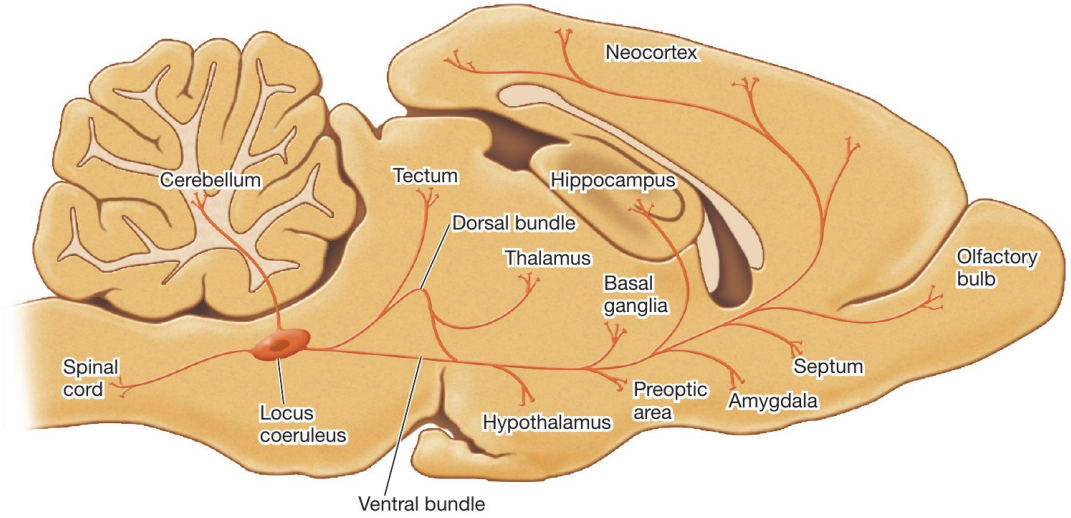
Singh et al. (2015), Video 1.

Locus coeruleus neurons distribute norepinephrine

The **locus coeruleus** is a small brainstem nucleus whose neurons release **norepinephrine** across many brain regions. Their widely branching axons allow a compact source population to alter cortical, hippocampal, and other circuits. Norepinephrine changes how receiving neurons respond to inputs through several classes of adrenergic receptor.

Locus coeruleus activity is generally high during alert waking, lower during NREM sleep, and very low during REM sleep. Brief increases can accompany important sensory events or transitions toward waking. The transmitter can therefore change both the ongoing state of a network and its response to an event occurring within that state.

The effect is determined by the target. Norepinephrine can alter neuronal membrane currents, transmitter release, glial signaling, and vascular activity through receptors on different cell types. A change in locus coeruleus firing does not mean every receiving neuron changes in the same direction or that the entire brain receives an identical electrical command.



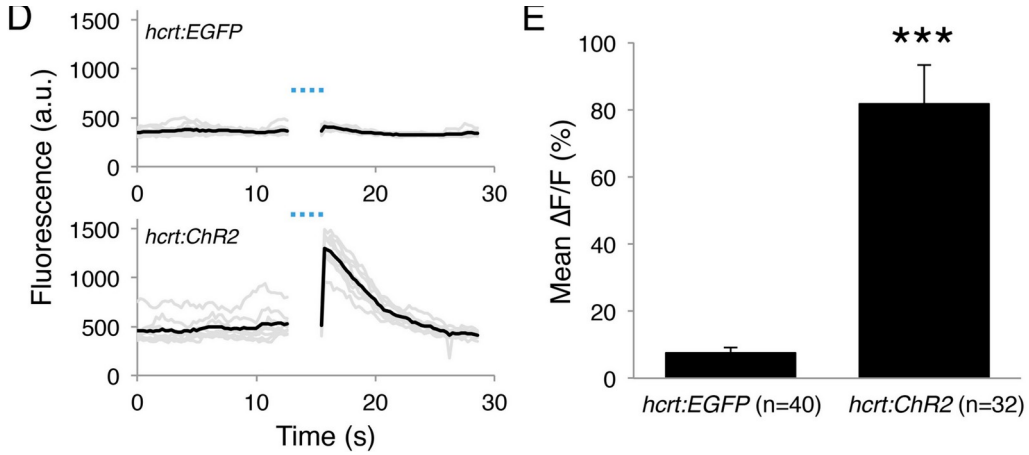
Carlson and Birkett (2022), Fig. 9.12, p. 281.

Hypocretin stimulation recruits locus coeruleus neurons

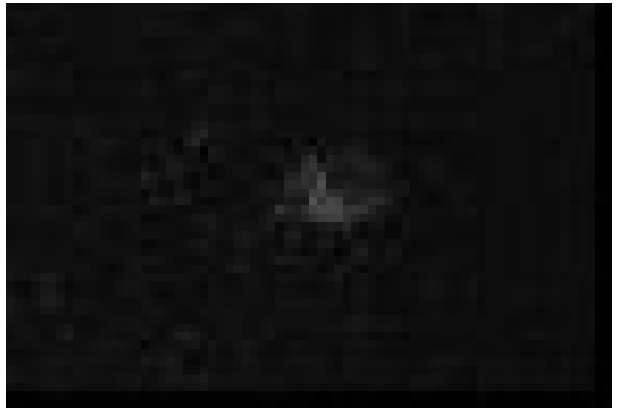
In the channelrhodopsin condition, light stimulation of hypocretin neurons increased calcium signals in locus coeruleus neurons. The original recording displays this response after the stimulation period. Together with the light-only control, it establishes a functional route by which increasing one arousal population's activity can recruit another population with broad norepinephrine projections.

The investigators also studied fish unable to synthesize normal norepinephrine because of a mutation in its production pathway. These fish slept more overall, but their arousal threshold was lower: relatively weak stimulation could wake them. The amount of sleep and resistance to awakening therefore changed in different directions within the same animals.

Increasing hypocretin signaling normally promoted waking, but this effect was reduced when norepinephrine signaling was unavailable. The downstream system was needed for much of the behavioral response to the upstream manipulation. Measuring both the locus coeruleus response and the sleep phenotype identifies separate steps between activation of hypothalamic neurons and sustained waking behavior.



Singh et al. (2015), Fig. 7, selected panels.



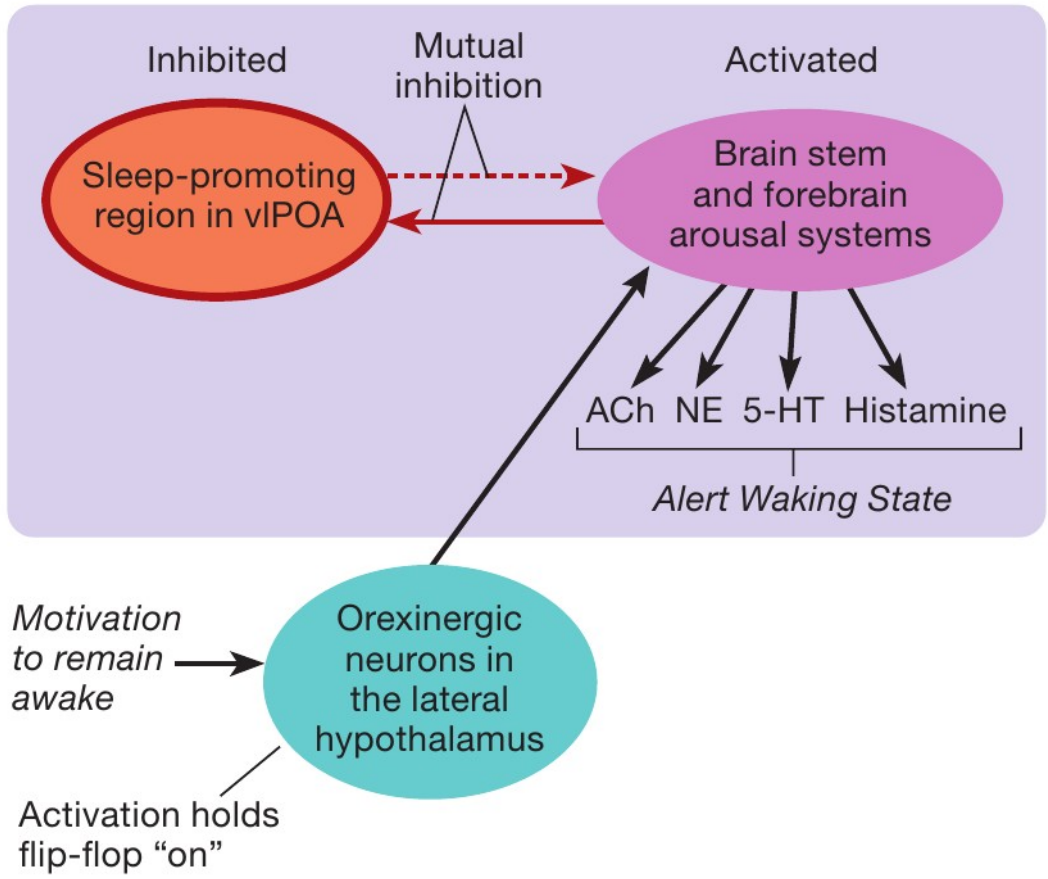
Singh et al. (2015), Video 2.

Orexin helps sustain a stable waking state

Orexin neurons lie mainly in the lateral hypothalamus and send axons to several arousal systems, including the locus coeruleus and other brainstem targets. Their peptide acts through orexin receptors on receiving cells. Increasing the responsiveness of these arousal systems helps an animal maintain waking during active behavior.

Orexin provides stability to a network already containing mutually inhibitory sleep and wake circuits. When arousal is needed, its excitation makes the wake-promoting side less vulnerable to brief fluctuations. This stabilization helps explain how an animal can remain awake while searching for food or responding to a salient event despite an accumulating need for sleep.

Loss of orexin signaling can make boundaries between states unstable. A person may become overwhelmingly sleepy during the day or experience sudden loss of muscle tone while still conscious. These symptoms involve different components of sleep and waking entering an unusual combination, rather than the whole brain simply turning off at once.



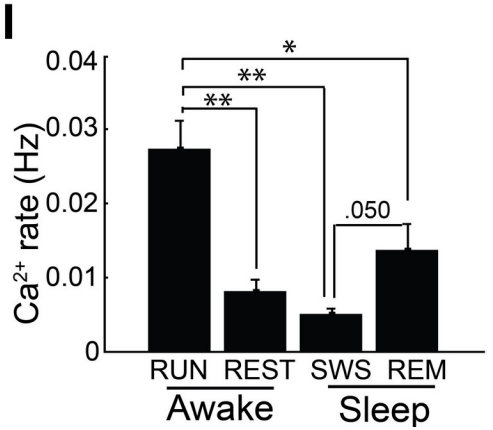
Carlson and Birkett (2022), Fig. 9.18, p. 285.

Hippocampal activity differs across sleep and waking

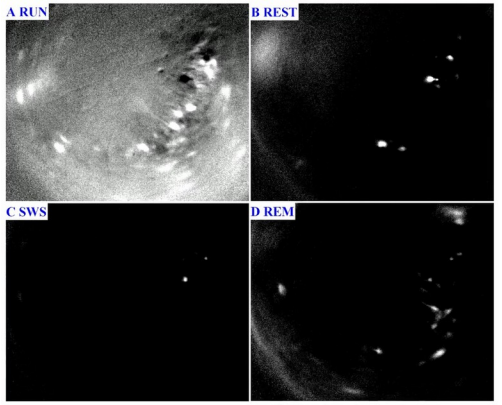
The **hippocampus** produces different population patterns during exploration, quiet waking, NREM sleep, and REM sleep. Zhou and colleagues imaged calcium in CA1 pyramidal neurons while recording electrical activity and movement in mice. Aligning the measurements allowed each fluorescent event to be assigned to an independently identified behavioral state.

Calcium events were more frequent during exploration and REM sleep than during quiet waking or slow-wave sleep. The recording displays the same preparation across these states. A fluorescent event reflects a cellular calcium change associated with activity, while the local electrical signal describes a different aspect of the surrounding population's organization.

During quiet waking and NREM sleep, the hippocampus can generate **sharp-wave ripples**, brief coordinated events containing rapid electrical oscillations. Exploration and REM instead often contain a slower theta rhythm. Similar amounts of visible movement therefore do not predict identical neural patterns, and a relatively active REM calcium signal does not establish that the animal is awake.



Zhou et al. (2019), Fig. 1, selected panels.



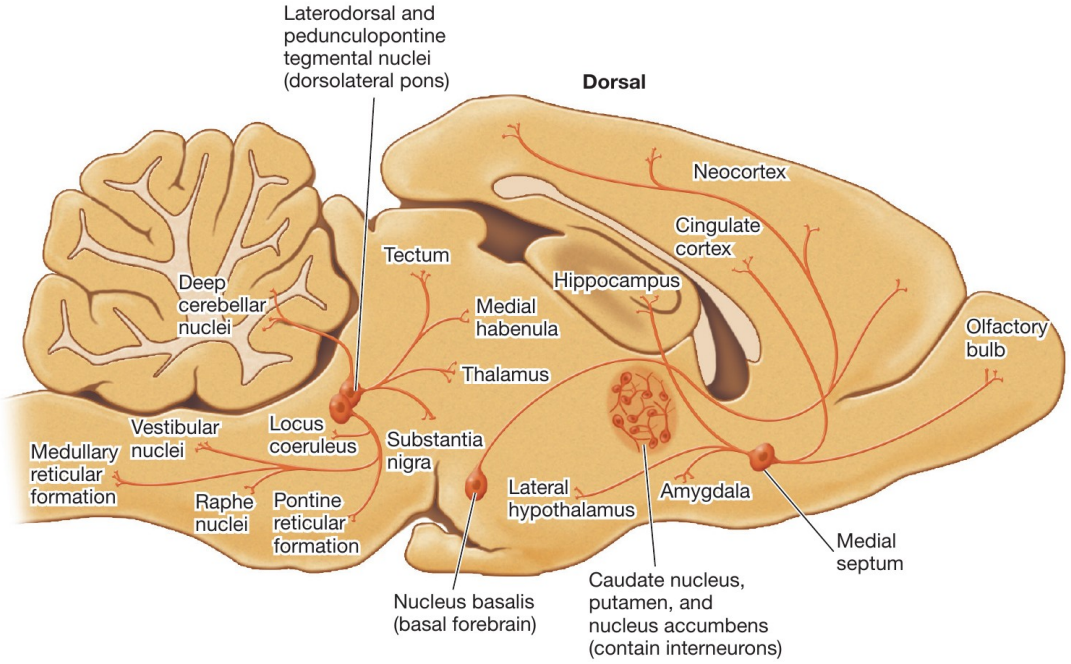
Zhou et al. (2019), Video 1.

Acetylcholine changes cortical and hippocampal activity

Acetylcholine changes neural activity through two major receptor families. Nicotinic receptors are ion channels that can produce rapid electrical responses. Muscarinic receptors act through intracellular signaling pathways and often change excitability or transmitter release over longer periods. The receiving receptor determines how a given acetylcholine input affects its target.

Cholinergic neurons in the basal forebrain project to cortex, while neurons in the medial septal region provide important inputs to the hippocampus. Additional cholinergic populations in the brainstem influence sleep-related circuits. These pathways can change activity across several structures without replacing the specific information carried by each structure's local connections.

Acetylcholine release is often relatively high during both active waking and REM sleep. Those states still differ in movement, sensory responsiveness, and the activity of other transmitter systems. The resulting network pattern depends on the combination of active inputs, the receptors they reach, and the electrical condition of the receiving neurons.



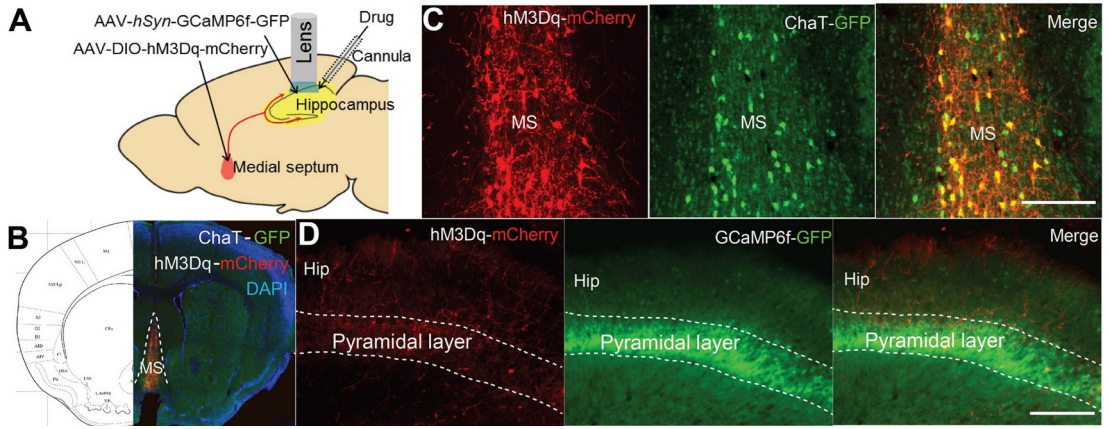
Carlson and Birkett (2022), Fig. 9.10, p. 280.

Activating cholinergic neurons changes population activity

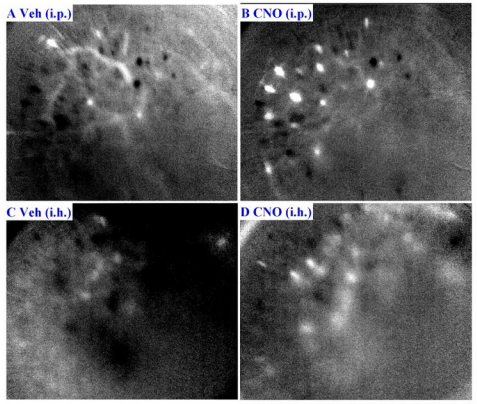
Zhou and colleagues expressed an engineered activating receptor in cholinergic neurons of the medial septum. Giving its activating drug increased calcium-event frequency in hippocampal CA1 neurons compared with the control condition. The source recording follows the receiving hippocampal population, so the observed fluorescence comes from cells downstream of the manipulated cholinergic neurons.

Sharp-wave ripple occurrence decreased after cholinergic activation. This change concerns the organization of hippocampal activity, not merely the number of fluorescent events. A population can produce more calcium events while becoming less likely to enter a particular synchronized electrical pattern, because those measurements describe different features of its operation.

The researchers did not observe a corresponding increase in overall multiunit firing, movement, or mean calcium-event amplitude under these conditions. Acetylcholine altered selected population properties rather than increasing every available measure of activity. Comparing the simultaneous signals reveals which features changed when a defined chemical input to the hippocampus was strengthened.



Zhou et al. (2019), Fig. 3, selected panels.



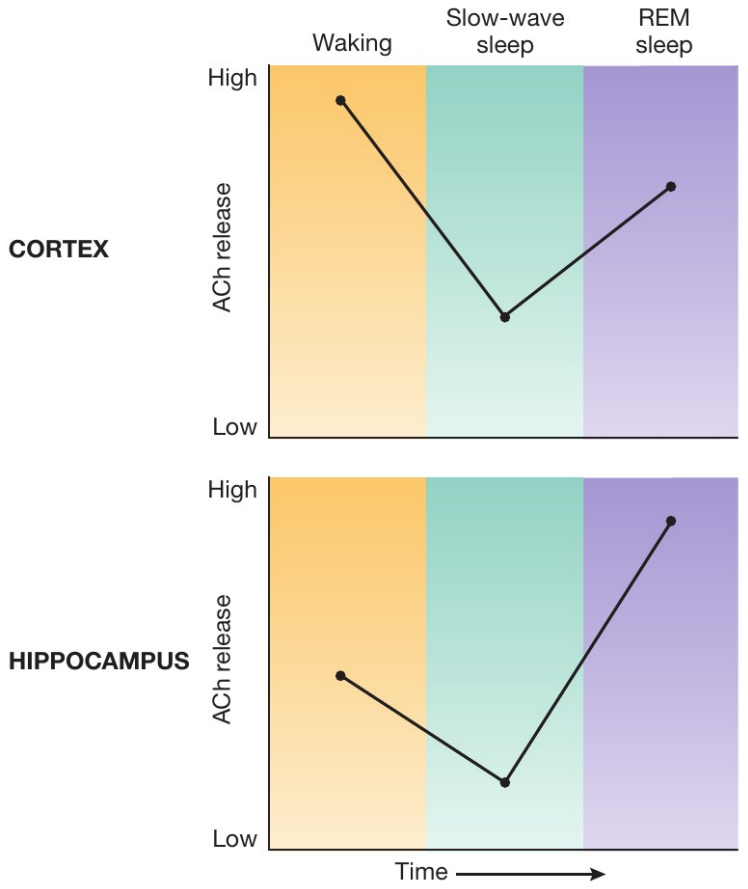
Zhou et al. (2019), Video 2.

A transmitter can support different activity patterns

A transmitter's behavioral effect depends on the other signals present at the same time. During waking, acetylcholine can act alongside norepinephrine, serotonin, and orexin-related arousal input. During REM sleep, cholinergic activity can remain high while several of those other systems become much less active, producing a different operating condition for the receiving networks.

Receptors also differ across cells and across parts of a neuron. Opening a membrane channel near the cell body can change spike probability, while changing transmitter release at an axon terminal alters the input reaching another cell. Even one chemical can therefore influence several stages of a circuit through different physical actions.

Electrical recordings distinguish the resulting patterns. A low-amplitude cortical EEG, a hippocampal theta rhythm, and a sharp-wave ripple are different population events with different timing and anatomical sources. A chemical manipulation can favor one pattern and suppress another without making the entire brain uniformly more active or uniformly less active.



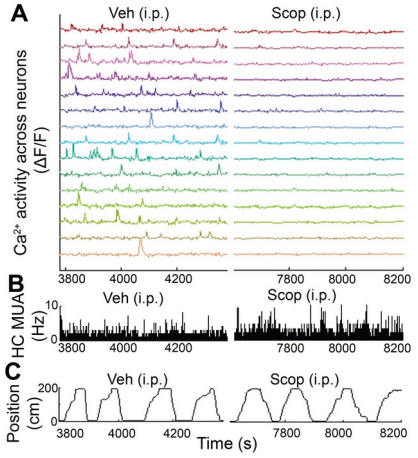
Carlson and Birkett (2022), Fig. 9.11, p. 280.

Muscarinic blockade changes hippocampal signals

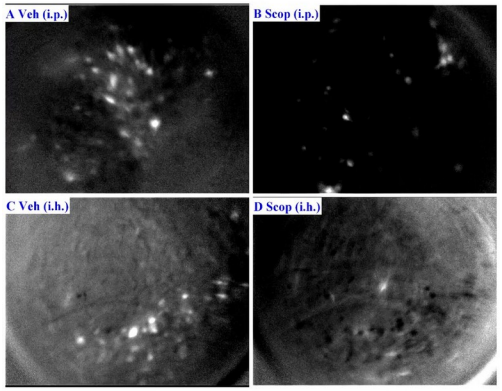
Scopolamine blocks muscarinic acetylcholine receptors. Zhou and colleagues used it to test whether these receptors contributed to hippocampal calcium activity and electrical organization. Compared with a control injection, the drug reduced calcium-event frequency in CA1 neurons and increased the occurrence of sharp-wave ripples under the measured conditions.

This result complements the experiment that increased cholinergic input. Strengthening the septal cholinergic pathway raised calcium-event frequency and suppressed ripples, while blocking receiving-cell muscarinic receptors shifted those measures in the opposite directions. The source recording compares control and drug conditions using the same kinds of cellular and electrical measurements.

The researchers also delivered scopolamine locally into the hippocampus. A local receptor blockade helps test whether the relevant action can occur at the receiving structure, rather than requiring the drug to change every cholinergic target throughout the brain. The experiment places part of the mechanism at hippocampal muscarinic receptors themselves.



Zhou et al. (2019), Fig. 4, selected panels.



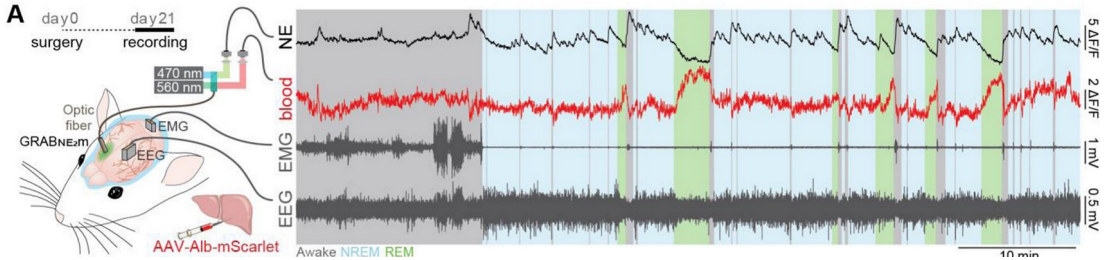
Zhou et al. (2019), Video 3.

Norepinephrine fluctuations help drive fluid movement

Cerebrospinal fluid, or CSF, surrounds the brain and enters spaces associated with its blood vessels. Hauglund and colleagues measured norepinephrine, blood-volume changes, and CSF-related signals in sleeping mice. During NREM sleep, these measurements showed coordinated fluctuations much slower than the individual electrical waves usually used to identify sleep stages.

A rise in norepinephrine can constrict blood vessels, reducing the volume of blood they contain; declining norepinephrine allows relaxation and expansion. Because the skull contains a limited volume, vascular changes can accompany movement of fluid in nearby spaces. The measurements followed these coupled changes across naturally occurring NREM periods.

The researchers linked the slow vascular rhythm to norepinephrine activity through additional manipulations and tracer measurements. Brief arousal-related fluctuations within NREM helped organize the rhythm even when they did not become long waking periods. Averaging an entire sleep episode into one value would hide this recurring sequence of chemical, vascular, and fluid changes.



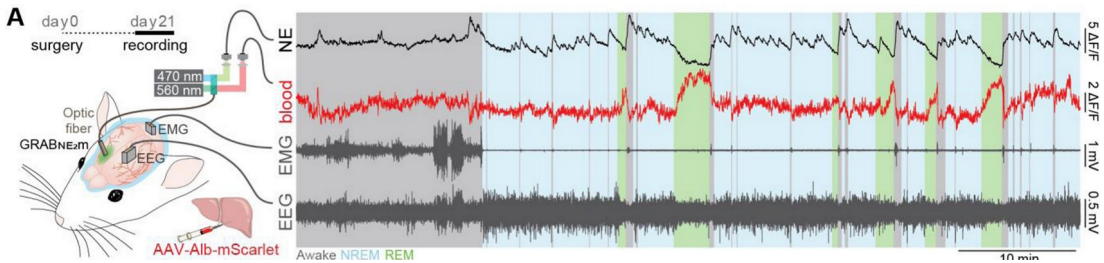
Hauglund et al. (2025), Fig. 1, selected panels.

Vascular contractions can move cerebrospinal fluid

Vasomotion is a recurring change in blood-vessel diameter. To test whether these changes could help move CSF, Hauglund and colleagues expressed a light-sensitive channel in vascular smooth-muscle cells. Light could then contract the vessel wall directly, allowing a vascular component of the proposed mechanism to be manipulated separately from ordinary neuronal signaling.

Repeated vascular stimulation increased entry of a fluorescent CSF tracer into brain tissue. The tracer made fluid movement measurable rather than inferring it solely from a change in vessel diameter. Alternating contraction and relaxation provided a mechanical route for fluid displacement through spaces surrounding the vessels during the experimental condition.

Tracer entry and removal of a substance already inside brain tissue are separate measurements. This experiment tested how vascular motion influences incoming CSF, while other parts of the study examined clearance using different tracers and recording periods. Following the injection location and the direction of movement is necessary to interpret what each assay actually measures.



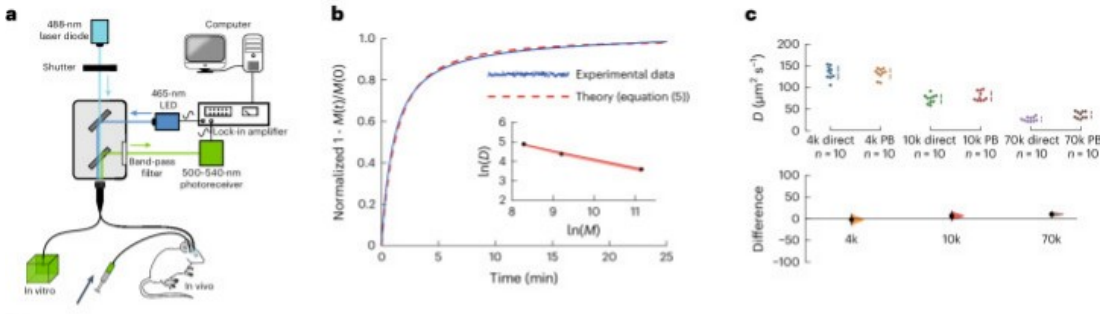
Hauglund et al. (2025), Fig. 1, selected panels.

Fluid entry and solute clearance are different measurements

A molecule can spread through brain tissue without immediately leaving the brain. **Diffusion** describes movement down a concentration gradient, whereas **clearance** describes removal from the measured tissue or compartment. Miao and colleagues injected fluorescent molecules into mouse brain tissue and tracked their movement and disappearance during waking, sleep, and anesthesia.

The measured diffusion of the tracer did not change substantially across the tested states, but its clearance was reduced during sleep and anesthesia. The result concerns material placed inside tissue and measured as it left that compartment. It differs from an experiment that introduces a tracer into CSF and measures how much subsequently enters the brain.

These assays can be interpreted by drawing the actual physical path: CSF into tissue, tracer spreading within tissue, and tracer leaving tissue. Increasing movement at one step does not automatically increase every other step. The direction of tracer delivery, its molecular properties, and the measured time interval determine which part of that path is observed.



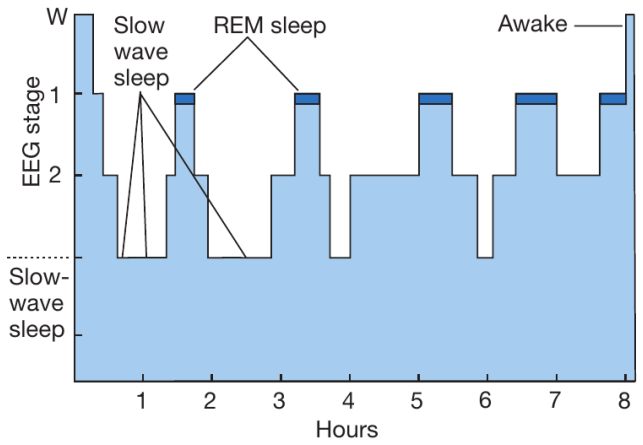
Miao et al. (2024), Fig. 1, selected panels.

Sleep state also changes circulation and body activity

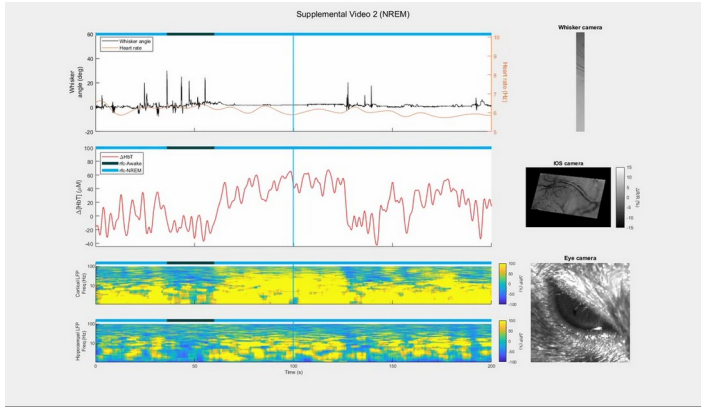
Sleep changes circulation as well as cortical electrical activity. Blood vessels adjust their diameter, the heart changes its timing, and breathing becomes organized differently across states. These body measurements help distinguish sleep stages and reveal processes that an EEG alone cannot identify, even when the animal remains in a similar posture.

The mouse recording shows a naturally occurring NREM episode while the head is stationary. Its eyes can remain open, illustrating why eye closure alone is not a reliable sleep-stage measure across species and preparations. Electrical activity, muscle signals, and behavior together establish the state represented during the recording.

The accompanying optical signal is sensitive to blood in cortical vessels. Increased hemoglobin absorbs more of the illuminating light, so a darker reflectance image can correspond to greater blood volume. Interpreting this image requires that physical relationship; brightness cannot simply be read as the amount of neuronal firing or the animal's level of awareness.



Carlson and Birkett (2022), Fig. 9.3, p. 271.



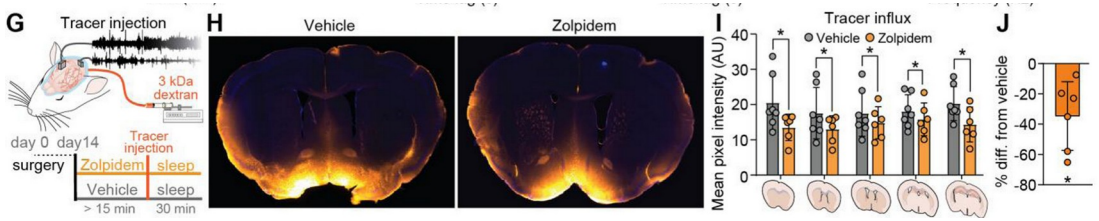
Turner et al. (2020), Video 2.

A sleep drug can alter vascular and fluid dynamics

Zolpidem promotes sleep by increasing the action of GABA at particular GABA-A receptors. Hauglund and colleagues tested how the drug affected the slow norepinephrine and vascular fluctuations observed during mouse sleep. Producing a sleeping behavioral state did not ensure that the same internal chemical and vascular rhythms were preserved.

After zolpidem, the slow norepinephrine oscillations and associated vascular dynamics were reduced. The researchers also measured less CSF-tracer influx under the tested drug condition. These measurements describe specific physiological changes in the mice rather than establishing that every sleep-promoting treatment has the same effect on brain-fluid movement.

A drug can therefore alter sleep quantity and sleep physiology through different parts of its action. GABA receptor modulation changes neuronal communication, which can change the timing of arousal-related norepinephrine signals and the vascular responses that follow them. Measuring those intermediate steps distinguishes increased time asleep from preservation of a particular naturally occurring NREM rhythm.



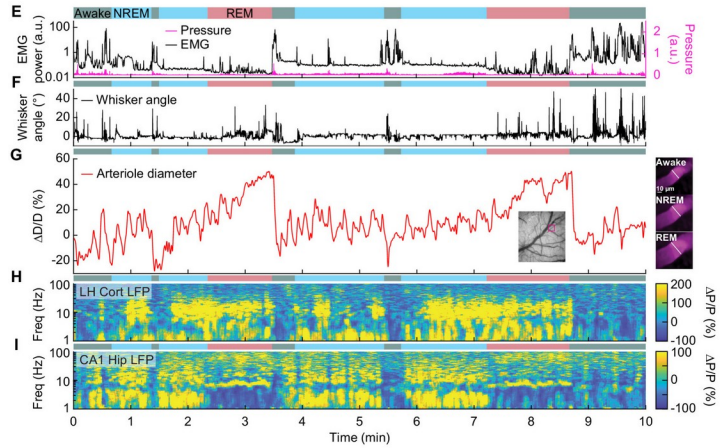
Hauglund et al. (2025), Fig. 6, selected panels.

REM sleep has a distinctive vascular signature

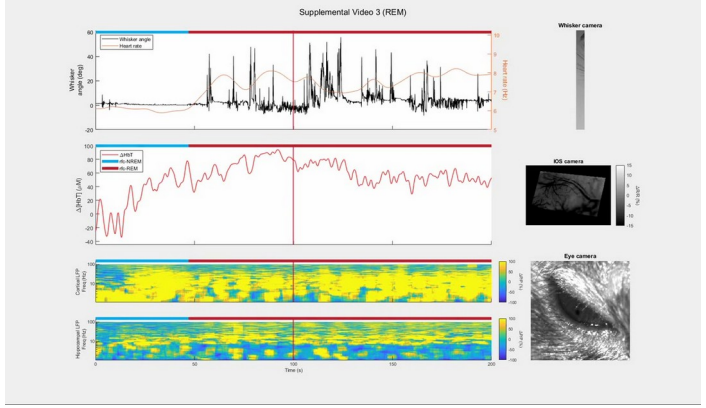
Turner and colleagues measured cortical blood-volume changes together with neural and behavioral signals in mice that could move between waking, NREM, and REM sleep. REM episodes produced a distinctive vascular response with substantial changes in blood volume and vessel diameter. These changes could be larger than responses evoked by an ordinary sensory stimulus.

The original REM recording includes small whisker movements and physiological state information. The animal can show a wake-like aspect of brain activity while its body remains largely still. State classification therefore uses the combination of hippocampal rhythms, muscle activity, and behavior rather than treating a movement or an open eye as sufficient evidence of waking.

Blood-volume signals and electrical activity were coordinated differently across sleep states. This matters when interpreting functional brain imaging, because vascular measurements depend on both neural activity and the blood vessels' responses. A larger hemodynamic signal during REM does not by itself identify a stronger sensory response or the content of a dream.



Turner et al. (2020), Fig. 3, selected panels.



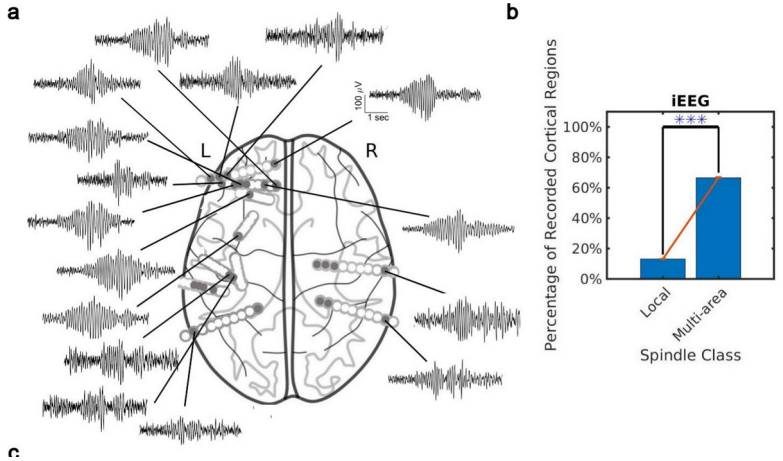
Turner et al. (2020), Video 3.

Known simulated waves can test a spindle detector

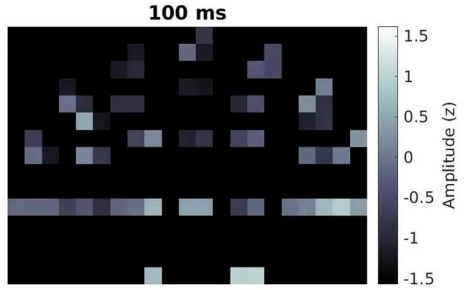
A **spindle detector** is a computational procedure that identifies short rhythmic events in an electrical recording. Mofrad and colleagues trained a neural-network detector to recognize spindle waveforms even when their amplitudes varied. A detector based only on a fixed amplitude threshold can miss an event that is clear in timing but relatively small at one electrode.

The source animation contains **simulated data**, constructed by the researchers with a known rotating-wave pattern. It is not a recording of a sleeping person's brain. Because the intended direction is known in advance, the simulation allows the analysis to be tested against a pattern whose timing and spatial arrangement are already specified.

The researchers also compared detections across scalp, cortical-surface, and intracranial recordings. Their method identified more events occurring across multiple areas than a conventional threshold method. An apparent absence of a spindle at one site can therefore reflect the detector's sensitivity, rather than establishing that the site did not participate in the event.



Mofrad et al. (2022), Fig. 2, selected panels.



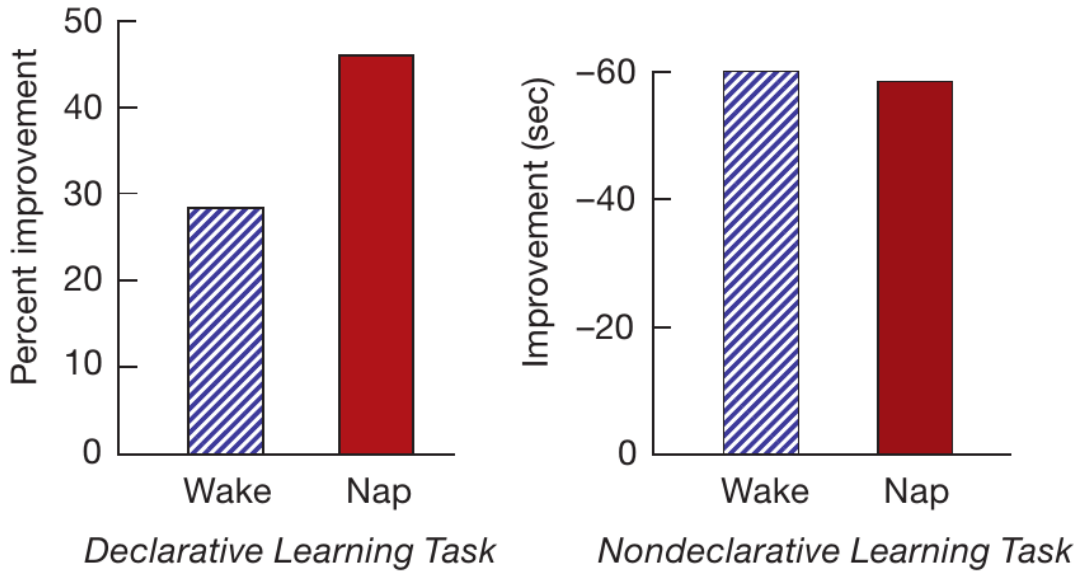
Mofrad et al. (2022), Video 2.

Sleep rhythms organize communication for memory

A cortical **slow oscillation** alternates between population states with relatively little firing and states with greater firing. A thalamocortical **spindle** contains a faster burst of repeated cycles, while a hippocampal **sharp-wave ripple** is a brief coordinated event with an even faster oscillation. These rhythms arise in different circuits and operate on different timescales.

During NREM sleep, their timing can become coordinated. Spindles can occur at particular phases of the cortical slow oscillation, while hippocampal ripples occur within selected parts of that combined pattern. This arrangement creates recurring windows in which activity from the hippocampus can reach cortical circuits while their responsiveness is changing.

Repeated activation can alter synapses in the participating networks through activity-dependent plasticity. Which cells fire, and when their inputs arrive relative to postsynaptic activity, influences whether a connection strengthens or weakens. Coordinating sleep rhythms provides one proposed mechanism for changing memory representations through repeated communication among hippocampal, thalamic, and cortical populations.



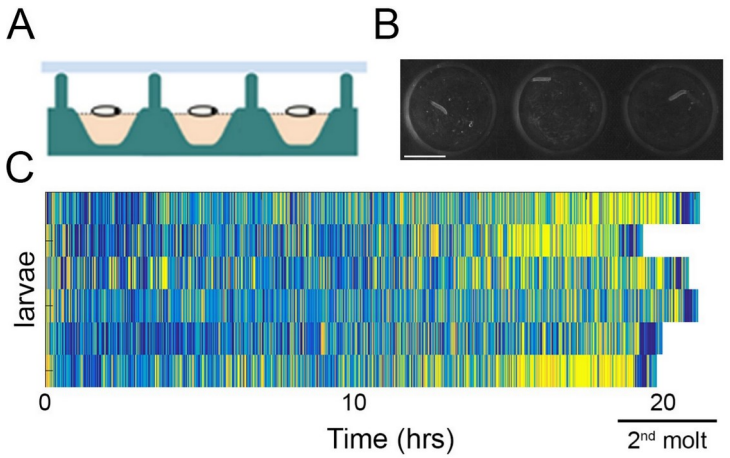
Carlson and Birkett (2022), Fig. 9.7, p. 277.

Young fly larvae alternate between activity and sleep

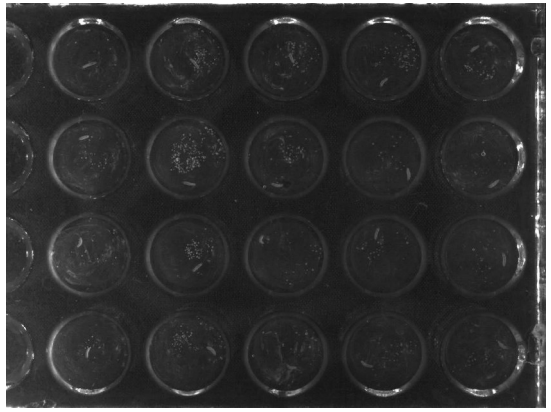
Szuperak and colleagues developed **LarvaLodge** chambers to follow individual fruit-fly larvae over extended periods. The chambers provided food and a restricted space in which movement could be measured. The original recording shows multiple larvae, allowing prolonged pauses and active periods to be distinguished without repeatedly handling each animal.

Larvae entered recurring periods of immobility with a characteristic withdrawn head position. They were less responsive to weak stimulation during those periods, but stronger stimulation could restore movement. After sleep was disrupted, subsequent rest increased, providing a homeostatic criterion in addition to the visible posture and reduced responsiveness.

These animals were still developing their nervous systems, including production of new neurons from dividing precursor cells. Measuring sleep at this stage made it possible to ask about a process largely completed before birth in many mammals. The behavioral assay established when a larva slept so that later cellular measurements could be related to a defined sleep history.



Szuperak et al. (2018), Fig. 1, selected panels.



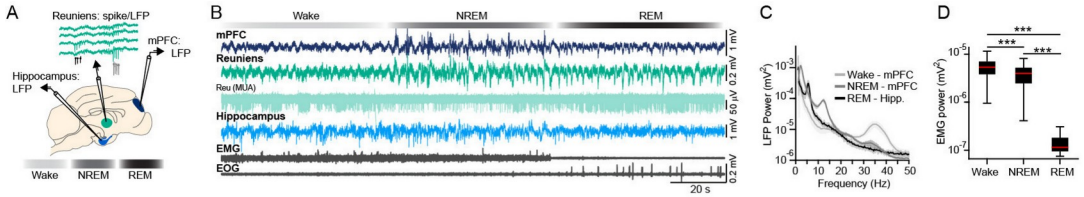
Szuperak et al. (2018), Video 1.

Sleep events align hippocampus, thalamus, and cortex

The **reuniens nucleus** is a thalamic structure with connections to the hippocampus and medial prefrontal cortex. Basha and colleagues recorded these regions together during natural sleep in cats. Simultaneous recording allowed a hippocampal event, thalamic firing, and a cortical spindle to be compared on the same timeline.

During NREM sleep, hippocampal sharp-wave ripples and reuniens activity tended to precede the onset of prefrontal spindles. Cortical transitions into active UP states also changed the probability of subsequent hippocampal ripples. The measured sequence therefore contained influences traveling through a recurrent network, rather than a single fixed message moving once from hippocampus to cortex.

During REM sleep, reuniens activity was more strongly organized around hippocampal theta, while its relationship with prefrontal activity differed from NREM. The anatomical connections remained present across states, but the timing of communication changed. A circuit's wiring and its current physiological state jointly determine which populations become coordinated at a particular moment.



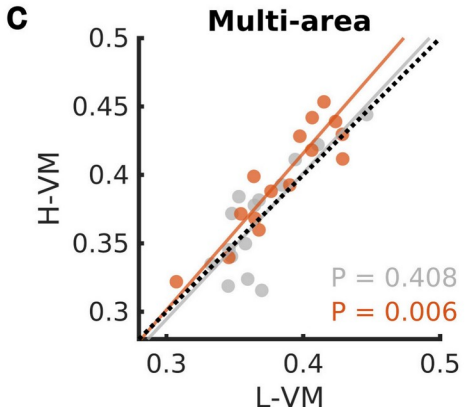
Basha et al. (2025), Fig. 2, selected panels.

Widespread spindles change after a memory task

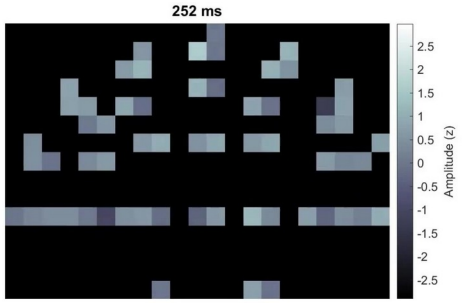
Mofrad and colleagues compared sleep after visual-memory tasks with different memory loads. Participants had previously encoded either more or fewer visual scenes. The investigators then examined the spatial organization of detected spindles, asking whether the direction of a traveling pattern differed according to the preceding learning condition.

The source movie displays an actual EEG spindle, with the filtered signal mapped across scalp locations. Its changing colors represent the phase of the electrical waveform, not the movement of a remembered picture. Comparing neighboring locations reveals a rotating pattern with activity reaching different recording sites at successive moments.

After the higher memory load, one direction of rotation became more prominent in widespread, multi-area spindles. The same change was not evident when only local spindles were analyzed. The effect therefore involved how a sleep rhythm was coordinated across regions, beyond a simple count of whether spindles occurred somewhere in the recording.



Mofrad et al. (2022), Fig. 4, selected panels.



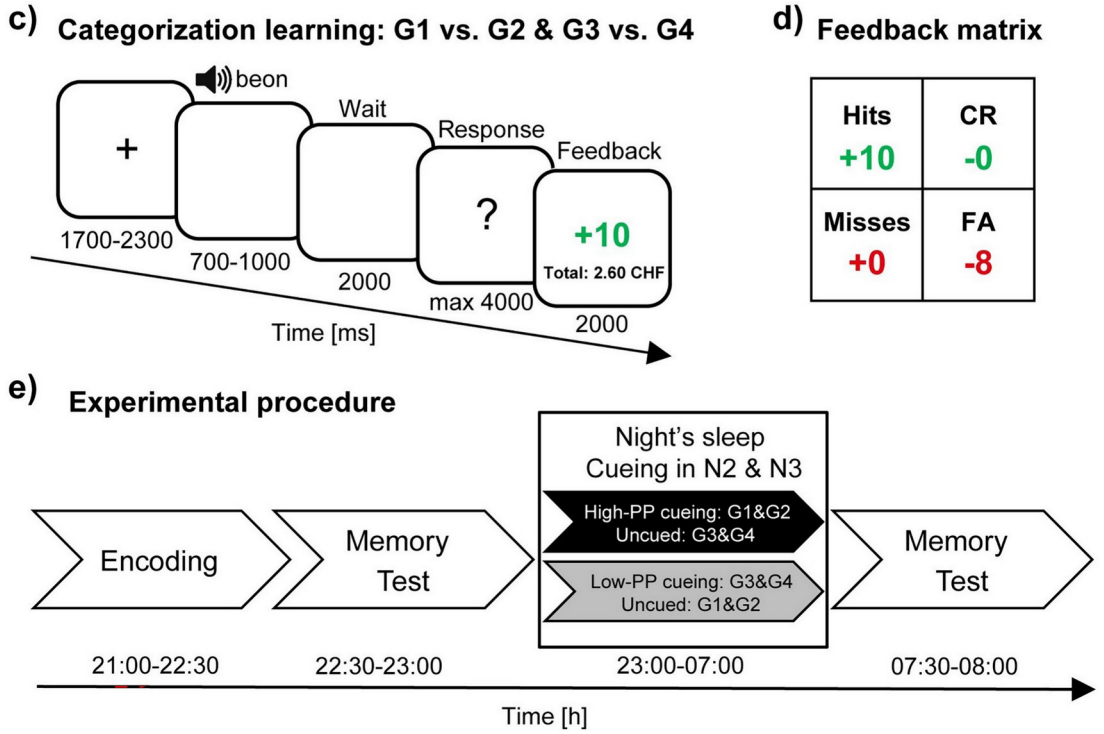
Mofrad et al. (2022), Video 1.

Sounds during sleep can reactivate learned material

Targeted memory reactivation uses a cue associated with previously learned material to influence its processing during sleep. Klaassen and Rasch trained participants to classify artificial spoken words while awake. Some word forms were easier to learn because their sound sequences were more familiar, allowing prior learning difficulty to be varied within the experiment.

Selected words were quietly presented again during NREM sleep. Memory for the easier words benefited from this cueing, whereas the more difficult words did not show the same improvement. The participants had already learned the word classifications before sleeping; the procedure tested reactivation of those representations rather than acquisition of an entirely new vocabulary while asleep.

Both kinds of word cues increased slow-wave occurrence, but effective cues produced stronger spindle-band activity within slow-wave UP states. The electrical response therefore differed according to how well the material had been encoded. An incoming sound can evoke a sleep response without necessarily producing the same later memory benefit for every associated item.



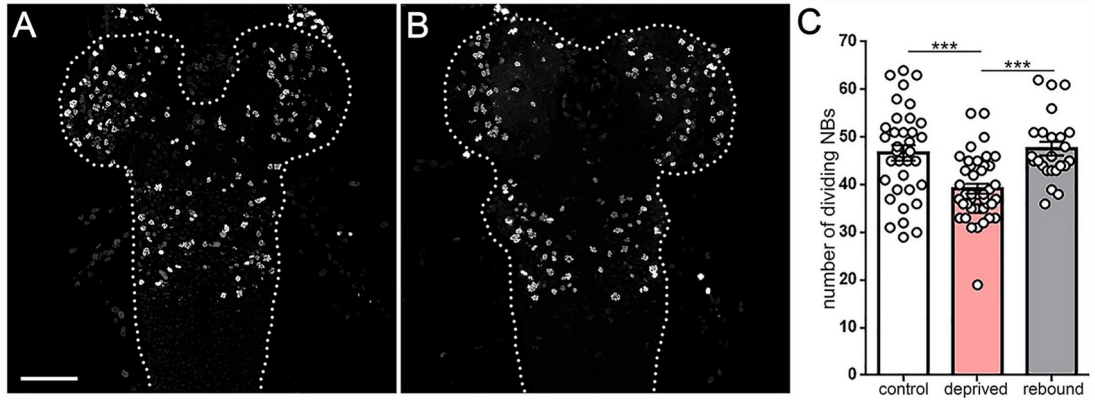
Klaassen and Rasch (2024), Fig. 1, selected panels.

Early sleep loss reduces neural progenitor division

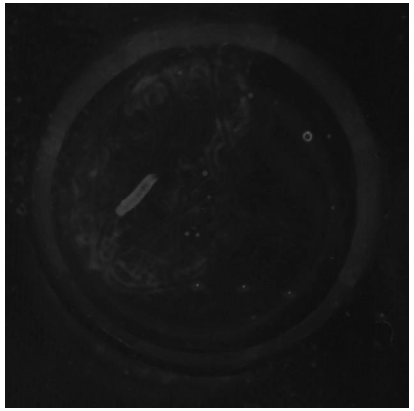
A **neural progenitor** is a precursor cell capable of dividing to produce cells of the nervous system. Szuperak and colleagues labeled dividing cells in larval fly brains after normal sleep or sleep deprivation. Deprived larvae had fewer labeled divisions, indicating reduced ongoing proliferation during this early period of nervous-system development.

A closed-loop system detected when a larva became inactive and delivered a stimulus to interrupt that rest. Comparison animals received stimulation during waking instead. This arrangement separated stimulation itself from stimulation that specifically prevented sleep. The source movie demonstrates rapid behavioral reversibility when light awakens an inactive larva, a property needed for the interruption procedure.

Recovery sleep restored the division measurement toward the control condition. The total pool of neural progenitors did not disappear after deprivation; fewer of those cells were dividing at the measured time. The result therefore concerns the timing or rate of precursor-cell proliferation, rather than evidence that the brief deprivation destroyed all cells capable of making new neurons.



Szuperak et al. (2018), Fig. 6, selected panels.



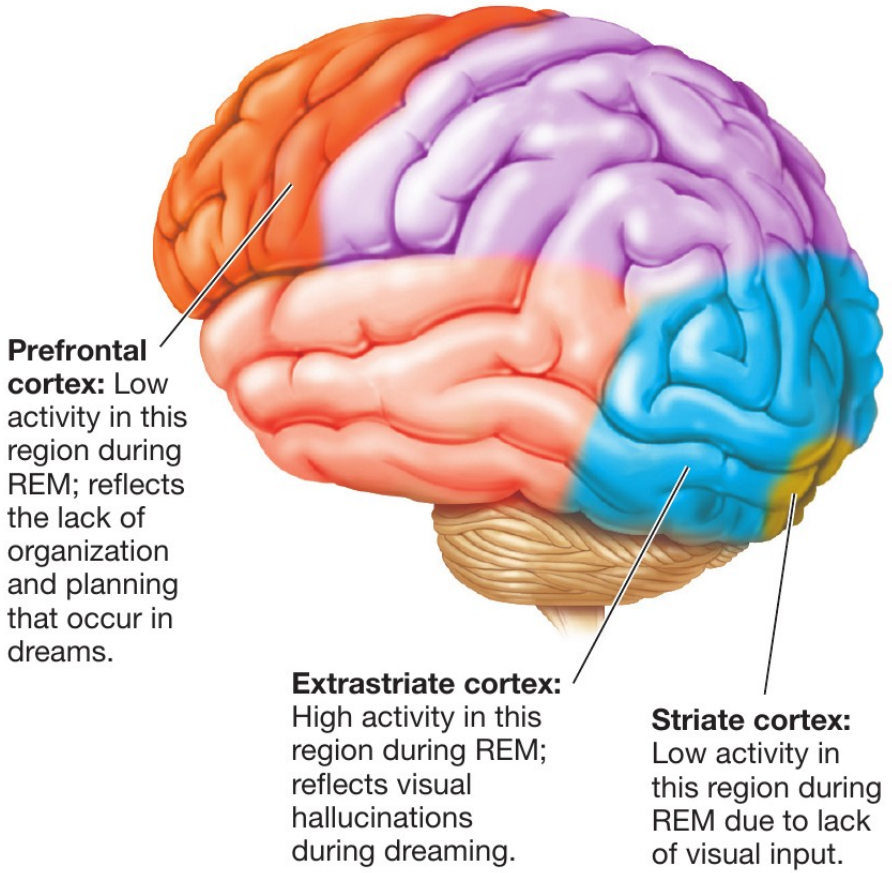
Szuperak et al. (2018), Video 5.

Dreaming involves active perception and altered control

Dream experiences are assembled through brain activity that can generate scenes without the corresponding sensory events occurring outside the body. Visual association regions contribute representations of objects and places, while emotion-related systems can contribute urgency or fear. Activity in movement-related circuits can accompany dreamed actions even when postural muscles remain inhibited.

A dreamer's ability to evaluate an event can also differ from waking. Reduced engagement of some prefrontal functions can make contradictions or abrupt changes less likely to be questioned. A scene may therefore feel convincing while it happens, even though its sequence would be recognized as unusual when recalled after waking.

In a **lucid dream**, the sleeper recognizes that the experience is a dream. This awareness does not necessarily terminate REM sleep or restore ordinary movement. It creates an experimental opportunity: a trained person may deliberately produce an agreed signal with a muscle system that remains usable, such as the eyes, while the surrounding physiological recording still indicates REM.



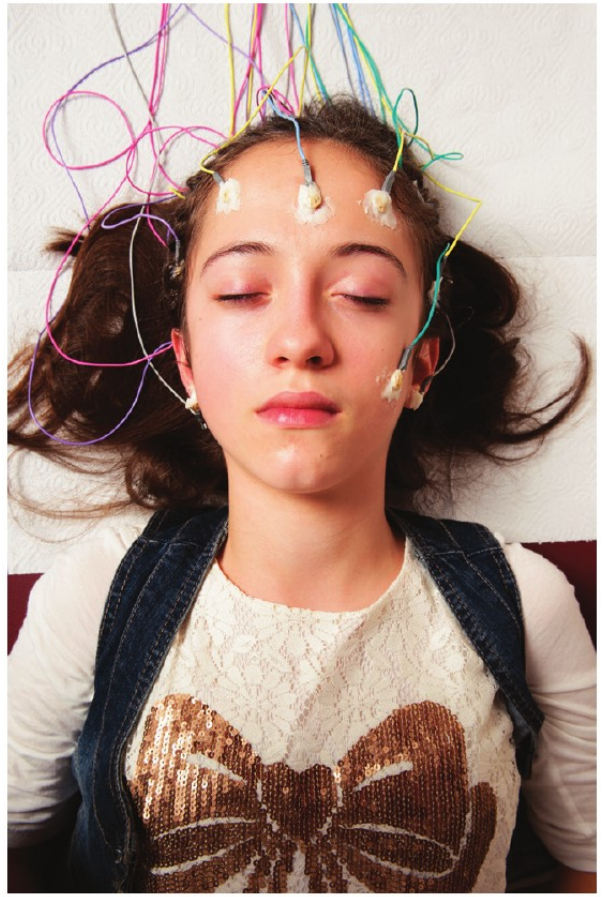
Carlson and Birkett (2022), Fig. 9.4, p. 271.

Some lucid dreamers can answer questions during REM

Konkoly and colleagues tested communication with people experiencing lucid dreams during REM sleep. Participants learned signals before sleep, such as a specific sequence of left-right eye movements. EEG, eye, and muscle recordings allowed researchers to determine whether a signal occurred during verified REM rather than relying only on the person's later description.

Experimenters then presented questions requiring an immediate answer. In one documented trial, a spoken subtraction problem, eight minus six, was followed by two eye signals. Other participants used selected facial-muscle responses. Correct answers occurred in a subset of trials and sleepers, demonstrating a capacity that was available intermittently rather than throughout every REM episode.

The external question had to be perceived, held briefly in mind, processed, and translated into the agreed response. Those operations can therefore sometimes occur while the person remains in REM sleep. An answer recorded at the time also provides information that can be compared with the later dream report, which may omit or misremember parts of the interaction.



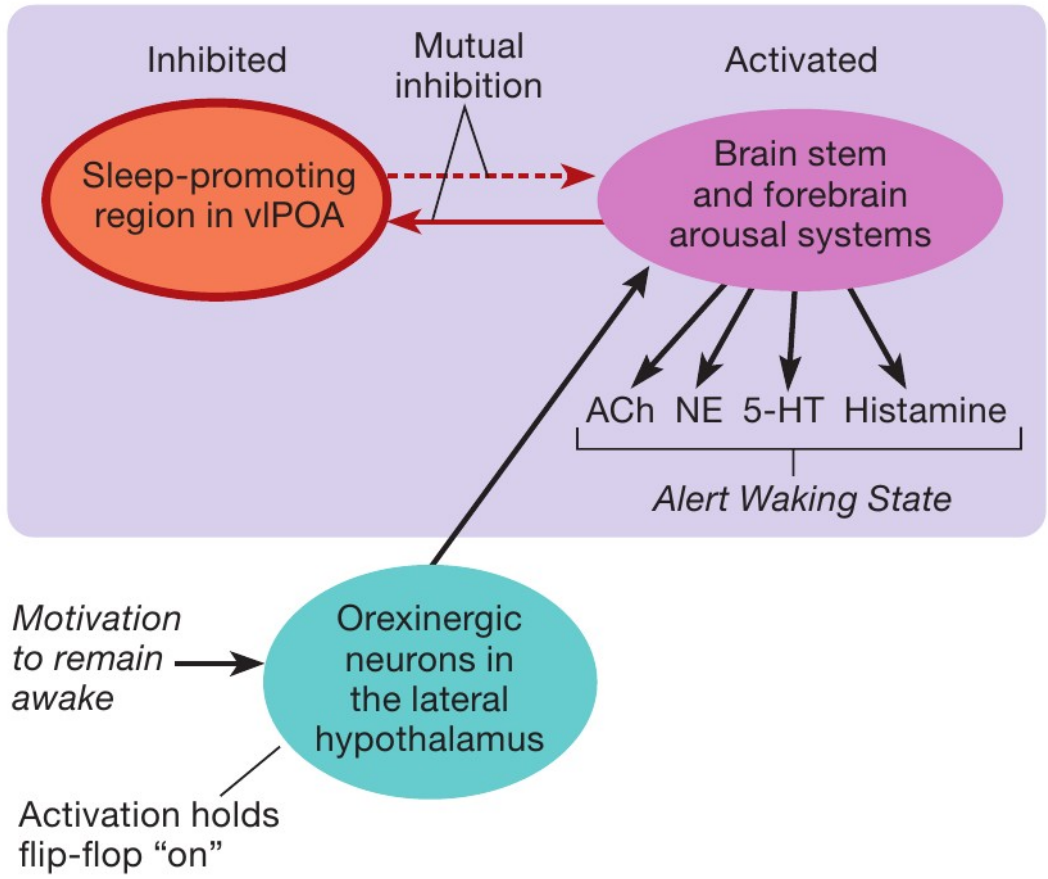
Carlson and Birkett (2022), Fig. 9.1, p. 268.

Orexin loss can destabilize boundaries between states

Narcolepsy type 1 commonly involves loss of the hypothalamic neurons that produce orexin. Without their stabilizing input, wake-promoting systems have greater difficulty maintaining a continuous waking period. Affected people can experience intense daytime sleepiness and unplanned transitions into sleep despite intending to stay awake.

Cataplexy is a sudden reduction or loss of muscle tone during waking, often triggered by emotion. Consciousness can be preserved while the person becomes unable to maintain posture. This symptom resembles the motor inhibition associated with REM sleep entering a waking state, so loss of tone and loss of awareness must be described separately.

Other symptoms can include vivid experiences at sleep onset and temporary inability to move when awakening. These events involve incomplete alignment of sensory experience, awareness, muscle control, and sleep-stage physiology. The normal boundaries between components of sleep and waking become less stable when orexin input no longer supports the arousal network.



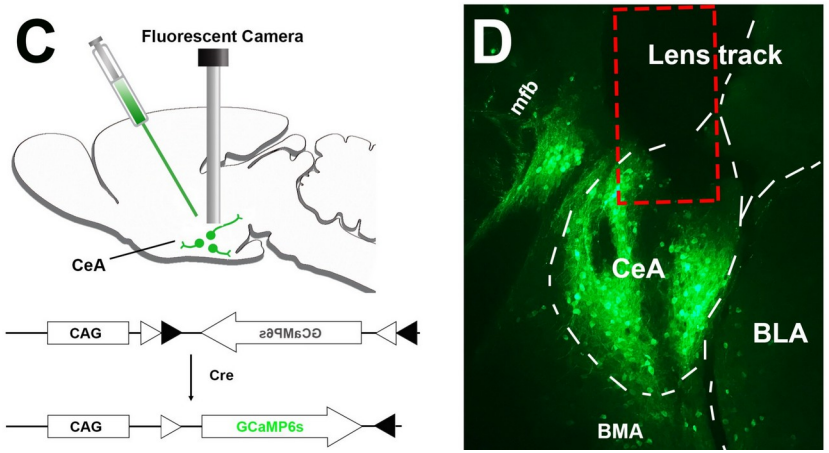
Carlson and Birkett (2022), Fig. 9.18, p. 285.

Amygdala activity can be recorded during cataplexy

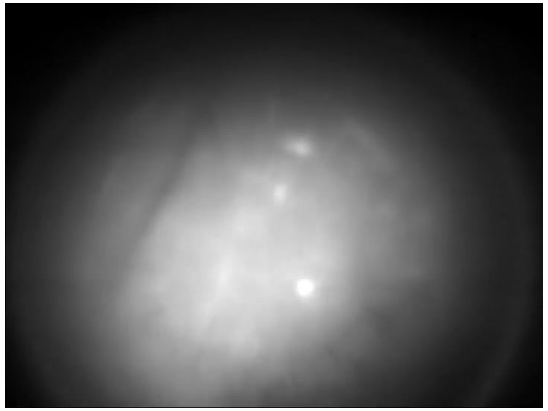
The **central amygdala** contains neurons involved in processing emotionally significant events and influencing defensive behavior. Sun and colleagues expressed a calcium indicator in its GABA-producing neurons in mice lacking normal orexin neurons. A small implanted lens and camera allowed those deep-brain cells to be followed while behavior was recorded.

The source movie contains the original calcium images, played at the authors' accelerated speed. Brightening indicates a change in the indicator's fluorescence associated with intracellular calcium; it is not a direct picture of GABA molecules leaving a terminal. Individual cells could be tracked to compare their activity across waking, sleep, and cataplexy episodes.

Different neurons showed different state preferences. Some became active during ordinary waking or REM sleep, while other groups were particularly active during cataplexy. Recording the same identified cells across these conditions allowed the investigators to distinguish an event-specific response from a general increase shared by every GABAergic neuron in the amygdala.



Sun et al. (2019), Fig. 1, selected panels.



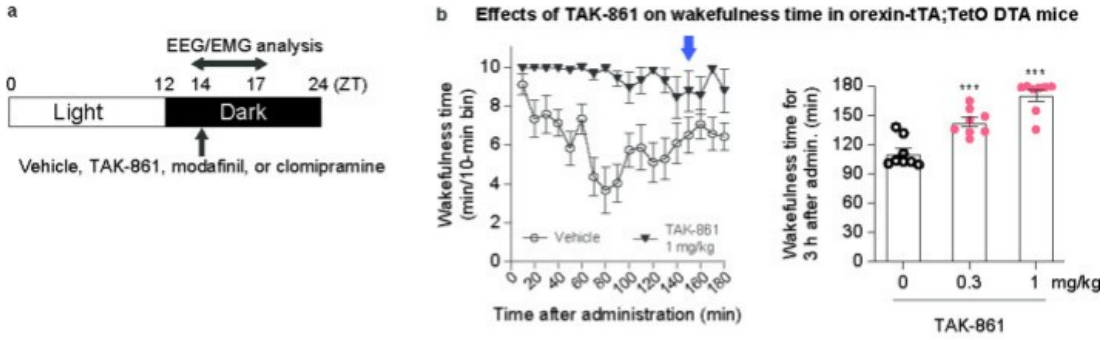
Sun et al. (2019), Video 1.

An orexin agonist restores wake-promoting signals

An **agonist** activates a receptor by binding to it. Mitsukawa and colleagues studied TAK-861, a small molecule designed to activate orexin receptor 2. Because it acts on the receiving receptor, the compound can provide an orexin-like signal even when the neurons that normally release the peptide have been lost.

In mouse models with loss of orexin neurons, oral TAK-861 increased waking and reduced fragmentation of waking periods. It also reduced cataplexy-like episodes under the tested conditions. EEG and muscle recordings allowed the researchers to measure both time awake and episodes of abnormal muscle-tone loss rather than combining them into one behavioral score.

The researchers compared these effects with drugs acting through other mechanisms, including modafinil and clomipramine. Increasing alertness and suppressing cataplexy are separable treatment actions, so the same comparison drug need not improve both equally. Activating an orexin receptor directly tests a way to replace a missing signal at its downstream target.



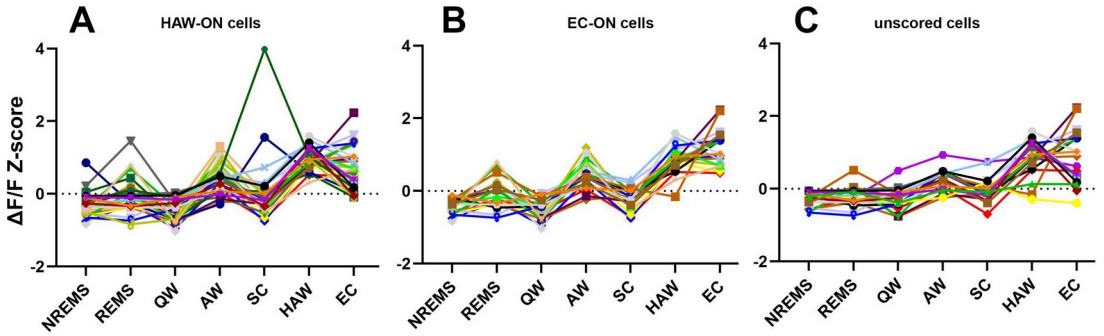
Mitsukawa et al. (2024), Fig. 4, selected panels.

Predator odor acts differently in narcoleptic mice

Sun and colleagues exposed control and orexin-deficient mice to coyote urine, a predator-associated odor. The original side-by-side recording shows their behavioral responses to the same kind of stimulus. The odor created an emotionally significant condition in which the investigators could compare ordinary arousal with the transition into cataplexy.

The narcoleptic mice showed stronger and more widespread calcium responses in the recorded central-amygdala GABAergic population. Activity in some neurons remained elevated as the animals entered odor-associated cataplexy. In control mice, the odor-related response did not produce the same combination of extreme population activation and subsequent loss of muscle tone.

A GABA-producing neuron inhibits the cells reached by its own axon terminals, but its activation does not mean the whole brain becomes inhibited. If its targets normally prevent muscle-tone suppression, reducing those targets' activity could permit atonia. Identifying the active amygdala population helps locate an emotion-sensitive entry point into circuits regulating the motor component of state.



Sun et al. (2019), Fig. 3, selected panels.



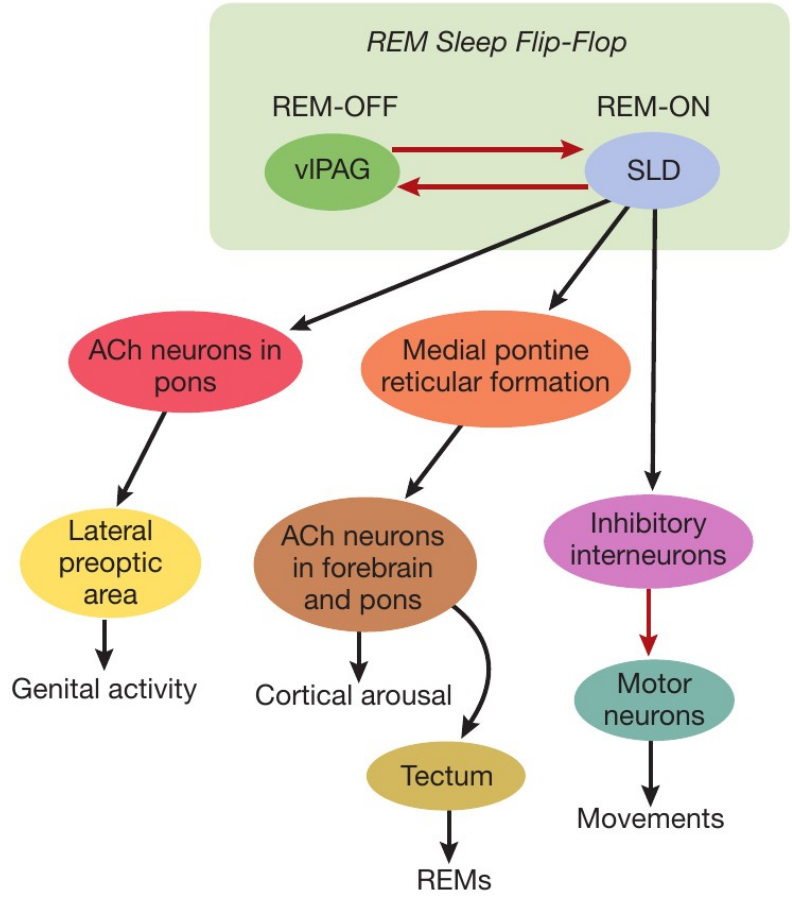
Sun et al. (2019), Video 2.

Failure of REM muscle inhibition permits dream enactment

During normal REM sleep, brainstem circuits reduce activity in motor pathways that maintain posture. Inhibitory neurons acting through GABA and glycine contribute to this **REM atonia**. Eye movements and breathing continue because their control systems are organized differently from the motor pools responsible for most voluntary limb and trunk movements.

In **REM sleep behavior disorder**, this motor inhibition is incomplete, allowing a sleeper to move while experiencing a dream. Movements may include reaching, kicking, or vocalizing. A sleep recording can reveal excessive muscle activity during REM, connecting the observed behavior to a physiological failure of the usual state-related inhibition.

Cataplexy and REM sleep behavior disorder involve opposite errors in the timing of muscle control. Cataplexy introduces atonia during waking, whereas REM sleep behavior disorder permits activity during a state that normally suppresses it. Distinguishing these conditions requires simultaneous information about consciousness, sleep stage, muscle activity, and the timing of the behavior.



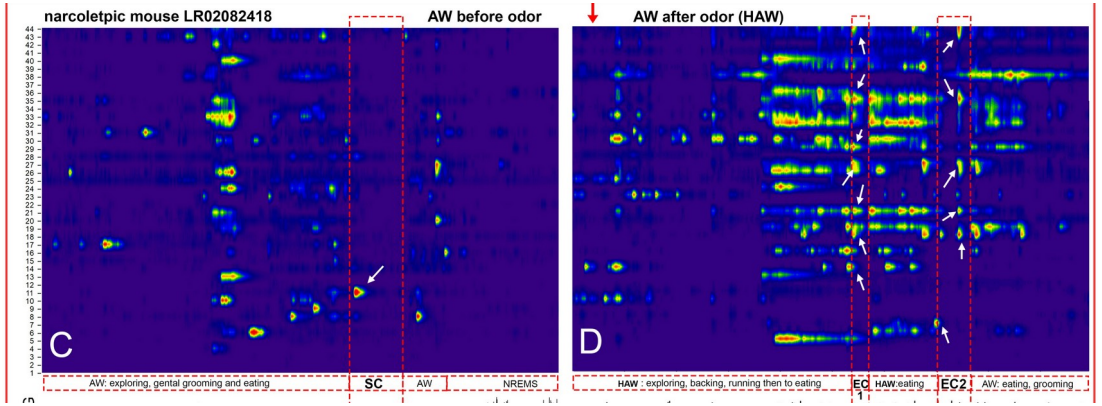
Carlson and Birkett (2022), Fig. 9.24, p. 288.

Spontaneous cataplexy recruits a particular neuronal group

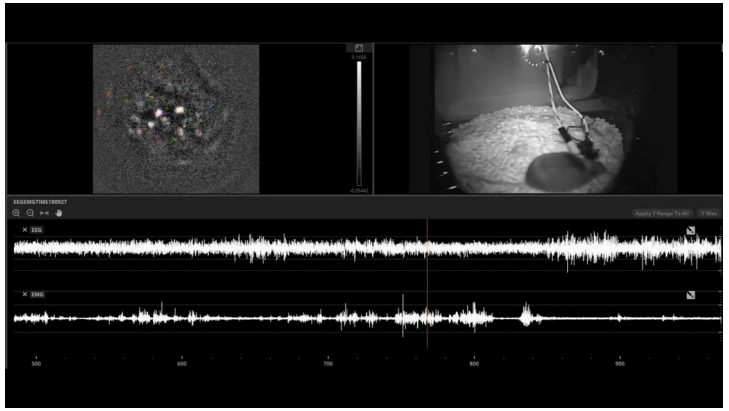
The amygdala recordings also captured **spontaneous cataplexy**, episodes occurring without the experimental predator odor. The source movie follows an episode aligned with the cellular recording. Its playback is accelerated by the original authors, so the duration of an event must be read from the source timing rather than estimated from screen time.

Some central-amygdala GABAergic neurons became especially active during these spontaneous episodes. Their activity differed from that of neurons most strongly recruited by the predator-odor condition. Classifying cells by their activity across repeated events separated populations that would have been mixed together in an average signal from the entire region.

The behavioral episode involved loss of postural muscle tone, while the recording identified which measured cells changed activity at that time. Calcium imaging establishes this temporal pattern; a separate targeted intervention would be needed to determine which cells are necessary to initiate the episode. The observed population was therefore an identified candidate circuit component, not a direct measurement of motor inhibition itself.



Sun et al. (2019), Fig. 4, selected panels.



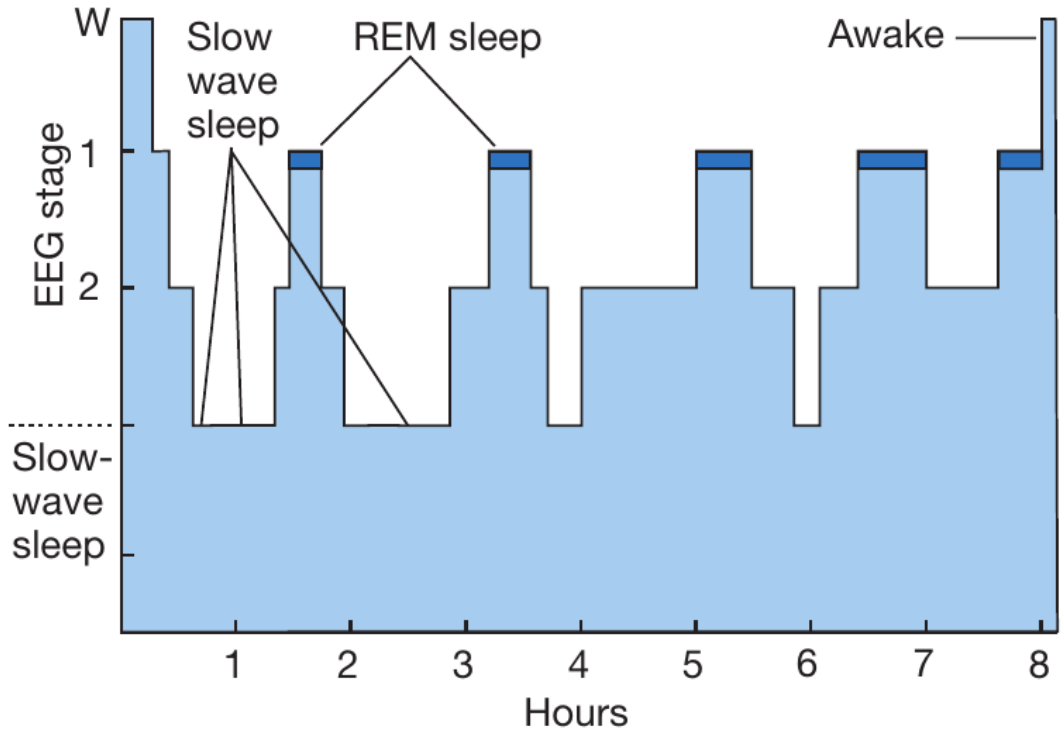
Sun et al. (2019), Video 3.

NREM parasomnias differ from REM dream enactment

Parasomnias are unwanted behaviors or experiences occurring during sleep or transitions around it. Sleepwalking and sleep terrors commonly arise from NREM sleep, especially deep sleep early in the night. The person can perform movements while remaining only partly responsive, and later recall may be limited or absent.

These events differ from ordinary REM dream enactment in their sleep-stage context. In a NREM arousal episode, some systems supporting movement and autonomic activity become active while other aspects of wakefulness remain incomplete. The resulting behavior can look purposeful without the flexible awareness and memory expected during an ordinary waking conversation.

Sleep paralysis involves another combination: awareness returns while REM-related muscle inhibition temporarily persists. A person can perceive the bedroom and feel unable to move, sometimes with dreamlike sensory experiences. The episode ends as motor control and waking awareness become aligned, illustrating why sleep-stage transitions must be considered separately from stable periods of sleep.



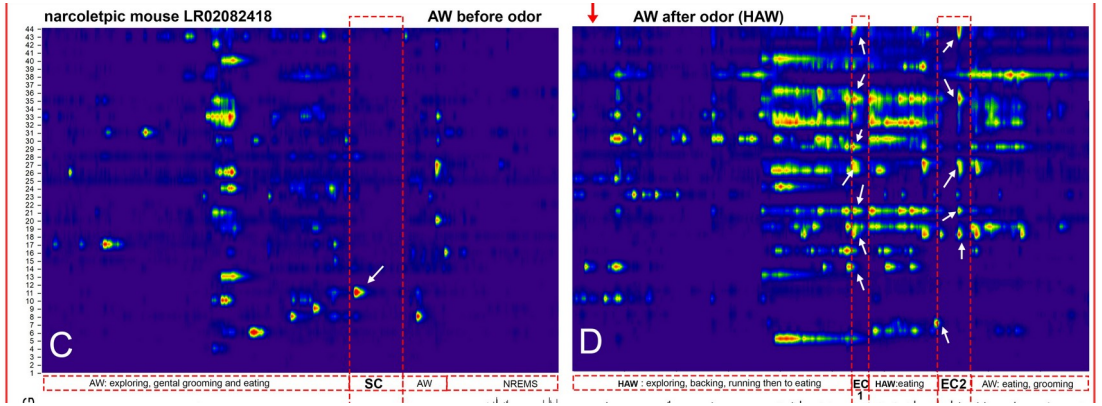
Carlson and Birkett (2022), Fig. 9.3, p. 271.

Emotion-related cataplexy recruits another activity pattern

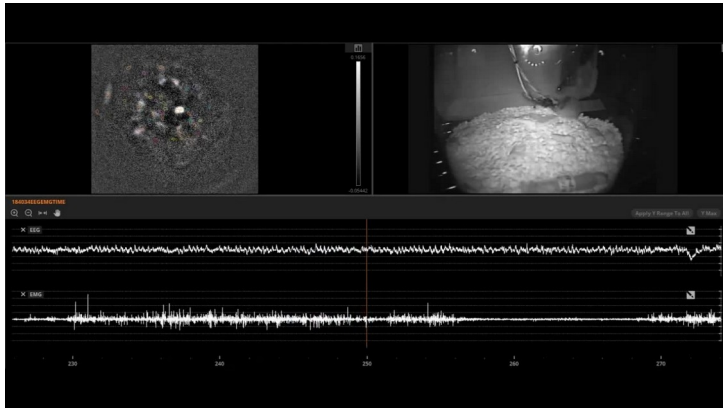
Predator-odor-associated cataplexy recruited a partly different central-amygdala population from spontaneous cataplexy. In Sun and colleagues' recordings, many of these cells were relatively inactive during ordinary sleep-wake cycles. Their activity rose during the emotionally significant odor condition and remained high into the subsequent cataplexy episode.

The recorded neurons also became more synchronized during odor-associated cataplexy. Synchronization means their calcium events occurred with more coordinated timing, adding information beyond an increase in the activity of each cell considered alone. A downstream target could receive a more concentrated inhibitory input when many connected neurons become active together.

The original behavior movie is aligned with the odor-associated episode, while the figure follows the cellular response across the transition. Comparing those timelines separates the initial aroused behavior from the later loss of tone. The same animal can move from strong emotional activation into atonia without first passing through a normal extended NREM sleep period.



Sun et al. (2019), Fig. 4, selected panels.



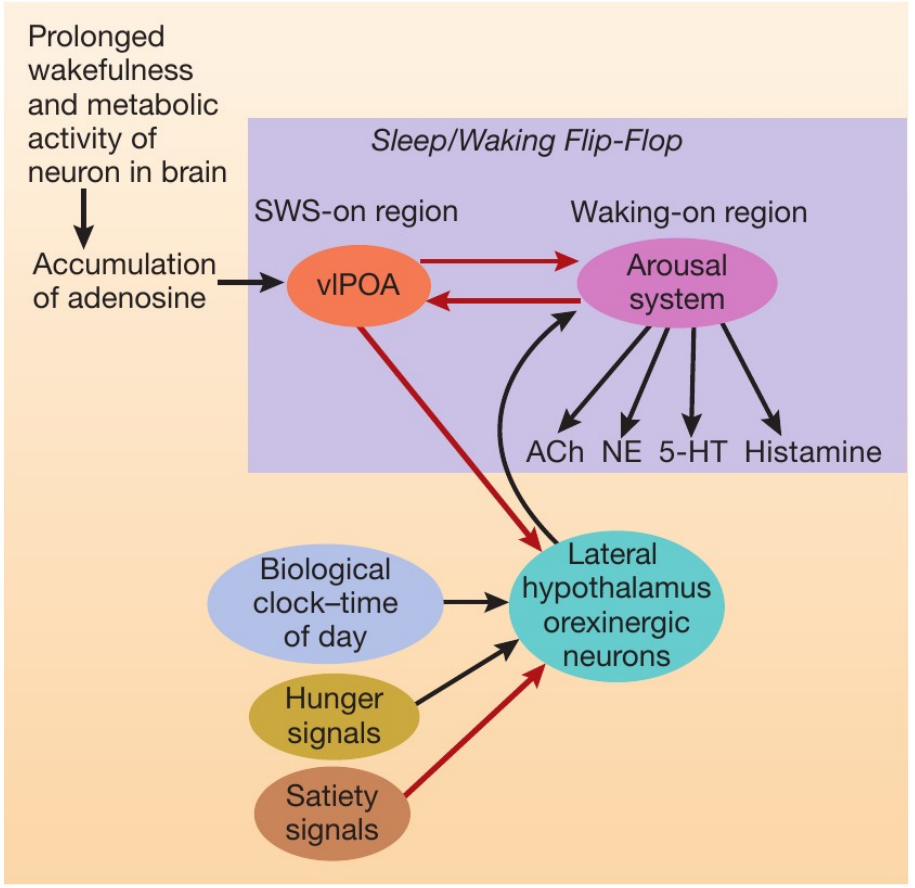
Sun et al. (2019), Video 4.

Insomnia involves disrupted sleep and persistent arousal

Insomnia involves difficulty initiating or maintaining sleep, or waking earlier than intended, together with daytime consequences despite an adequate opportunity to sleep. The problem is not defined by one universal number of hours. The timing of the difficulty, its persistence, and its effects on daytime functioning all contribute to the clinical description.

Persistent cognitive or physiological arousal can interfere with sleep-promoting processes. Worry about sleeping may sustain attention to bodily sensations and the passage of time, while stress-related systems increase alertness. A person can consequently feel exhausted yet remain unable to enter or maintain the coordinated state changes needed for sleep.

Behavioral treatment can change associations between the bed and prolonged waking, while adjusting time in bed can strengthen the pressure for consolidated sleep. Circadian misalignment requires attention to timing as well. A person whose internal night occurs unusually late may struggle to sleep at an imposed early bedtime even though sleep is more continuous at a later preferred time.



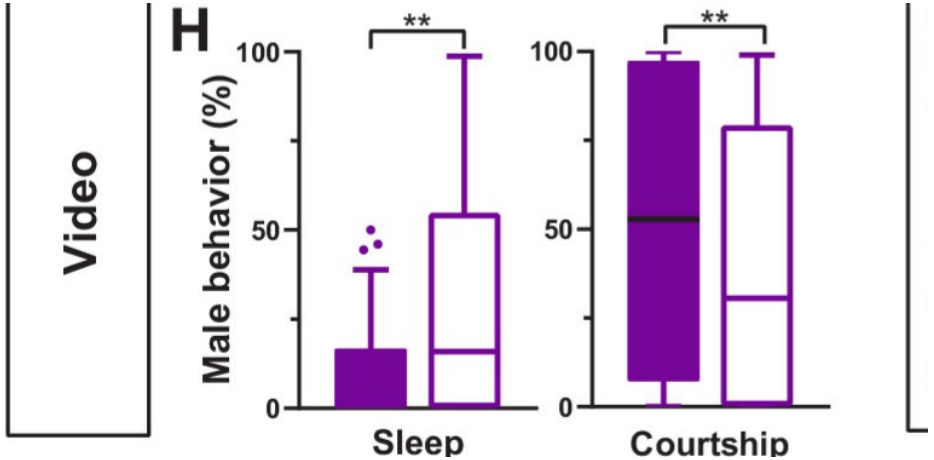
Carlson and Birkett (2022), Fig. 9.20, p. 286.

Nutrition changes competition between sleep and courtship

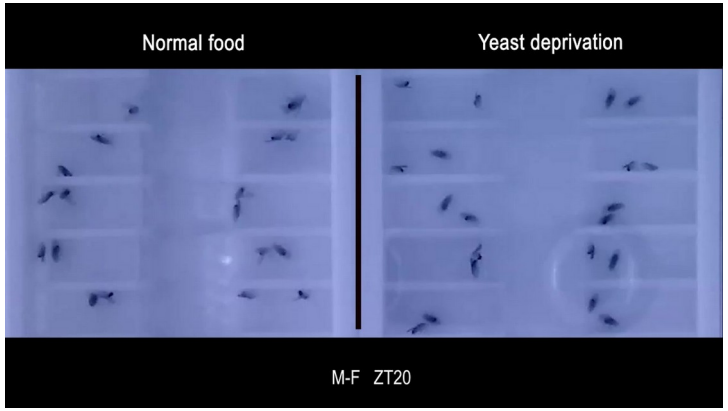
Sleep competes with other behaviors that require waking. Duhart and colleagues tested how nutrition changes this competition in male fruit flies housed with female partners. Normally fed males reduced nighttime sleep while courting, whereas males deprived of dietary yeast spent more of that period asleep, despite being physically capable of courtship.

The source recording compares these social conditions during the night under infrared illumination. The researchers separately scored sleeping and courting, rather than treating every movement as the same behavior. Yeast-deprived males still showed daytime courtship, indicating that the nighttime difference involved behavioral allocation rather than complete loss of the courtship repertoire.

The study identified dopamine neurons downstream of courtship-related P1 neurons that could reduce sleep in normally fed males. Activating this pathway did not produce the same sleep loss in yeast-deprived males. Nutritional state therefore changed how a social circuit influenced sleep, placing metabolic information within the pathway that selects whether the animal remains active.



Duhart et al. (2020), Fig. 1, selected panels. Filled: normally fed; open: yeast deprived.



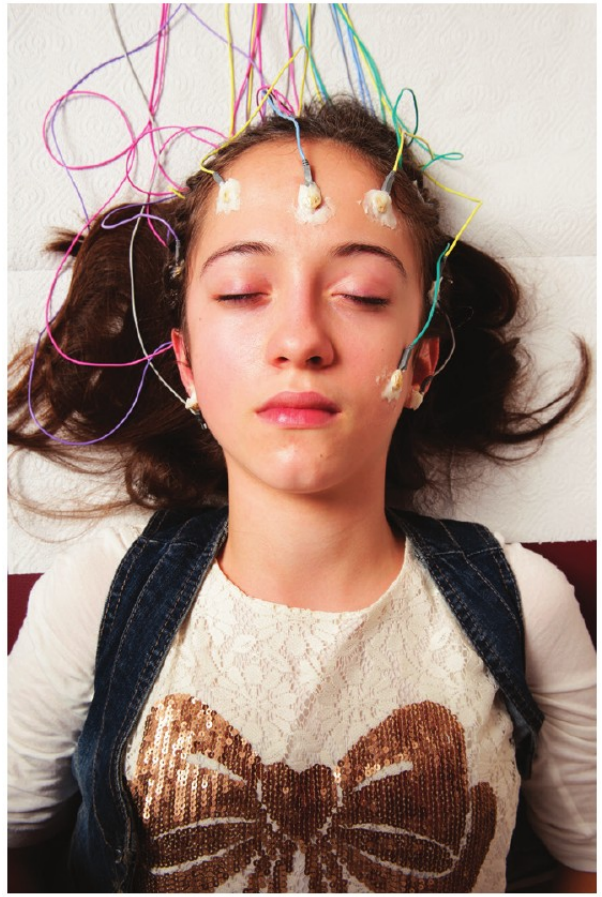
Duhart et al. (2020), Video 1.

Sleep apnea repeatedly interrupts breathing and sleep

Obstructive sleep apnea occurs when the upper airway repeatedly narrows or closes during sleep despite continued breathing effort. Airflow falls, oxygen can decline, and the accumulating respiratory disturbance can trigger a brief arousal. Restoring muscle activity may reopen the airway, after which the cycle can recur when sleep resumes.

In **central sleep apnea**, breathing effort itself is temporarily reduced or absent because the respiratory control signal changes. Recording airflow together with chest and abdominal movement distinguishes an obstructed effort from a pause in respiratory drive. Oxygen measurements and EEG show how the breathing event affects the body and interrupts the sleeping brain.

Repeated brief arousals can fragment sleep even when the person remembers few awakenings and spends many hours in bed. A continuous positive airway-pressure device treats obstruction by using pressure to keep the airway open. Its immediate physical action targets airflow, allowing sleep to continue without the same repeated sequence of obstruction and arousal.



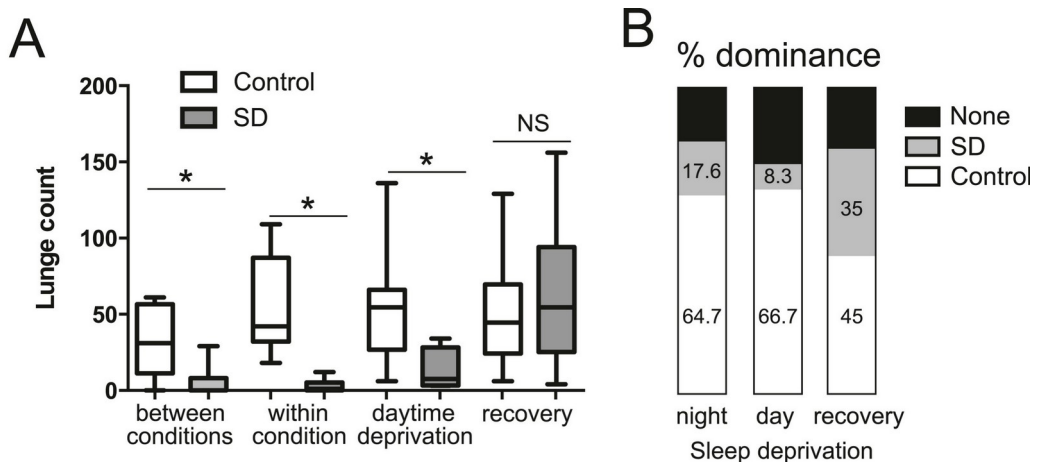
Carlson and Birkett (2022), Fig. 9.1, p. 268.

Sleep deprivation suppresses aggression in fruit flies

Kayser and colleagues measured aggression in male fruit flies after sleep deprivation. They scored **lunges**, recognizable attacks directed at another fly, rather than equating general movement with aggression. Sleep-deprived males lunged less and were less likely to establish dominance when competing with rested males in the same arena.

The source recording shows a sleep-deprived pair in the behavioral assay. Other social actions, including courtship, were relatively preserved under the tested conditions, and time spent interacting was not simply eliminated. The behavioral effect therefore involved a reduction in aggressive actions rather than a complete inability to walk, encounter another fly, or perform social behavior.

Aggression recovered after an opportunity for recovery sleep. Manipulations of **octopamine**, a fly transmitter with functions resembling parts of mammalian norepinephrine signaling, also influenced the deprivation-related reduction. The study connects recent sleep history to a particular social behavior through a chemically modulated circuit, without assuming that sleep loss increases aggression in every species or situation.



Kayser et al. (2015), Fig. 1, selected panels.



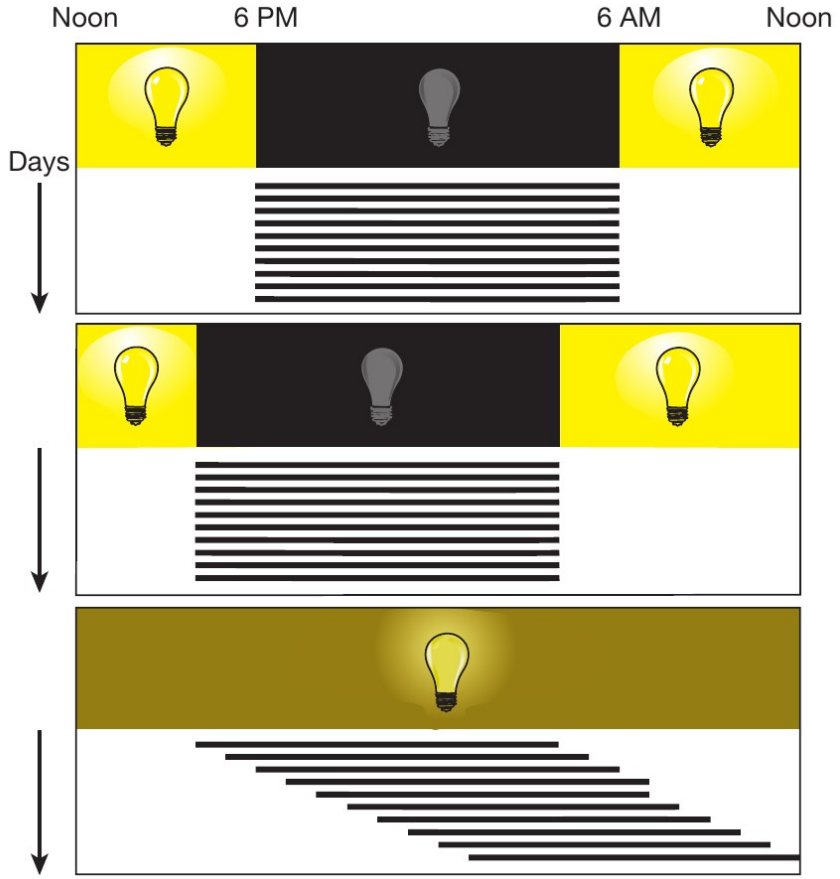
Kayser et al. (2015), Video 2.

Circadian rhythms persist without a daily light signal

A **circadian rhythm** is an internally generated cycle with a period close to one day. An animal can continue showing a daily activity rhythm in constant darkness, even though no new sunrise tells it when to become active. This persistence distinguishes a biological clock from a response that occurs only when a light is switched on.

An **actogram** displays activity across successive days, with clock time along the horizontal axis. If activity begins slightly later each day in constant conditions, the internal period is longer than twenty-four hours. If onset moves earlier, the period is shorter. The diagonal progression makes the difference between internal and environmental time visible.

Daily environmental signals normally adjust the clock so that its rhythm matches the external cycle. This process is called **entrainment**. A rhythm continuing under constant conditions is called **free-running**, because its timing reflects the organism's own period without the usual daily resetting signal keeping it aligned with the environment.



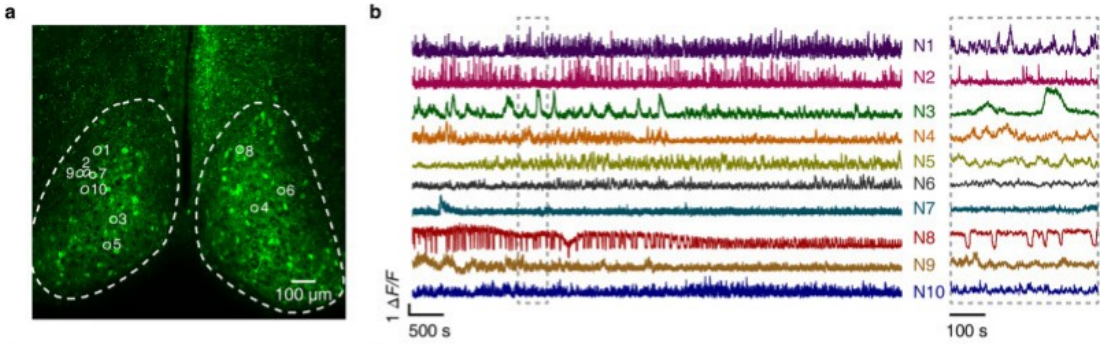
Carlson and Birkett (2022), Fig. 9.25, p. 293.

Thousands of SCN neurons carry diverse calcium signals

The **suprachiasmatic nucleus**, or SCN, is a paired hypothalamic structure located above the optic chiasm. Its neurons contain molecular clocks and communicate with one another. Wang and colleagues used large-scale calcium imaging in adult mouse SCN slices to examine how the population represents time across many individual cells.

The recordings contained calcium signals with different amplitudes, durations, and temporal patterns. Pooling information from many neurons allowed a computational decoder to estimate the time represented by the preparation more accurately than relying on one cell. A decoder is an analysis that predicts a variable from measured activity, not a literal reader placed inside the animal's brain.

The source video displays a four-dimensional scan through the SCN volume, including time and depth. It shows the authors' acquired calcium data across the preparation rather than a hand-drawn clock mechanism. Volumetric imaging captures populations that would be missed in a single thin optical plane, allowing spatial position to be compared with the signals carried by different cells.



Wang et al. (2024), Fig. 1, selected panels.



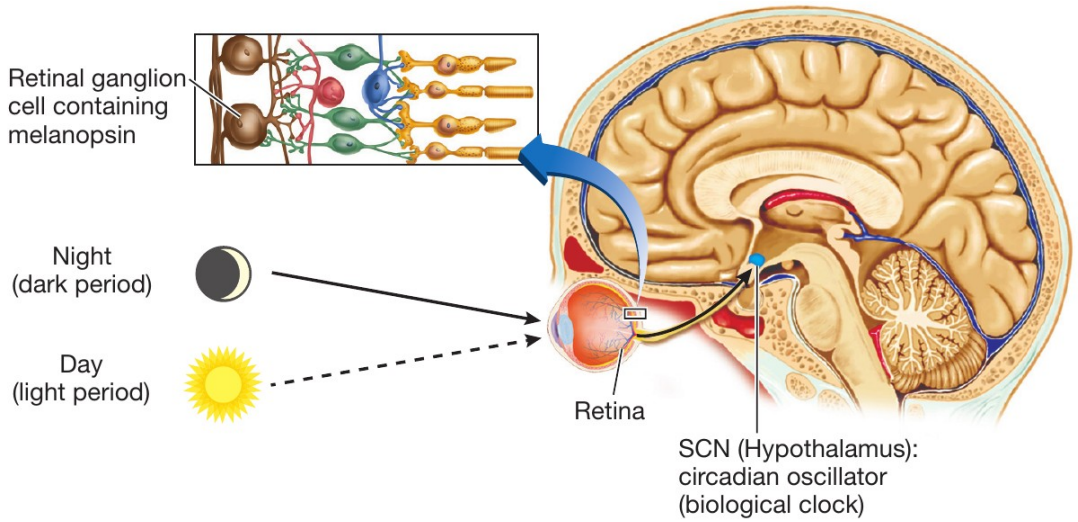
Wang et al. (2024), Video S1.

Retinal signals synchronize the brain's central clock

Specialized retinal ganglion cells contain **melanopsin**, a light-sensitive pigment, and send axons toward the SCN through the **retinohypothalamic tract**. These cells also receive information originating from rods and cones. Their output provides the clock with a measure of environmental illumination, including light that does not require conscious recognition of a visual object.

At SCN terminals, retinal input releases transmitters including glutamate. Receiving neurons change electrical activity and intracellular signaling, which can alter expression of clock-related genes. Light can therefore shift the molecular process generating the rhythm instead of merely producing a brief movement or a conscious sensation of brightness.

The clock's response depends on its current phase. The same light exposure can have different effects at different biological times, because the molecular and electrical state of the receiving network changes over the day. Recording clock timing before and after exposure distinguishes resetting from the immediate alerting or behavioral response to the light itself.



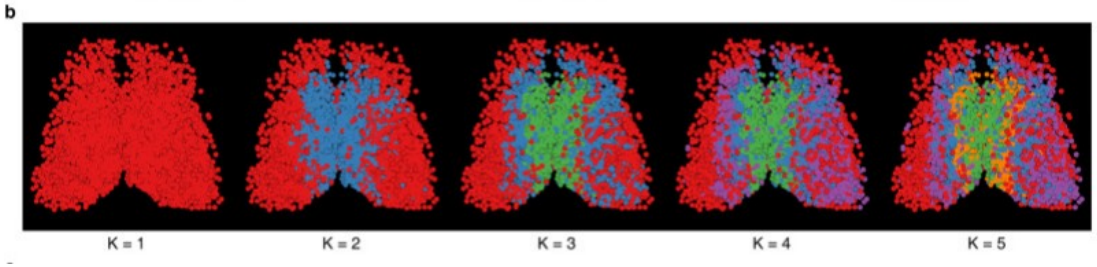
Carlson and Birkett (2022), Fig. 9.27, p. 295.

SCN modules represent different time features

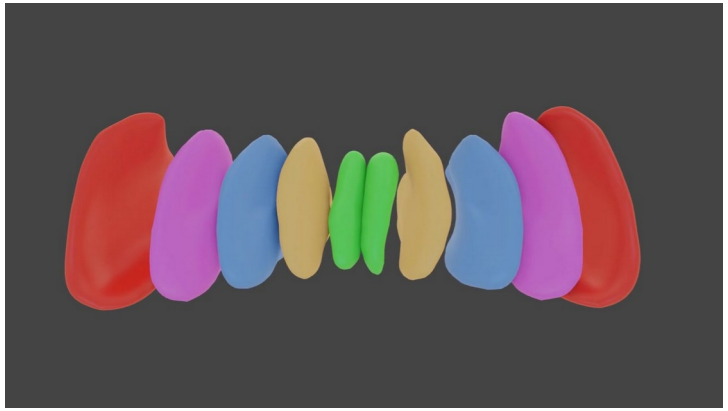
Wang and colleagues grouped SCN neurons according to features of their calcium signals without first assigning them to an anatomical region. Cells with similar signal features tended to occupy related spatial locations. The resulting groups formed approximately bilateral modules, indicating that functional differences were organized within the tissue rather than scattered completely at random.

The source movie is a three-dimensional **analytical reconstruction** of these classified cells. Its colors identify computationally defined groups; they do not represent a live wave of firing passing through the SCN. Rotating the reconstruction reveals how similarly classified neurons are arranged across the depth and sides of the preparation.

A time decoder trained on the whole population performed worse when it sampled only one module. A decoder trained specifically within a module could still recover timing information from that module. Different groups therefore contained time-related signals with different features, while successful interpretation depended on the relationship between the sampled cells and the decoder's training.



Wang et al. (2024), Fig. 4, selected panels.



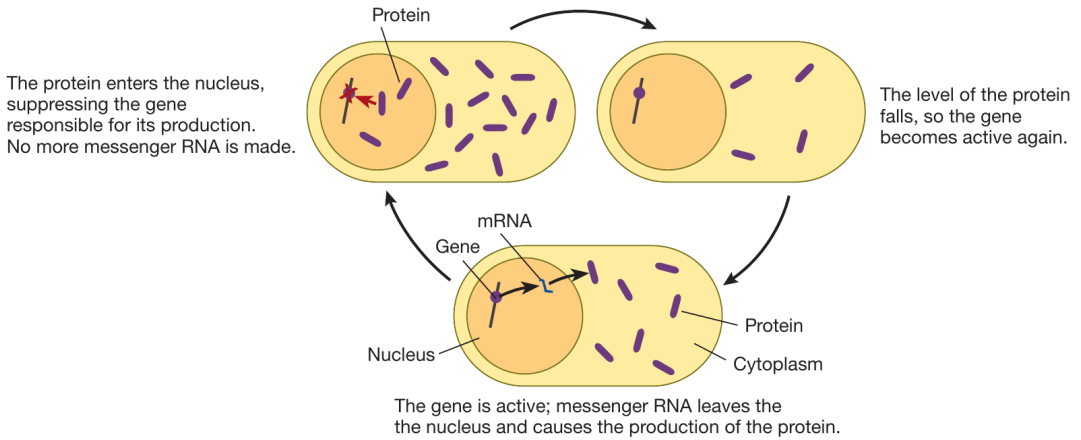
Wang et al. (2024), Video S2.

Clock proteins create a delayed negative-feedback cycle

The mammalian molecular clock contains a **delayed negative-feedback loop**. Proteins including CLOCK and BMAL1 promote transcription of genes such as Period and Cryptochrome. Transcription produces RNA, which is then translated into PER and CRY proteins. These steps introduce time between activation of the genes and accumulation of their protein products.

As PER and CRY accumulate and enter the nucleus, they suppress the transcriptional activity that helped produce them. Their own production consequently declines. Later, modification and degradation remove enough inhibitory protein for transcription to rise again, beginning another cycle of synthesis, accumulation, inhibition, and removal within the cell.

The delay is essential to the rhythm. Immediate feedback would tend to hold the system near a steady level, whereas time-consuming production, transport, and degradation allow it to cycle. Changes in protein stability can lengthen or shorten the interval, connecting a molecular alteration to the period observed in an animal's daily behavior.



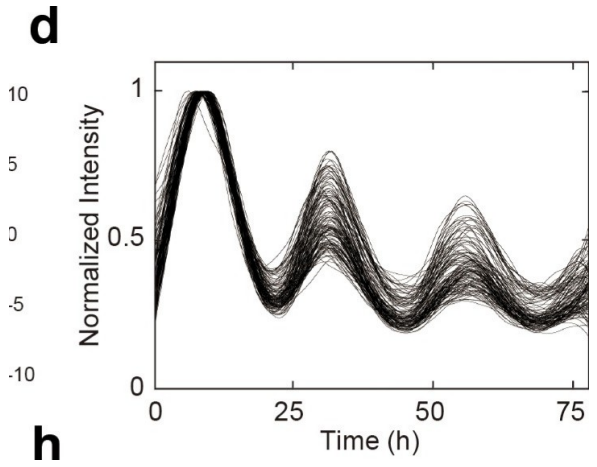
Carlson and Birkett (2022), Fig. 9.29, p. 297.

Individual clock cells normally keep coordinated rhythms

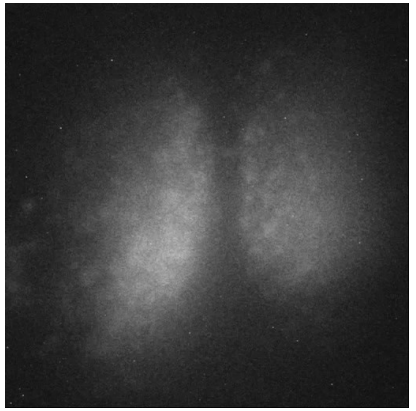
Hayasaka and colleagues studied the clock protein PER2 using a **luciferase reporter**. Luciferase produces light through a chemical reaction, allowing changes in a linked clock signal to be followed over time. Cultured SCN slices continued producing rhythmic light signals after removal from the animal, demonstrating that the preparation retained an internal timing process.

The source recording shows a normal mouse SCN slice across successive cycles. Individual cells change their light output, and their timing is coordinated enough to produce an organized population rhythm. The changing brightness reports the molecular reporter rather than ordinary neural electrical spikes or a flash of calcium associated with a single action potential.

The investigators measured both **period**, the time between cycles, and **phase**, the position of a cell within its cycle. Two cells can have similar periods while reaching their peaks at different times. Following single cells therefore reveals whether a weak population rhythm arises from weak individual oscillators or from oscillators whose peaks no longer align.



Hayasaka et al. (2017), Fig. 2, selected panels.



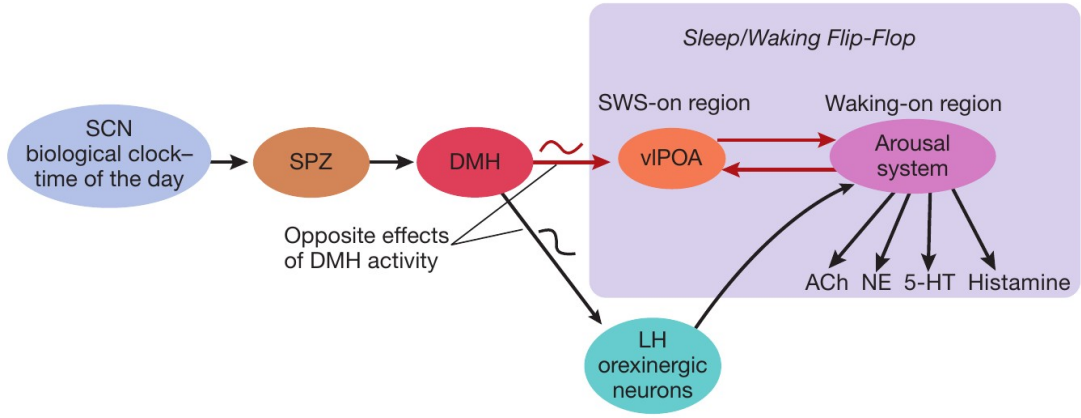
Hayasaka et al. (2017), Figure 2, video 1.

Light can advance or delay a circadian clock

A **phase advance** moves a rhythm earlier, while a **phase delay** moves it later. In many circumstances, light late in the biological night advances the human clock, whereas light earlier in the biological night delays it. These effects depend on internal phase, so civil clock time alone cannot specify the response for every person.

Repeated light exposures can entrain a clock whose natural period differs slightly from the environmental day. Small daily adjustments compensate for that difference, keeping sleep-related and hormonal rhythms near a stable relationship to sunrise and sunset. A brief light-induced change in alertness is separate from this lasting adjustment of the clock's phase.

A **phase-response curve** plots the direction and size of resetting against the biological time of the stimulus. Its advance and delay regions explain why identical evening and morning exposures can shift timing in opposite directions. When the external schedule changes, the clock must undergo a sequence of these adjustments before its outputs realign.



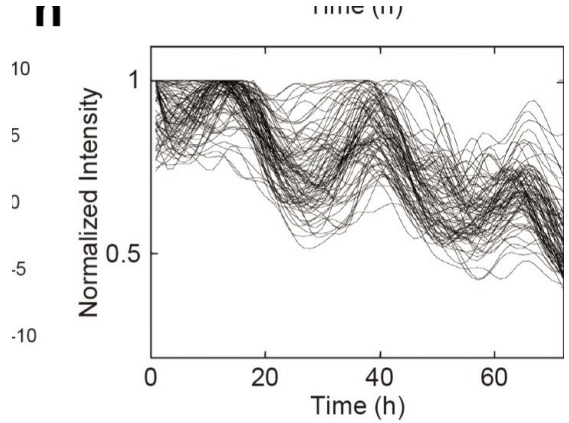
Carlson and Birkett (2022), Fig. 9.28, p. 296.

SIK3 loss destabilizes molecular clock coordination

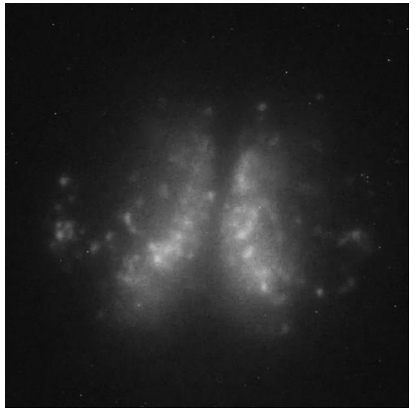
SIK3 is an enzyme that contributes to regulation of clock-protein stability. Hayasaka and colleagues compared SCN slices from normal mice with slices lacking SIK3. The mutant cells displayed a broader distribution of periods and phases, producing less coordinated molecular rhythms across the tissue despite the continued presence of many oscillating cells.

The source movie follows the SIK3-deficient preparation using the same PER2-linked light reporter as the normal slice. A population image can lose a coherent rhythm when individual cells peak at different times. Single-cell tracking showed that the altered signal involved instability and desynchronization rather than simply disappearance of all cellular clock activity.

Additional experiments found increased PER2 abundance when SIK3 was reduced and lower PER2 abundance when SIK3 was increased. The effect involved phosphorylation-dependent destabilization of PER2, directly or through intermediate steps. Slowing removal of an inhibitory clock protein changes feedback timing, providing a molecular route to the altered daily rhythms observed in the animals.



Hayasaka et al. (2017), Fig. 2, selected panels.



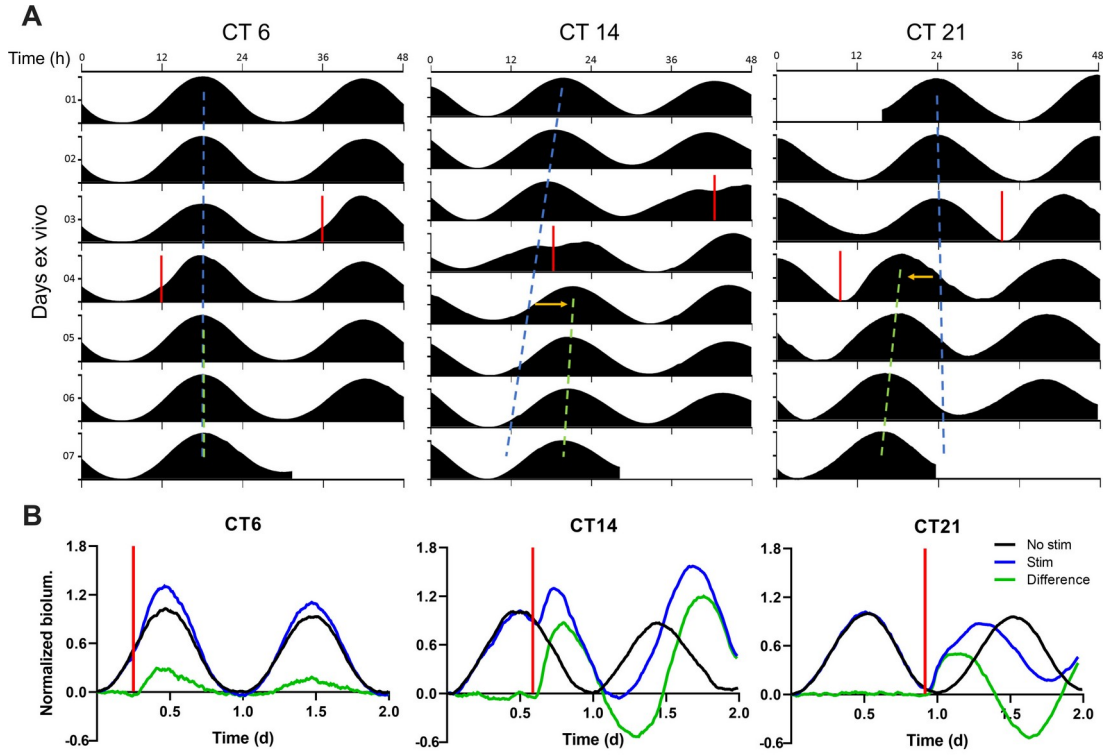
Hayasaka et al. (2017), Figure 2, video 2.

Dawn and dusk reshape molecular clock waveforms

Kim and McMahon simulated light input directly in cultured SCN tissue using an optogenetic channel. They monitored PER2-linked bioluminescence before and after stimulation at different circadian phases. Because the preparation was outside the animal, the experiment could examine clock resetting without requiring a change in locomotion, feeding, or the timing of sleep.

Stimulation altered the shape of the molecular waveform as well as its peak time. Rising and falling phases changed in duration, producing an asymmetric cycle during resetting. Advancing or delaying the clock therefore involved changing the progression of the molecular process, rather than moving an otherwise unchanged curve along the time axis.

Different SCN regions showed different responses. Acute waveform shifts were strongest in the ventrolateral region, while period effects were greater dorsomedially. Repeated stimulation resembling different seasonal day lengths also reshaped the rhythms, providing a cellular mechanism through which the clock network can represent both daily timing and the duration of environmental light.



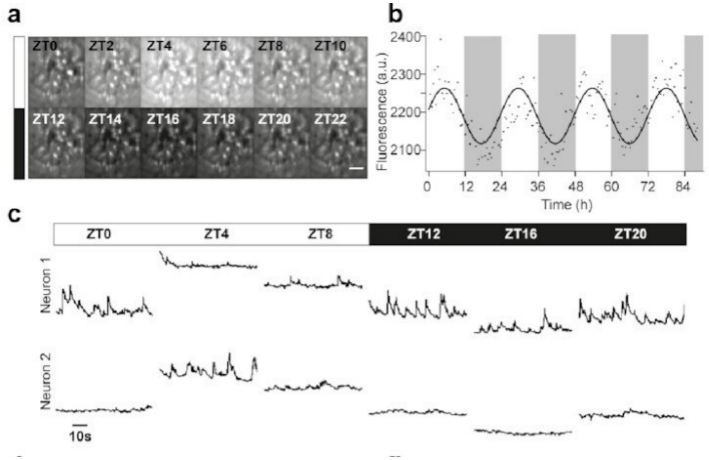
Kim and McMahon (2021), Fig. 2, selected panels.

A living SCN contains fast and slow calcium rhythms

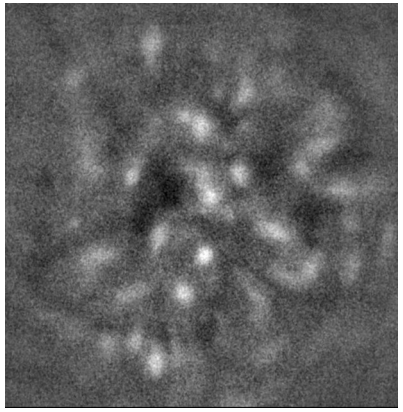
El Cheikh Hussein and colleagues recorded calcium in SCN neurons of freely behaving mice using a miniature imaging system. Each signal contained fast calcium events superimposed on a more slowly changing baseline. Measuring those components separately allowed the investigators to compare rapid cellular activity with changes distributed across the daily cycle.

The source movie shows an acquired SCN recording at the authors' accelerated playback speed. Individual brightening events occur within a tissue field whose overall fluorescence also changes more slowly. The fast event rate and the baseline are different features of the same measurement, so averaging them together would remove information about how each neuron operates.

Daily changes in baseline calcium were relatively coordinated across neurons, while the times of peak fast-event activity were more diverse. A cell could therefore share the population's broad daily organization while producing its rapid events at a different time. The living clock combined temporal unity at one scale with diversity at another.



El Cheikh Hussein et al. (2022), Fig. 1, selected panels.



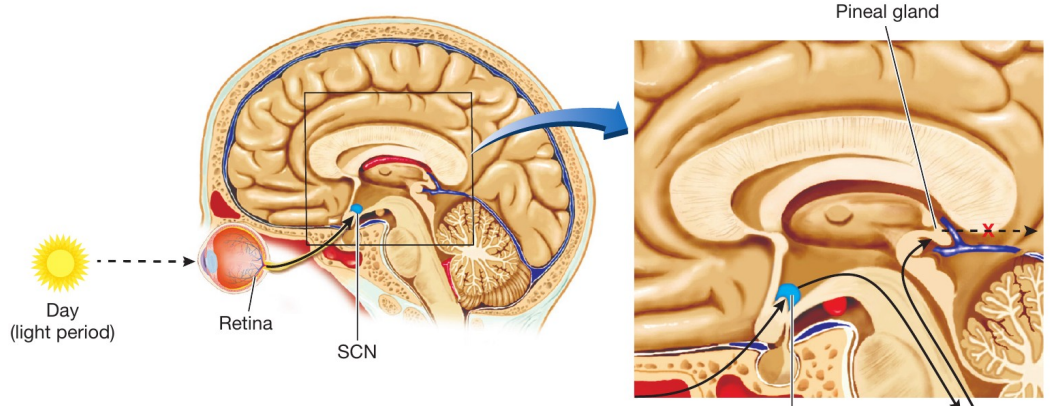
El Cheikh Hussein et al. (2022), Movie S1.

Melatonin communicates the timing of darkness

Melatonin is a hormone released by the pineal gland, with secretion normally increasing during the biological night. It communicates information about darkness and circadian phase to tissues containing melatonin receptors. Its rise is a useful timing signal, but it is not an electrical command that forces every brain region into sleep.

The SCN influences the pineal gland through a pathway involving the hypothalamus, spinal cord, and sympathetic neurons in the superior cervical ganglion. Norepinephrine released at the pineal gland promotes the biochemical steps needed for nighttime melatonin synthesis. The route therefore converts clock-network output into a circulating hormonal signal through several identifiable neural stages.

Light at night can suppress melatonin secretion through retinal input to this system. Researchers often measure **dim-light melatonin onset**, the evening rise recorded under controlled low illumination, to estimate internal circadian phase. Keeping light low during measurement reduces immediate light suppression that could otherwise obscure when the hormone's nightly rise would occur.



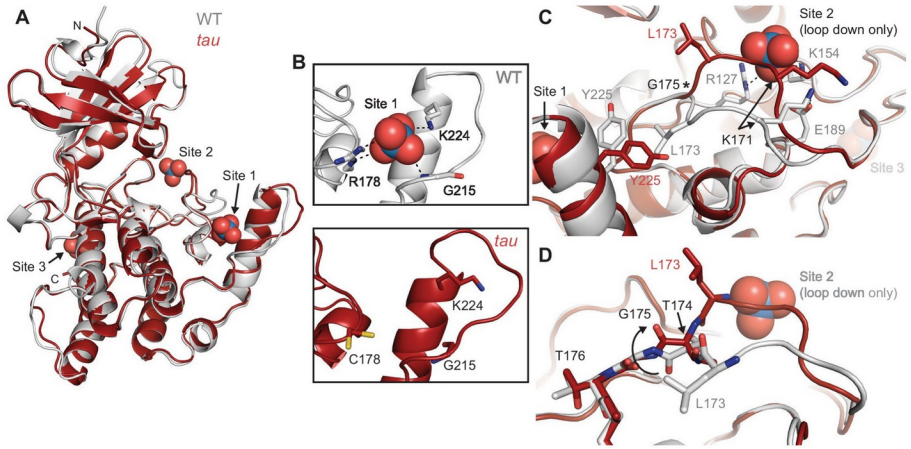
Carlson and Birkett (2022), Fig. 9.31, p. 299.

Protein phosphorylation changes the lifetime of PER2

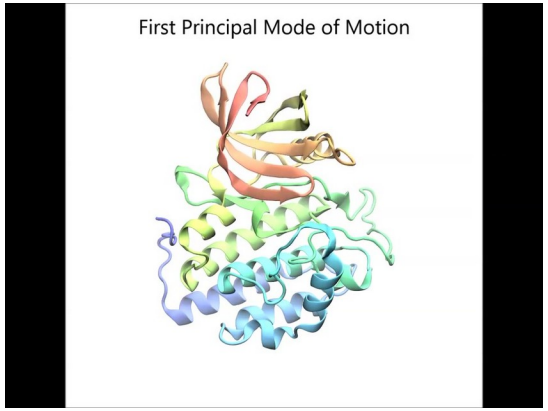
A **kinase** is an enzyme that adds phosphate groups to proteins. Philpott and colleagues studied casein kinase 1, which phosphorylates the clock protein PER2 at different sites. Modification at one group of sites can stabilize PER2, while modification at another can favor its recognition by the machinery that removes proteins from the cell.

The kinase's three-dimensional shape influences which sites it modifies efficiently. The source animation represents the authors' computed modes of motion, including opening and twisting of parts of the enzyme. It is a molecular model derived from structural analysis, rather than a microscope recording of one enzyme moving inside a sleeping brain.

Mutations that changed the kinase's structural behavior also shifted its relative action on stabilizing and degradation-related sites. This **phosphoswitch** alters how long PER2 remains available to inhibit the clock's transcriptional machinery. Changing the lifetime of that inhibitory protein changes the delay in the feedback cycle and can consequently change the circadian period.



Philpott et al. (2020), Fig. 2, selected panels.



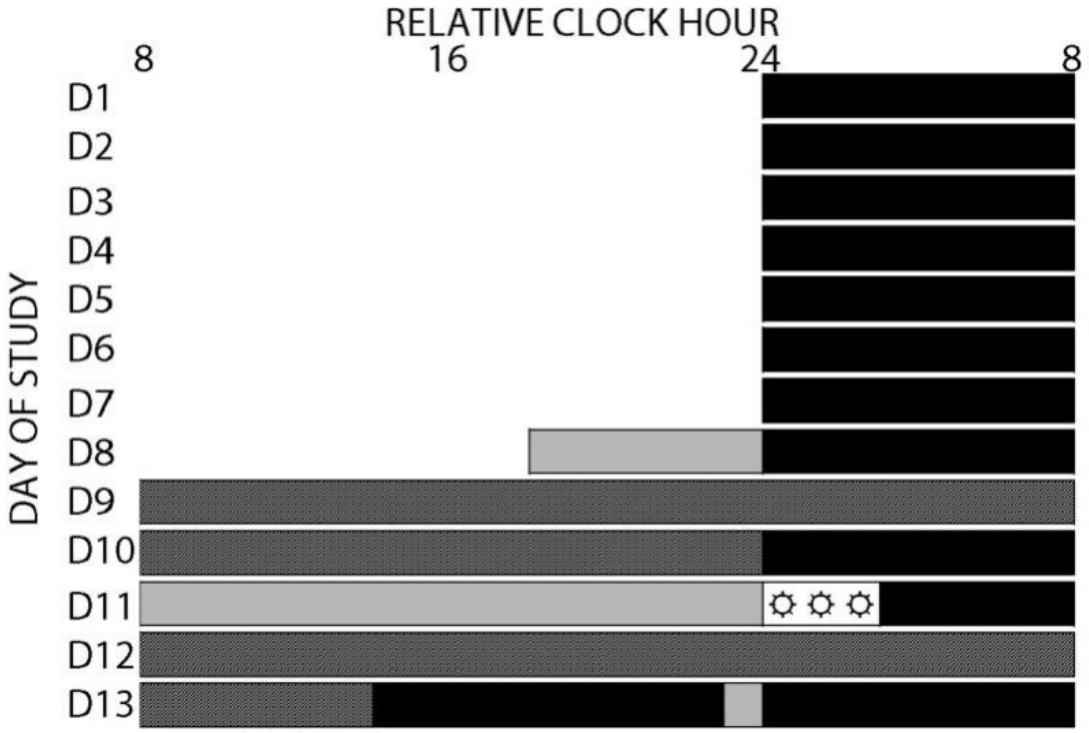
Philpott et al. (2020), Video 1.

Evening caffeine can delay the human melatonin rhythm

Burke and colleagues tested whether evening caffeine changes circadian timing in addition to increasing alertness. Participants received caffeine or a placebo before their habitual bedtime under controlled laboratory conditions. The researchers measured the melatonin rhythm before and after treatment, separating a shift in internal timing from the person's immediate feeling of being more awake.

Caffeine taken three hours before habitual bedtime delayed the subsequent melatonin rhythm by about forty minutes under the tested conditions. An evening bright-light condition also delayed the rhythm. These comparisons show why an intervention's timing is part of its biological action: the same exposure can affect tonight's alertness and the placement of the internal night.

In cultured human cells carrying clock reporters, caffeine lengthened molecular rhythms through mechanisms involving adenosine-receptor signaling and cyclic AMP, an intracellular messenger. A cell-level effect did not require a person to stay awake later by choice. The experiments therefore identified both a human phase shift and a plausible receiving-cell pathway through which caffeine can alter clock function.



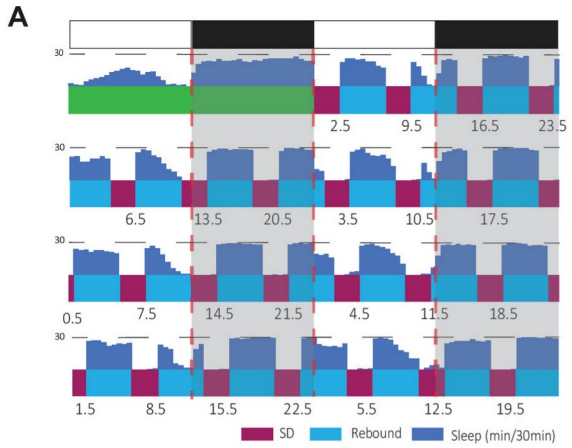
Burke et al. (2015), Fig. 1, selected panels.

Morning sleep loss produces strong rebound in flies

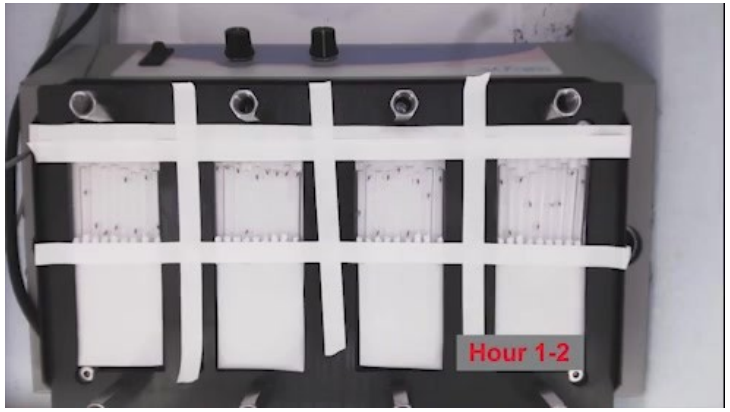
Andreani and colleagues deprived flies of sleep for the same short duration at different times of day. After deprivation ending in the morning, the animals showed strong recovery sleep. The original recording follows the hours after that morning interruption, with source time labels showing how the accelerated movie corresponds to the behavioral observation period.

The researchers measured rebound relative to baseline sleep at the same time of day. This comparison matters because an animal may normally sleep more in the morning than in the evening. Additional sleep after deprivation must be separated from the daily sleep pattern that would have occurred even without the experimental interruption.

Neurons in the **ellipsoid body**, a central fly-brain structure, participate in tracking sleep need. After morning sleep loss, a defined group showed increased calcium and changes in presynaptic markers. The clock could therefore influence the cellular response to recent waking, rather than contributing only a separate signal added to an otherwise time-independent homeostat.



Andreani et al. (2022), Fig. 1, selected panels.



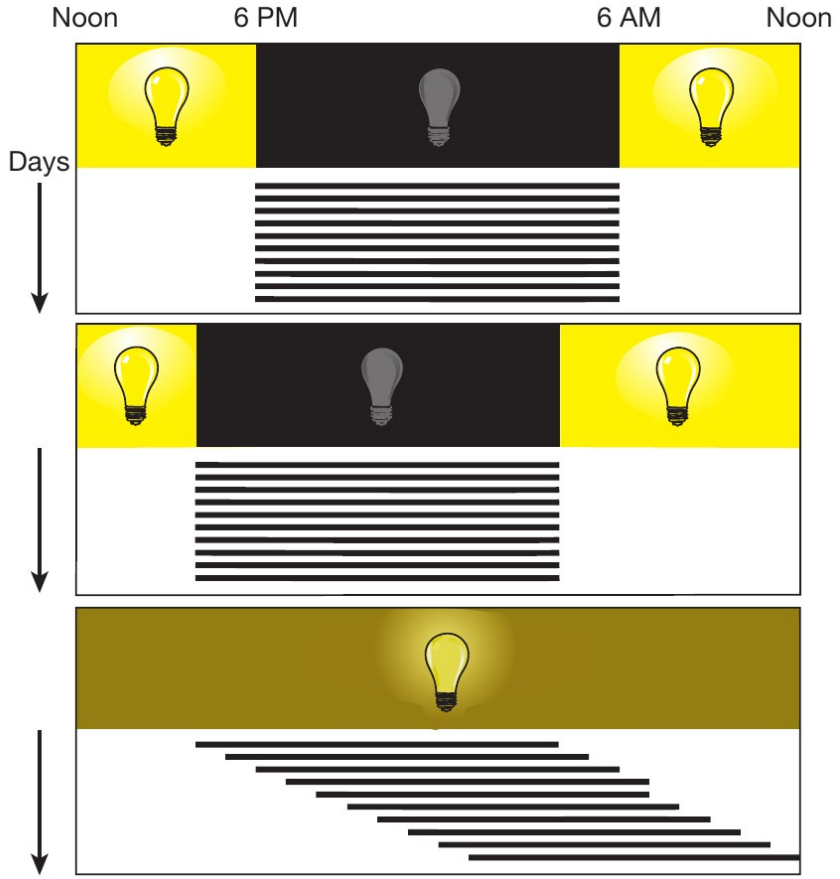
Andreani et al. (2022), Video 1.

Jet lag and shift work misalign internal and external time

Jet lag occurs when the internal clock remains aligned with the previous time zone while the environment has changed. Local bedtime can then arrive during the traveler's biological day, or waking obligations can occur during the internal night. Sleepiness, appetite, and temperature rhythms may initially occur at inconvenient local times.

Different body systems need not reset at the same rate. Light can change SCN timing, while meal timing influences clocks in metabolic tissues. During adjustment, central and peripheral rhythms may temporarily adopt different relationships, so returning to a regular bedtime does not instantly realign every physiological cycle.

Night-shift work can create recurring misalignment when a person must be active during the biological night and sleep during daylight. Daytime light, social schedules, and days off can oppose adaptation to the shift. The resulting sleep pattern depends on both accumulated pressure and the clock's continued promotion of alertness during part of the intended sleep period.



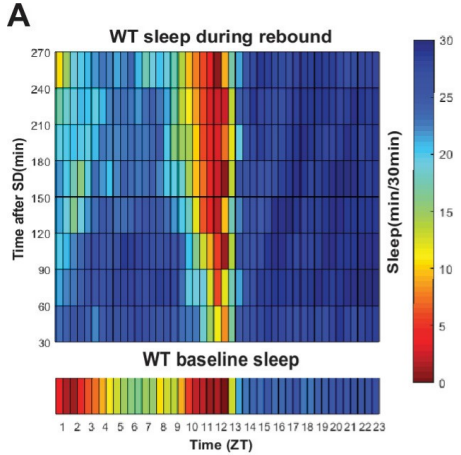
Carlson and Birkett (2022), Fig. 9.25, p. 293.

The same sleep loss produces less rebound in the evening

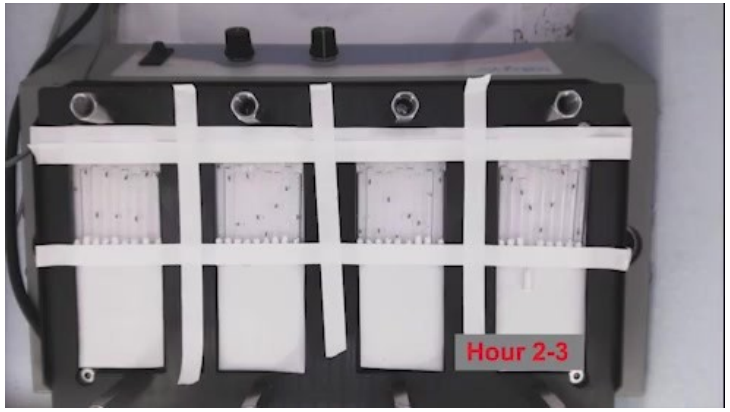
When the same duration of deprivation ended in the evening, Andreani and colleagues observed less rebound than after morning deprivation. The source movie shows the greater activity during the later recovery interval. Equal time kept awake therefore did not produce equal immediate recovery sleep at every circadian phase.

The ellipsoid-body neurons also responded differently: the calcium increase observed after morning sleep loss was not present in the same way after evening loss. Clock-neuron manipulations changed this time-dependent regulation. The homeostatic circuit's cellular response was therefore being programmed by circadian input, rather than responding only to the number of minutes awake.

The investigators distinguished clock populations associated with morning and evening timing. Their influence on rebound could be separated from effects on the ordinary daily locomotor rhythm. A clock circuit can regulate how sleep need is expressed without producing an identical change in every behavior that normally varies across the day.



Andreani et al. (2022), Fig. 2, selected panels.



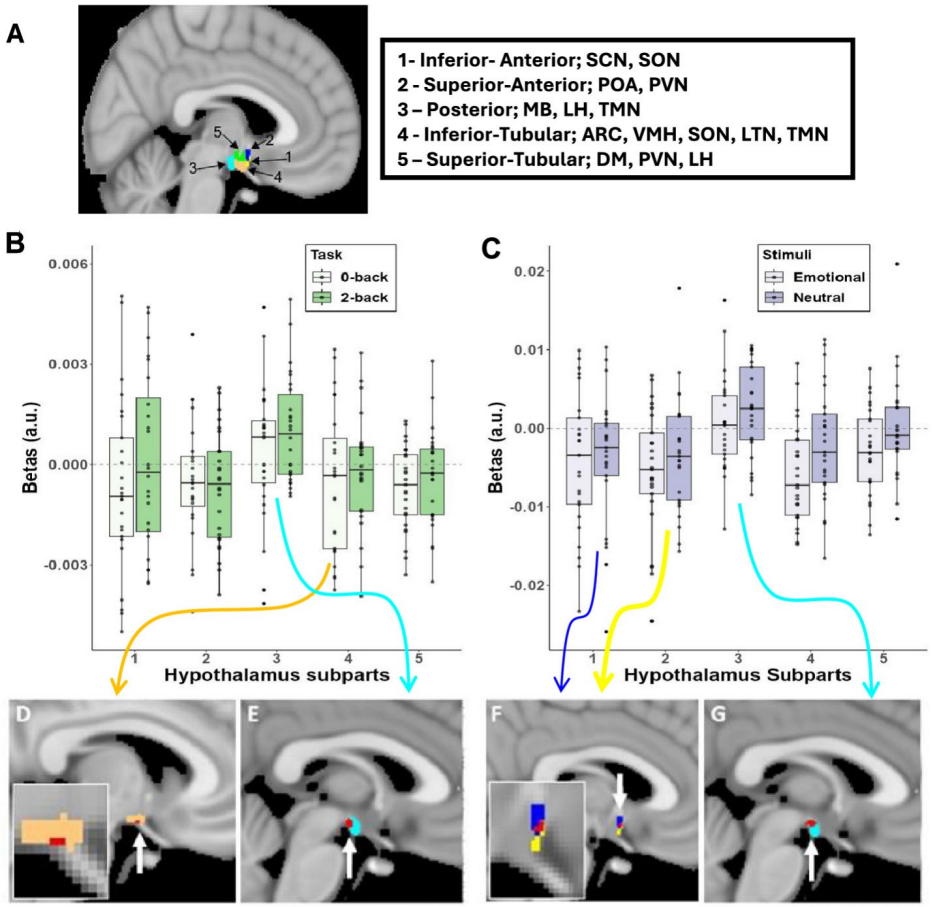
Andreani et al. (2022), Video 2.

Light changes activity across the human hypothalamus

Light can influence alertness and cognition in addition to resetting circadian phase. Campbell and colleagues used high-field functional MRI while participants performed auditory tasks under different illumination levels. The measurement detects blood-oxygenation-related changes, allowing small hypothalamic regions to be compared while visual illumination changes during an ongoing cognitive task.

Brighter illumination increased the measured response in posterior hypothalamic regions while decreasing it in some anterior and ventral regions. The hypothalamus therefore did not respond as one uniform block. Areas containing parts of arousal-related and clock-related systems showed different response patterns within the same broad anatomical structure.

Executive-task performance also improved under brighter conditions, but an anatomical MRI region contains several cell types and small nuclei. Its blood-related signal cannot identify the firing of a specific orexin or histamine neuron. The experiment measures a regional human response to light, complementing cellular studies that can isolate particular receiving neurons and receptors.



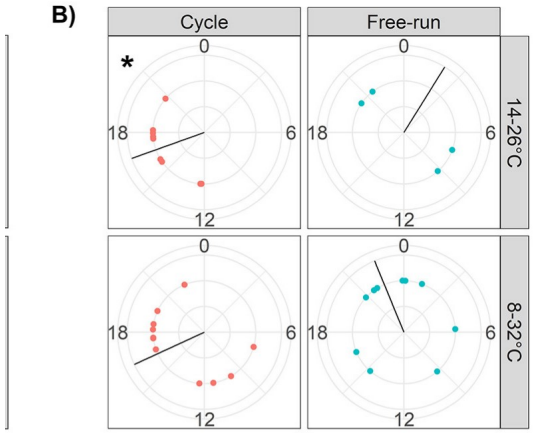
Campbell et al. (2024), Fig. 2, selected panels.

Conflicting time cues can disrupt an animal's rhythms

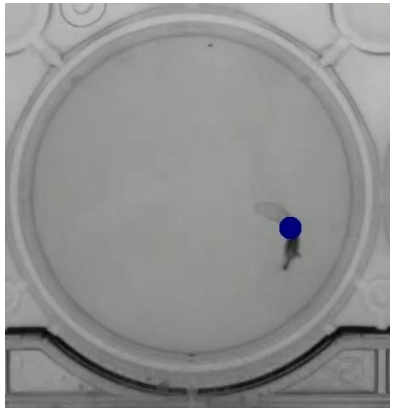
A **zeitgeber** is an environmental timing cue that can synchronize a biological rhythm. Berger and Tarrant studied light and temperature cues in the sea anemone Nematostella. Temperature cycles organized daily locomotor behavior, allowing the investigators to test what happens when more than one environmental signal supplies information about time.

The source recording follows real anemone movement with a computer-tracked center point. The authors accelerated playback, so the visible motion spans a much longer behavioral interval than its screen duration. Tracking converts changes in position into a time series that can be compared with the light and temperature schedules experienced by the animal.

When the light and temperature cycles were chronically misaligned, behavioral and gene-expression rhythms became disrupted. Some metabolic genes nevertheless retained rhythms related to temperature. Conflicting cues therefore affected parts of the rhythmic system differently, rather than requiring every measured process to lose its daily organization at the same moment.



Berger and Tarrant (2023), Fig. 2, selected panels.



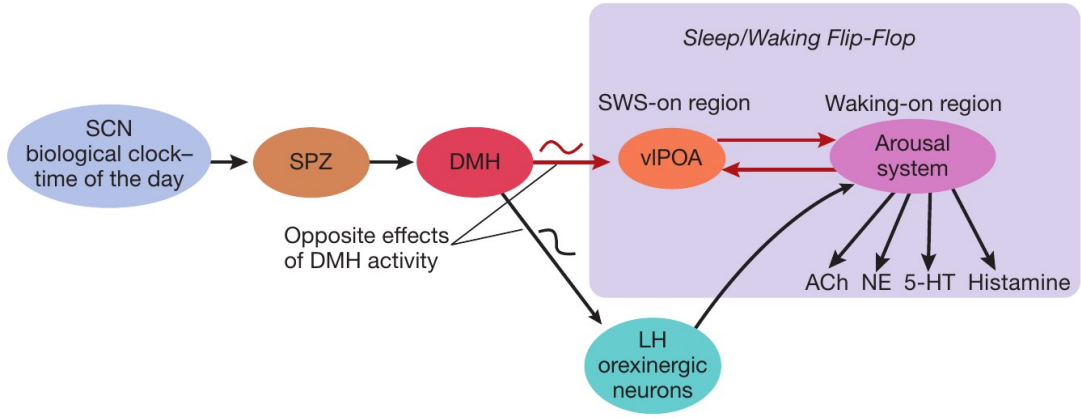
Berger and Tarrant (2023), Video 1.

Brain and body clocks coordinate different daily functions

Many tissues contain cellular clocks, including the liver, pancreas, and other organs involved in metabolism. Their rhythms can organize when enzymes, transporters, and hormone responses are most active. The SCN coordinates these systems through neural and hormonal outputs while the tissues also respond to local signals such as the timing of food intake.

Coordination does not require every organ to peak at the same hour. A useful relationship may place one process before another, such as preparing metabolic pathways before a predictable feeding period. **Internal synchrony** means maintaining appropriate phase relationships among rhythms, not making every physiological measurement rise and fall together.

A change in external schedule can temporarily alter those relationships. The light-sensitive central clock, feeding-sensitive peripheral tissues, and sleep-pressure system receive different information and operate through different mechanisms. Recording several outputs, such as melatonin, activity, meals, and temperature, reveals which parts have shifted and which remain aligned with the earlier schedule.



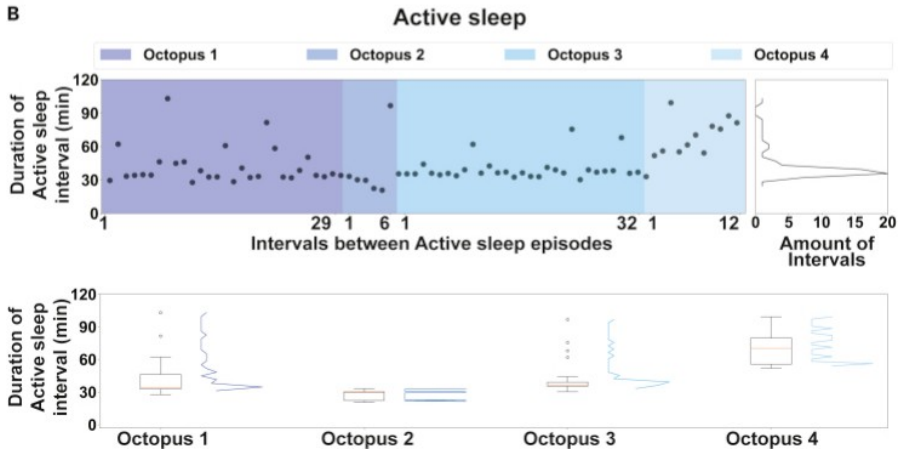
Carlson and Birkett (2022), Fig. 9.28, p. 296.

Sleep cycles can repeat within the daily rhythm

An **ultradian rhythm** repeats more often than once per day. Medeiros and colleagues observed recurring active-sleep episodes in octopuses, commonly separated by longer quiet-sleep periods. Many intervals between active episodes were roughly half an hour, providing a shorter cycle nested within the animal's broader daily pattern of behavior.

The source recording shows active sleep at night, with changing skin patterns and small movements. Comparing this bout with daytime examples establishes that the visible state can occur in more than one part of the daily cycle. Its classification depends on the combined state features, rather than the time of day alone.

Circadian timing and ultradian state transitions therefore describe different levels of organization. A daily rhythm can influence when sleep is likely, while the sleeping animal alternates between shorter recurring states. Counting active episodes without their quiet-sleep intervals would miss the sequence that gives those bouts their place within the overall sleep pattern.



Medeiros et al. (2021), Fig. 6, selected panels.



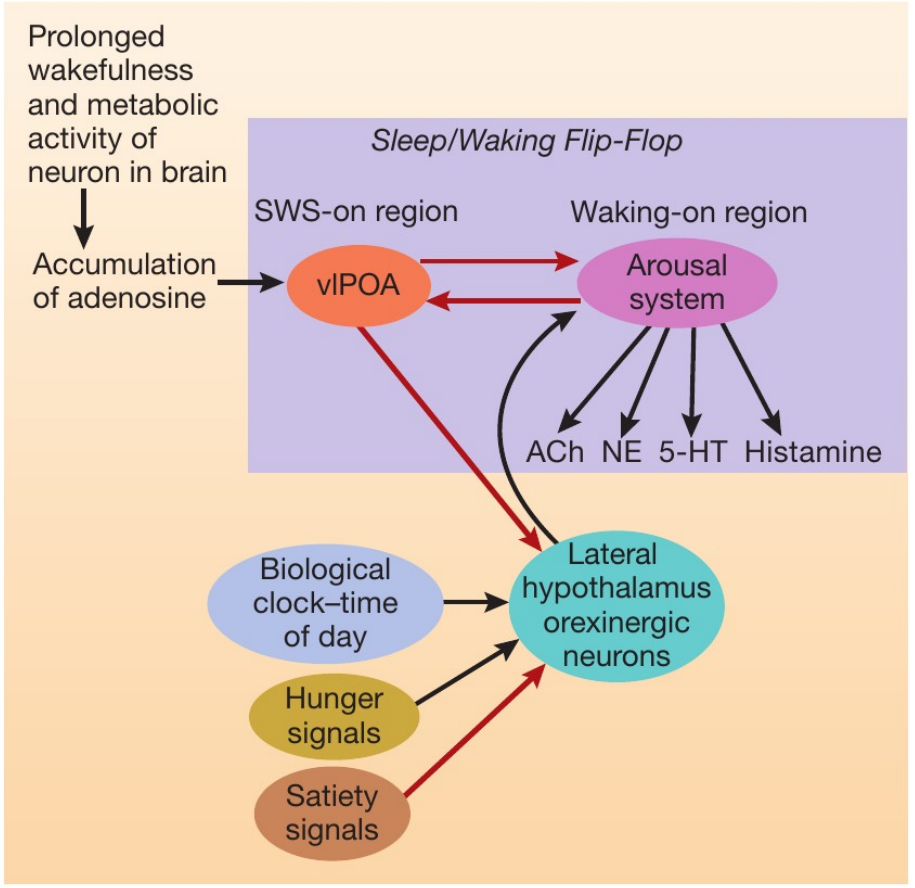
Medeiros et al. (2021), Video S8.

Measure sleep pressure and clock timing separately

Sleep can be measured along several independent dimensions: when it occurs, how long it lasts, how its stages are arranged, and how strongly the animal responds to stimulation. Electrical intensity adds another dimension. A manipulation that increases slow-wave activity has changed a different quantity from one that merely lengthens the opportunity to remain asleep.

Homeostatic experiments alter recent sleep history, while circadian experiments measure or manipulate biological time. Comparing deprivation at several phases can reveal interactions between these systems. Measuring melatonin or another clock marker helps determine whether a delayed sleep period reflects changed internal timing, accumulated waking, or a combination of both influences.

After prolonged waking, stronger early-night slow-wave activity can indicate elevated sleep pressure. If melatonin onset also occurs later, the clock has shifted as well. A longer sleep period alone cannot distinguish these changes: the electrical intensity and phase marker provide separate measurements needed to identify the contributions of recent waking and internal biological time.



Carlson and Birkett (2022), Fig. 9.20, p. 286.

Key takeaways

Sleep is an active, reversible state. Combine behavior, responsiveness, and electrical or physiological measurements to identify its organization.

Pressure and biological time interact. Recent waking changes homeostatic need, while circadian circuits regulate when that need is expressed.

State control is distributed. Inhibitory sleep circuits, arousal transmitters, orexin, and glial signaling change distinct properties of sleep.

Sleep rhythms organize communication. Slow oscillations, spindles, and hippocampal ripples coordinate activity relevant to memory processing.

State components can separate. Narcolepsy, cataplexy, parasomnias, and apnea disrupt different combinations of awareness, muscle control, and physiology.

Cellular clocks require timed feedback. Light resets SCN networks, while protein turnover and tissue coordination determine daily phase and period.

Carlson & Birkett (2022); primary sources in slide notes